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M3dmod functions

Synopsis

The functions prefixed with M3dmod make up the 3D model finder module.

Functions

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M3dmodAlloc

Synopsis

Allocate a 3D model finder context.

Syntax

```
AIL_ID M3dmodAlloc(
    AIL_ID SysId,             //in
    AIL_INT64 ContextType,    //in
    AIL_INT64 ControlFlag,    //in
    AIL_ID *Context3dmodIdPtr //out
)
```

Description

This function allocates a 3D model finder context on the specified system. A 3D model finder context contains information needed to perform a M3dmodFind() search. You can also use **M3dmodAlloc**() to allocate a draw 3D model finder context for drawing results using M3dmodDraw3d().

Define and add a model to the model finder context using M3dmodControl(). You can define one model per model finder context.

When the 3D model finder context is no longer required, release it using M3dmodFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the 3D model finder context's lifetime and you must not manually free it.

Parameters

SysId

Specifies the identifier of the system on which to allocate the 3D model finder context. Set this parameter to one of the values below:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ContextType

Specifies the type of 3D model finder context to allocate. Set this parameter to one of the values below:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of 3D model context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DRAW_3D_GEOMETRIC_CONTEXT	Allocates a draw geometric 3D model finder context for use with M3dmodDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_3D_SURFACE_CONTEXT	Allocates a draw surface 3D model finder context for use with M3dmodDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_BOX_CONTEXT	Allocates a find box 3D model finder context for use with M3dmodFind().	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_CYLINDER_CONTEXT	Allocates a find cylinder 3D model finder context for use with M3dmodFind().	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_PLANAR_SURFACE_CONTEXT	Allocates a find planar surface 3D model finder context for use with M3dmodFind().	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_RECTANGULAR_PLANE_CONTEXT	Allocates a find rectangular plane 3D model finder context for use with M3dmodFind().	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_SPHERE_CONTEXT	Allocates a find sphere 3D model finder context for use with M3dmodFind().	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_SURFACE_CONTEXT	Allocates a find surface 3D model finder context for use with M3dmodFind().	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Context3dmodIdPtr

Specifies the address of the variable in which to write the 3D model finder context identifier or specifies the data type that the function should use to return the 3D model finder context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D model finder context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b

M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D model finder context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_3DMOD_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the 3D model finder context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the draw geometric 3D model finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated draw geometric 3D model finder context identifier. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the draw surface 3D model finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated draw surface 3D model finder context identifier. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the find box 3D model finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated box 3D model finder context identifier. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Address in which to write the find cylinder 3D model finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated cylinder 3D model finder context identifier. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the find planar surface 3D model finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated planar surface 3D model finder context identifier. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the find rectangular plane 3D model finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated rectangular plane 3D model finder context identifier. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the find sphere 3D model finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated sphere 3D model finder context identifier. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the find surface 3D model finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated surface 3D model finder context identifier. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the 3D model finder context identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_3DMOD_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodAllocResult

Synopsis

Allocate a 3D model finder result buffer.

Syntax

```
AIL_ID M3dmodAllocResult(
    AIL_ID SysId,            //in
    AIL_INT64 ResultType,    //in
    AIL_INT64 ControlFlag,   //in
    AIL_ID *Result3dmodIdPtr //out
)
```

Description

This function allocates a 3D model finder result buffer, on the specified system, to store results obtained from an M3dmodFind() operation.

When the 3D model finder result buffer is no longer required, release it using M3dmodFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the 3D model finder result buffer's lifetime and you must not manually free it.

Parameters

SysId

Specifies the system on which to allocate the result buffer. This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ResultType

Specifies the type of result buffer to allocate.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of result buffer to allocate																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FIND_BOX_RESULT	Specifies to allocate a box 3D model finder result buffer used to store M3dmodFind() results.	c		e	f	g	a		i		j	k	l	m	n	b

M_FIND_CYLINDER_RESULT	Specifies to allocate a cylinder 3D model finder result buffer used to store M3dmodFind() results.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_PLANAR_SURFACE_RESULT	Specifies to allocate a planar surface 3D model finder result buffer used to store M3dmodFind() results.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_RECTANGULAR_PLANE_RESULT	Specifies to allocate a rectangular plane 3D model finder result buffer used to store M3dmodFind() results.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_SPHERE_RESULT	Specifies to allocate a sphere 3D model finder result buffer used to store M3dmodFind() results.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_SURFACE_RESULT	Specifies to allocate a surface 3D model finder result buffer used to store M3dmodFind() results.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion. This parameter must be set to **M_DEFAULT**.

Result3dmodIdPtr

Specifies the address of the variable in which to write the 3D model finder result buffer identifier or specifies the data type that the function should use to return the 3D model finder result buffer identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D model finder result buffer ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b

M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D model finder result buffer ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_3DMOD_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the 3D model finder result buffer (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the box 3D model finder result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated box 3D model finder result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the cylinder 3D model finder result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated cylinder 3D model finder result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the planar surface 3D model finder result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated planar surface 3D model finder result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the rectangular plane 3D model finder result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated rectangular plane 3D model finder result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Address in which to write the sphere 3D model finder result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated sphere 3D model finder result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the surface 3D model finder result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated surface 3D model finder result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the 3D model finder result buffer identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_3DMOD_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodControl

Synopsis

Control a setting of a find 3D model finder context or result buffer.

Syntax

```
void M3dmodControl(
    AIL_ID ContextOrResult3dmodId, //out
    AIL_INT64 Index,               //in
    AIL_INT64 ControlType,         //in
    AIL_DOUBLE ControlValue        //in
)
```

Description

This function controls a specified setting of a find 3D model finder context or result buffer. These settings control the execution of M3dmodFind(). You can inquire about most of these settings using M3dmodInquire().

> Note that changing control type settings of a 3D model finder context requires preprocessing the context again, using M3dmodPreprocess().

To control draw 3D model finder context settings, use M3dmodControlDraw() instead.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **M3dmodControl()** internally calls the *AIL_DOUBLE* version of this function (M3dmodControlDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (M3dmodControlInt64()).

Parameters

ContextOrResult3dmodId

Specifies the identifier of the find 3D model finder context or result buffer to control. The find 3D model finder context or result buffer must have been previously allocated on the required system using M3dmodAlloc() with M_FIND_..._CONTEXT or M3dmodAllocResult(), respectively.

See the Parameter associations section for possible values that can be returned.

Index

Specifies what to control. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a general context, result buffer or individual model																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies the default value. If a find 3D model finder context is specified, same as M_CONTEXT. If a find 3D model finder result buffer is specified, same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTEXT	Specifies to control a setting of a specified find 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b

M_GENERAL	Specifies to control a general setting of a specified find 3D model finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
0	Specifies to control the 3D model defined in the context. Note that the model must have been added to the context, using M3dmodDefine(), prior to calling **M3dmodControl**() with this value.	c		e	f	g	a		i		j	k	l	m	n	b

ControlType

Specifies the setting to change.

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the setting's new value.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The tables below list possible values for the **ControlType** and **ControlValue** parameters and possible values returned to the **ContextOrResult3dmodId** parameter.

- For controlling a find box, cylinder, rectangular plane, sphere, surface, or planar surface 3D model finder context
- For controlling a box, cylinder, rectangular plane, sphere, or surface 3D model
- For a 3D model finder result buffer

The following ContextOrResult3dmodId, ControlType, and ControlValue parameter settings can be specified for a find 3D model finder context when Index is set to M_CONTEXT or M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling a find box, cylinder, rectangular plane, sphere, surface, or planar surface 3D model finder context																	
ContextOrResult3dmodId - Possible values returned ControlType ControlValue	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Find box 3D model finder context ID	Specifies a find box 3D model finder context, allocated using M3dmodAlloc() with M_FIND_BOX_CONTEXT, and used in M3dmodFind() operations.															

M_DIRECTION_MODE	Sets how to interpret the reference direction (M_DIRECTION_REFERENCE_...). The reference direction typically matches the position of your 3D sensor or the direction of the 3D sensor's line of sight. When you know the exact position of the 3D sensor, set this control type to M_AWAY_FROM_POSITION or M_TOWARDS_POSITION, and set M_DIRECTION_REFERENCE_... to the 3D sensor's position. When you only know the direction of the 3D sensor's line of sight, set this control type to M_TOWARDS_DIRECTION, and set M_DIRECTION_REFERENCE_... to the components of the vector that points in this direction. The reference direction establishes when a single plane should be considered a box occurrence (when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1), given the line of sight of the 3D sensor and the specified completion tolerance (M_COMPLETION_ANGLE_TOLERANCE). M_DIRECTION_MODE specifies how to interpret the reference direction and the direction in which the box should be extruded (typically, M_AWAY_FROM_POSITION or M_TOWARDS_DIRECTION). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_TOWARDS_DIRECTION.	c		e	f	g	a		i		j	k	l	m	n	b
M_AWAY_FROM_POSITION	Specifies to interpret the reference direction as away from the position specified using M_DIRECTION_REFERENCE_....	c		e	f	g	a		i		j	k	l	m	n	b
M_TOWARDS_DIRECTION	Specifies to interpret the reference direction as towards the direction specified using M_DIRECTION_REFERENCE_....	c		e	f	g	a		i		j	k	l	m	n	b
M_TOWARDS_POSITION	Specifies to interpret the reference direction as towards the position specified using M_DIRECTION_REFERENCE_....	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION_REFERENCE_X	Sets the X-coordinate of the position, or the X-component of the vector, that determines the reference direction, depending on M_DIRECTION_MODE. The reference direction is used to identify the location of the 3D sensor when it acquired the target point cloud, so the reference direction should match the position of your 3D sensor or the direction of the 3D sensor's line of sight. For a find box 3D model finder context, this control type is only used when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the X-coordinate of the position or the X-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION_REFERENCE_Y		Sets the Y-coordinate of the position, or the Y-component of the vector, that determines the reference direction, depending on M_DIRECTION_MODE. The reference direction is used to identify the location of the 3D sensor when it acquired the target point cloud, so the reference direction should match the position of your 3D sensor or the direction of the 3D sensor's line of sight. For a find box 3D model finder context, this control type is only used when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-coordinate of the position or the Y-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION_REFERENCE_Z		Sets the Z-coordinate of the position, or the Z-component of the vector, that determines the reference direction, depending on M_DIRECTION_MODE. The reference direction is used to identify the location of the 3D sensor when it acquired the target point cloud, so the reference direction should match the position of your 3D sensor or the direction of the 3D sensor's line of sight. For a find box 3D model finder context, this control type is only used when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is -1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Z-coordinate of the position or the Z-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE		Sets the fit distance when M_FIT_DISTANCE_MODE is set to M_USER_DEFINED. The fit distance is used when determining whether points belong to a certain occurrence. Typically, to set an appropriate value, use the automatic fit distance as a baseline and adjust accordingly. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE_MODE		Sets how Aurora Imaging Library establishes the fit distance, which is used when determining whether points belong to a certain occurrence. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_AUTO.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO	Specifies to automatically calculate the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_USER_DEFINED	Specifies to use the value set with M_FIT_DISTANCE as the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT		Sets the sorting key for result retrieval. This is useful, for example, to retrieve the results of the highest occurrence first (the one with the highest Z-coordinate), instead of retrieving those with the highest score first. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SCORE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AREA	Specifies to sort results by the area of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_X	Specifies to sort the results by the X-coordinate of the occurrence's center point.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_Y	Specifies to sort the results by the Y-coordinate of the occurrence's center point.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_Z	Specifies to sort the results by the Z-coordinate of the occurrence's center point.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_X	Specifies to sort the results by the maximum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_Y	Specifies to sort the results by the maximum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_Z	Specifies to sort the results by the maximum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIN_X	Specifies to sort the results by the minimum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

M_MIN_Y	Specifies to sort the results by the minimum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Z	Specifies to sort the results by the minimum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS	Specifies to sort the results by the number of points in the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE	Specifies to sort the results by the score.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE_TARGET	Specifies to sort results by the target score.	c		e	f	g	a		i		j	k	l	m	n	b
M_VOLUME	Specifies to sort results by the volume of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_DIRECTION	Sets whether results are sorted in ascending or descending order. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_SORT_DOWN.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_DOWN	Specifies to sort the results in descending order.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_UP	Specifies to sort the results in ascending order.	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT	Sets the maximum amount of time for M3dmodFind() to complete the 3D model finder operation before generating a time-out error. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b
Find cylinder 3D model finder context ID	Specifies a find cylinder 3D model finder context, allocated using M3dmodAlloc() with M_FIND_CYLINDER_CONTEXT, and used in M3dmodFind() operations.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE	Sets the fit distance when M_FIT_DISTANCE_MODE is set to M_USER_DEFINED. The fit distance is used when determining whether points belong to a certain occurrence. Typically, to set an appropriate value, use the automatic fit distance as a baseline and adjust accordingly. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE_MODE		Sets how Aurora Imaging Library establishes the fit distance, which is used when determining whether points belong to a certain occurrence. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_AUTO.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO	Specifies to automatically calculate the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_USER_DEFINED	Specifies to use the value set with M_FIT_DISTANCE as the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_NORMALS_DISTANCE		Sets the acceptable deviation between a given point's normal vector and the normal vector of the model at the same point, to decide which points are used when fitting. Model points whose normal vector falls outside the specified deviation are not used to perform the occurrence's fit operation. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0 degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 90.0	Specifies the acceptable deviation from the normal vector of the model at the same point, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_PERSEVERANCE		Sets the algorithm's search perseverance when searching for occurrences. This affects the number of times the algorithm tries to find occurrences before giving up. Increasing the perseverance will increase robustness and accuracy, but will increase search time. You should try increasing the perseverance if the search is not finding the specified number of occurrences or if the target is complex. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 20.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the algorithm's search perseverance, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b

M_SORT	Sets the sorting key for result retrieval. This is useful, for example, to retrieve the results of the highest occurrence first (the one with the highest Z-coordinate), instead of retrieving those with the highest score first. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_SCORE.	c		e	f	g	a		i		j	k	l	m	n	b
M_AREA	Specifies to sort results by the area of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_X	Specifies to sort the results by the X-coordinate of the occurrence's center point. The center point of the cylinder occurrence is the center point on the cylinder's central axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Y	Specifies to sort the results by the Y-coordinate of the occurrence's center point. The center point of the cylinder occurrence is the center point on the cylinder's central axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Z	Specifies to sort the results by the Z-coordinate of the occurrence's center point. The center point of the cylinder occurrence is the center point on the cylinder's central axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_X	Specifies to sort the results by the maximum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_Y	Specifies to sort the results by the maximum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_Z	Specifies to sort the results by the maximum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_X	Specifies to sort the results by the minimum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Y	Specifies to sort the results by the minimum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Z	Specifies to sort the results by the minimum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS	Specifies to sort the results by the number of points in the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE	Specifies to sort the results by the score.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE_TARGET	Specifies to sort results by the target score.	c		e	f	g	a		i		j	k	l	m	n	b

M_VOLUME	Specifies to sort results by the volume of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_DIRECTION	Sets whether results are sorted in ascending or descending order. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_SORT_DOWN.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_DOWN	Specifies to sort the results in descending order.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_UP	Specifies to sort the results in ascending order.	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT	Sets the maximum amount of time for M3dmodFind() to complete the 3D model finder operation before generating a time-out error. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b
Find rectangular plane 3D model finder context ID	Specifies a find rectangular plane 3D model finder context, allocated using M3dmodAlloc() with M_FIND_RECTANGULAR_PLANE_CONTEXT, and used in M3dmodFind() operations.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE	Sets the fit distance when M_FIT_DISTANCE_MODE is set to M_USER_DEFINED. The fit distance is used when determining whether points belong to a certain occurrence. Typically, to set an appropriate value, use the automatic fit distance as a baseline and adjust accordingly. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE_MODE	Sets how Aurora Imaging Library establishes the fit distance, which is used when determining whether points belong to a certain occurrence. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_AUTO.	c		e	f	g	a		i		j	k	l	m	n	b

	M_AUTO	Specifies to automatically calculate the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_USER_DEFINED	Specifies to use the value set with M_FIT_DISTANCE as the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT		Sets the sorting key for result retrieval. This is useful, for example, to retrieve the results of the highest occurrence first (the one with the highest Z-coordinate), instead of retrieving those with the highest score first. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SCORE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AREA	Specifies to sort results by the area of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_X	Specifies to sort the results by the X-coordinate of the occurrence's center point. The center point of the rectangular plane occurrence is the center point of the rectangular plane's bounding box.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_Y	Specifies to sort the results by the Y-coordinate of the occurrence's center point. The center point of the rectangular plane occurrence is the center point of the rectangular plane's bounding box.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_Z	Specifies to sort the results by the Z-coordinate of the occurrence's center point. The center point of the rectangular plane occurrence is the center point of the rectangular plane's bounding box.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_X	Specifies to sort the results by the maximum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_Y	Specifies to sort the results by the maximum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_Z	Specifies to sort the results by the maximum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIN_X	Specifies to sort the results by the minimum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIN_Y	Specifies to sort the results by the minimum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

M_MIN_Z	Specifies to sort the results by the minimum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS	Specifies to sort the results by the number of points in the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE	Specifies to sort the results by the score.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE_TARGET	Specifies to sort results by the target score.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_DIRECTION	Sets whether results are sorted in ascending or descending order. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_SORT_DOWN.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_DOWN	Specifies to sort the results in descending order.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_UP	Specifies to sort the results in ascending order.	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT	Sets the maximum amount of time for M3dmodFind() to complete the 3D model finder operation before generating a time-out error. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b
Find sphere 3D model finder context ID	Specifies a find sphere 3D model finder context, allocated using M3dmodAlloc() with M_FIND_SPHERE_CONTEXT, and used in M3dmodFind() operations.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE	Sets the fit distance when M_FIT_DISTANCE_MODE is set to M_USER_DEFINED. The fit distance is used when determining whether points belong to a certain occurrence. Typically, to set an appropriate value, use the automatic fit distance as a baseline and adjust accordingly. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b

M_FIT_DISTANCE_MODE	Sets how Aurora Imaging Library establishes the fit distance, which is used when determining whether points belong to a certain occurrence. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_AUTO.	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO	Specifies to automatically calculate the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_DEFINED	Specifies to use the value set with M_FIT_DISTANCE as the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_ITERATIONS_MAX	Sets the maximum number of fit iterations to perform, when finding a sphere occurrence. More iterations increase fit accuracy, but increase search times. For a very tight tolerance, you can use a wider range with an initial search, and then repeat using a tighter range. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1	Specifies the maximum number of sphere fit iterations.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_NORMALS_DISTANCE	Sets the acceptable deviation between a given point's normal vector and the normal vector of the model at the same point, to decide which points are used when fitting. Model points whose normal vector falls outside the specified deviation are not used to perform the occurrence's fit operation. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 10.0 degrees.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 90.0	Specifies the acceptable deviation from the normal vector of the model at the same point, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_PERSEVERANCE	Sets the algorithm's search perseverance when searching for occurrences. This affects the number of times the algorithm tries to find occurrences before giving up. Increasing the perseverance will increase robustness and accuracy, but will increase search time. You should try increasing the perseverance if the search is not finding the specified number of occurrences or if the target is complex. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 20.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the algorithm's search perseverance, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT		Sets the sorting key for result retrieval. This is useful, for example, to retrieve the results of the highest occurrence first (the one with the highest Z-coordinate), instead of retrieving those with the highest score first. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SCORE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AREA	Specifies to sort results by the area of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_X	Specifies to sort the results by the X-coordinate of the occurrence's center point.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_Y	Specifies to sort the results by the Y-coordinate of the occurrence's center point.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_Z	Specifies to sort the results by the Z-coordinate of the occurrence's center point.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_X	Specifies to sort the results by the maximum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_Y	Specifies to sort the results by the maximum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_Z	Specifies to sort the results by the maximum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIN_X	Specifies to sort the results by the minimum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIN_Y	Specifies to sort the results by the minimum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIN_Z	Specifies to sort the results by the minimum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NUMBER_OF_POINTS	Specifies to sort the results by the number of points in the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE	Specifies to sort the results by the score.	c		e	f	g	a		i		j	k	l	m	n	b

	M_SCORE_TARGET	Specifies to sort results by the target score.	c		e	f	g	a		i		j	k	l	m	n	b
	M_VOLUME	Specifies to sort results by the volume of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_DIRECTION		Sets whether results are sorted in ascending or descending order. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SORT_DOWN.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SORT_DOWN	Specifies to sort the results in descending order.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SORT_UP	Specifies to sort the results in ascending order.	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT		Sets the maximum amount of time for M3dmodFind() to complete the 3D model finder operation before generating a time-out error. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b
Find surface or planar surface 3D model finder context ID		Specifies a find surface or planar surface 3D model finder context, allocated using M3dmodAlloc() with M_FIND_SURFACE_CONTEXT or M_FIND_PLANAR_SURFACE_CONTEXT, and used in M3dmodFind() operations.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONVERSION_GAMMA		Sets whether to remove gamma correction before the match. Gamma correction refers to color data that has been processed, typically during data acquisition (that is, by the camera doing the grab), to compensate for a non-linear transformation between a point's component value and its displayed intensity. If gamma correction has been applied on the RGB source color data, you must remove it. Note that this control type is only used and should only be enabled when M_USE_COLOR is set to M_ENABLE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies not to remove gamma correction. In this case, your RGB color data should not be in a corrected state (it should be linear). For example, the camera capturing the RGB color data has not applied a gamma correction.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to remove gamma correction. In this case, your RGB color data should be in a corrected state (it should be non-linear). For example, the camera capturing the RGB color data has applied a gamma correction.	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION_MODE		Sets how to interpret the reference direction (M_DIRECTION_REFERENCE_...). The reference direction typically matches the direction of the 3D sensor's line of sight. Set this control type to M_TOWARDS_DIRECTION, and set M_DIRECTION_REFERENCE_... to the components of the vector that points in this direction. For a find surface or planar surface 3D model finder context, this control type establishes which part of the model would be occluded at the position of an occurrence when calculating the score and M_SCENE_PROJECTION is enabled, given the line of sight of 3D sensor. M_DIRECTION_MODE specifies how to interpret the reference direction and the direction of projection. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_TOWARDS_DIRECTION.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TOWARDS_DIRECTION	Specifies to interpret the reference direction as towards the direction specified using M_DIRECTION_REFERENCE_....	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION_REFERENCE_X		Sets the X-component of the vector that determines the reference direction, depending on M_DIRECTION_MODE. The reference direction is used to identify the location of the 3D sensor when it acquired the target point cloud, so the reference direction should match the direction of the 3D sensor's line of sight. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the X-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b

M_DIRECTION_REFERENCE_Y	Sets the Y-component of the vector that determines the reference direction, depending on M_DIRECTION_MODE. The reference direction is used to identify the location of the 3D sensor when it acquired the target point cloud, so the reference direction should match the direction of the 3D sensor's line of sight. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION_REFERENCE_Z	Sets the Z-component of the vector that determines the reference direction, depending on M_DIRECTION_MODE. The reference direction is used to identify the location of the 3D sensor when it acquired the target point cloud, so the reference direction should match the direction of the 3D sensor's line of sight. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Z-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_EXHAUSTIVE_SEARCH	Sets whether to perform very robust but slow exhaustive searching. When exhaustive searching is enabled, all possibilities are tried, and M_PERSEVERANCE is ignored. Note that this control type is not available for find planar surface 3D model finder contexts. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies to disable exhaustive searching.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to enable exhaustive searching.	c		e	f	g	a		i		j	k	l	m	n	b

M_FIT_DISTANCE	Sets the fit distance when M_FIT_DISTANCE_MODE is set to M_USER_DEFINED. The fit distance is used when determining whether points belong to a certain occurrence. Typically, to set an appropriate value, use the automatic fit distance as a baseline and adjust accordingly. Note that for a find surface or planar surface 3D model finder context, this control type is only used when M_FIT_DISTANCE_MODE is set to M_USER_DEFINED and M_SAVE_FIT_INFO is enabled. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE_MODE	Sets how Aurora Imaging Library establishes the fit distance, which is used when determining whether points belong to a certain occurrence. Note that for a find surface or planar surface 3D model finder context, this control type is only used when M_SAVE_FIT_INFO is enabled. In this case, M_FIT_DISTANCE_MODE and, if applicable, M_FIT_DISTANCE only determine the fit distance when calculating the RMS error and the fit score. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_AUTO.	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO	Specifies to automatically calculate the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_DEFINED	Specifies to use the value set with M_FIT_DISTANCE as the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_NORMAL_SEARCH_MODE	Sets the search mode for calculating model normals when the defined model has no normals component (M_COMPONENT_NORMALS_AIL). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. For a find surface 3D model finder context, the default value is M_AUTO. For a find planar surface 3D model finder context, the default value is M_TREE.	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO	Specifies to automatically set the search mode based on the point cloud's organization; either M_ORGANIZED if organized or M_TREE if unorganized.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ORGANIZED	Specifies to use the point cloud's organizational structure to determine the model normals. This option is supported only for an organized point cloud.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TREE	Specifies to use a KD tree search mode to determine the model normals.	c		e	f	g	a		i		j	k	l	m	n	b
M_PERSEVERANCE		Sets the algorithm's search perseverance when searching for occurrences. This affects the number of times the algorithm tries to find occurrences before giving up. Increasing the perseverance will increase robustness and accuracy, but will increase search time. You should try increasing the perseverance if the search is not finding the specified number of occurrences or if the target is complex and increasing the setting of M_SCENE_COMPLEXITY is insufficient. You should also try increasing perseverance if the model is not well-defined (for example, it has multiple areas that look alike). Note that this control type is not available for find planar surface 3D model finder contexts. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 20.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the algorithm's search perseverance, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_REFINE_REGISTRATION		Sets whether and how to perform refine registration of each occurrence found. By default, for a surface model, Aurora Imaging Library uses the features of the model and target to find occurrences, but it doesn't do any fitting. To get more accurate score and pose information, you should enable refine registration. When enabled, 3D registration is internally performed between the model and each of the found occurrences. You can set M_REFINE_REGISTRATION to use a predefined or a custom registration context. Note that enabling refine registration increases processing time. To calculate and retrieve the root-mean square (RMS) error and fit score of the occurrences, enable M_SAVE_FIT_INFO. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to disable refine registration.	c		e	f	g	a		i		j	k	l	m	n	b

	M_FIND_SURFACE_REFINEMENT_FAST	Specifies to use a predefined registration context that prioritizes performance to fine-tune results.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIND_SURFACE_REFINEMENT_PRECISE	Specifies to use a predefined registration context that prioritizes accuracy to fine-tune results.	c		e	f	g	a		i		j	k	l	m	n	b
	M_USER_DEFINED_REGISTRATION	Specifies to use a custom registration context. Inquire the identifier of the internal registration context using M3dmodInquire() with M_USER_DEFINED_REGISTRATION_CONTEXT_ID, and then use M3dregControl() to configure the context. You can also use M3dmodCopy() with M_REFINE_REGISTRATION to initialize the internal context with the settings of a predefined registration context, and then adjust a few settings using M3dregControl().	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_BACKGROUND		Sets whether to remove background points from the target point cloud before performing the match. Background points include those of objects that are too big or too small to be part of an occurrence of the model. You should enable M_REMOVE_BACKGROUND for scenes that are very complex. Note that removing the background takes some time, but sometimes makes the match faster. You can use M3dmodDraw3d() to draw the points that were considered background points (M3dmodControlDraw() with M_DRAW_BACKGROUND_POINTS). Note that M_REMOVE_BACKGROUND does not currently use the floor plane nor the resting plane (M3dmodCopy() with M_FLOOR or M_RESTING_PLANE) to establish which points to remove. You can remove floor points, using M_REMOVE_FLOOR. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to remove background points.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to remove background points.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_BACKGROUND_SEARCH_MODE		Sets the search mode for finding the background points when M_REMOVE_BACKGROUND is enabled. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_AUTO.	c		e	f	g	a		i		j	k	l	m	n	b

	M_AUTO	Specifies to automatically set the search mode based on the organization of the point clouds; either M_ORGANIZED if organized or M_TREE if unorganized.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ORGANIZED	Specifies to use the organizational structure of the point clouds to determine the background points. This option is supported only if the model and target point clouds are organized.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TREE	Specifies to use a KD tree search mode to determine the background points.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_FLOOR		Sets whether to remove floor points from the target point cloud before performing the match. Points located either above or below (M_REMOVE_FLOOR_DIRECTION) the floor plane (M3dmodCopy() with M_FLOOR), that fall within the specified distance (M_REMOVE_FLOOR_OFFSET), are removed when this control type is enabled. Note that removing the floor points takes some time, but sometimes makes the match faster. You can use M3dmodDraw3d() to draw the points that were considered floor points (M3dmodControlDraw() with M_DRAW_FLOOR_POINTS). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to remove floor points.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to remove floor points.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_FLOOR_DIRECTION		Sets the direction, relative to the defined floor plane, in which to remove points when M_REMOVE_FLOOR is enabled. Use M_REMOVE_FLOOR_OFFSET to set the distance from the floor plane within which to remove points, or to have Aurora Imaging Library automatically establish the removal distance. You can use M3dmodDraw3d() to draw the points that were considered floor points (M3dmodControlDraw() with M_DRAW_FLOOR_POINTS). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_AUTO.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ABOVE	Specifies to remove points above the floor plane.	c		e	f	g	a		i		j	k	l	m	n	b

	M_AUTO	Specifies to remove points in the direction with the least number of valid points. The points on the side of the floor with the larger number of valid points are assumed to belong to actual objects, not noise.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BELOW	Specifies to remove points below the floor plane.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_FLOOR_EXPECTED_PERCENTAGE		Sets the percentage of total points in the target point cloud that are expected to be floor points when M_REMOVE_FLOOR is enabled and M_REMOVE_FLOOR_OFFSET is set to M_AUTO_VALUE. Aurora Imaging Library uses the expected percentage when automatically determining the distance from the floor plane within which to remove points. Use M_REMOVE_FLOOR_DIRECTION to set whether to remove points above or below the floor plane, or to have Aurora Imaging Library automatically establish the removal direction. You can use M3dmodDraw3d() to draw the points that were considered floor points (M3dmodControlDraw() with M_DRAW_FLOOR_POINTS). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b
	5.0 <= *Value* < 100.0	Specifies the expected floor points, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_FLOOR_OFFSET		Sets the offset, relative to the defined floor plane, within which to remove points when M_REMOVE_FLOOR is enabled. Use M_REMOVE_FLOOR_DIRECTION to set whether to remove points above or below the floor plane, or to have Aurora Imaging Library automatically establish the removal direction. You can use M3dmodDraw3d() to draw the points that were considered floor points (M3dmodControlDraw() with M_DRAW_FLOOR_POINTS). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_AUTO_VALUE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO_VALUE	Specifies to automatically determine the offset within which to remove points.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the offset within which to remove points.	c		e	f	g	a		i		j	k	l	m	n	b

M_REUSE_RESULT	Sets whether to reuse the result of the previous search at the beginning of the current search. If you expect occurrences to be found in similar locations to the occurrences found in a previous search, you can enable this control and pass the result buffer containing the results to M3dmodFind() to speed up the operation. The algorithm will search for occurrences at the previously found positions, and then resume with the search if more occurrences are expected. Note that if you want to reuse the same results for multiple searches, you must avoid passing that result buffer to M3dmodFind() again, so as not to change the stored results. You should set up two result buffers: one for the initial search and another for use with subsequent searches. Before each subsequent search, copy into its result buffer the results from the initial search, using M3dmodCopyResult() with M_RESULT, and use this result buffer for the search. You can use M3dmodModifyResult() to modify or delete results before performing the next search. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to reuse the previous result.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to reuse the previous result.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE_FIT_INFO	Sets whether to enable the calculation of the root-mean square (RMS) error and the fit score. When enabled, besides being able to obtain these results, you can filter out occurrences based on the minimum required fit score (M_FIT_SCORE_MIN) and the maximum fit distance (M_FIT_DISTANCE_MODE and M_FIT_DISTANCE). Note that M_SAVE_FIT_INFO and its related settings have no effect on M_REFINE_REGISTRATION, and vice versa. Even if refine registration is enabled, you must enable M_SAVE_FIT_INFO to calculate and retrieve the root-mean square (RMS) error and the fit score; when calculating these, M_FIT_DISTANCE_MODE and M_FIT_DISTANCE are used, and not the settings of the refine registration. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies to disable the calculation of the root-mean square (RMS) error and the fit score. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to enable the calculation of the root-mean square (RMS) error and the fit score.	c		e	f	g	a		i		j	k	l	m	n	b

M_SCENE_COMPLEXITY	Sets the complexity of the scene. It sets how much of the target data belongs to the occurrence. For example, if the target scene contains one occurrence of the model on a simple background, setting M_SCENE_COMPLEXITY to M_LOW should be sufficient to find the occurrence. Whereas, if the target scene is busy and contains many similar features as the model, you should set M_SCENE_COMPLEXITY to M_MEDIUM or M_HIGH. Higher settings result in a more robust operation, but typically take more time. Note that this control type is not available for find planar surface 3D model finder contexts. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_MEDIUM.	c		e	f	g	a		i		j	k	l	m	n	b
M_HIGH	Specifies that the scene has a high complexity.	c		e	f	g	a		i		j	k	l	m	n	b
M_LOW	Specifies that the scene has a low complexity.	c		e	f	g	a		i		j	k	l	m	n	b
M_MEDIUM	Specifies that the scene has a medium complexity.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCENE_PROJECTION	Sets whether to apply occlusion handling to the model. This discards points of the model that would be occluded at the position of an occurrence, given the line of sight of the 3D sensor, before comparing the model to the occurrence during the refine registration and score calculation. This prevents the occurrences' scores from being lowered due to how the scene was acquired. Aurora Imaging Library uses the reference direction (M_DIRECTION_REFERENCE_...) to determine the line of sight and direction of projection. The default (0,0,1), provides a projection along the 3D sensor's Z-axis. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies to disable occlusion handling.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to enable occlusion handling.	c		e	f	g	a		i		j	k	l	m	n	b

M_SCORE_MODE	Sets the mode with which to associate model and target points for score calculations. The score mode affects the model score, target score, and color score. Note that points are associated using a positional comparison between model and target points, relative to local volumes within a partitioned 3D space. When using the M_STRICT mode, a point is either fully associated with another point or not associated at all. For example, when calculating the model score, a model point is considered fully covered if there is a target point in the same volume, and uncovered if not. When using the M_FLEXIBLE mode, points can be partially associated. For example, when calculating the model score, if a model point's position is close to the boundary of a volume adjacent to the one it occupies and there is a target point in either of the volumes, the point is considered half covered; there must be target points in both volumes to fully cover the model point. Note that the M_FLEXIBLE mode does not always produce higher scores than the M_STRICT mode. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. For a find surface 3D model finder context, the default value is M_STRICT. For a find planar surface 3D model finder context, the default value is M_FLEXIBLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_FLEXIBLE	Specifies to use the flexible mode when calculating the scores.	c		e	f	g	a		i		j	k	l	m	n	b
M_STRICT	Specifies to use the strict mode when calculating the scores.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT	Sets the sorting key for result retrieval. This is useful, for example, to retrieve the results of the highest occurrence first (the one with the highest Z-coordinate), instead of retrieving those with the highest score first. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_SCORE.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_X	Specifies to sort the results by the X-coordinate of the occurrence's center point. The center point of the surface occurrence is the center of the surface's bounding box.	c		e	f	g	a		i		j	k	l	m	n	b

		M_CENTER_Y	Specifies to sort the results by the Y-coordinate of the occurrence's center point. The center point of the surface occurrence is the center of the surface's bounding box.	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER_Z	Specifies to sort the results by the Z-coordinate of the occurrence's center point. The center point of the surface occurrence is the center of the surface's bounding box.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX_X	Specifies to sort the results by the maximum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX_Y	Specifies to sort the results by the maximum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX_Z	Specifies to sort the results by the maximum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN_X	Specifies to sort the results by the minimum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN_Y	Specifies to sort the results by the minimum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN_Z	Specifies to sort the results by the minimum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NO_SORT	Specifies not to sort the results; found occurrences are returned in an arbitrary order.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NUMBER_OF_POINTS	Specifies to sort the results by the number of points in the occurrence. Note that this control value is only used when M_SAVE_FIT_INFO is set to M_ENABLE; otherwise, found occurrences are returned in an arbitrary order.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE	Specifies to sort the results by the score.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_COLOR	Specifies to sort results by the color score. Note that this control value is only used when M_USE_COLOR is set to M_ENABLE; otherwise, found occurrences are returned in an arbitrary order.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_FIT	Specifies to sort results by the fit score. Note that this control value is only used when M_SAVE_FIT_INFO is set to M_ENABLE; otherwise, found occurrences are returned in an arbitrary order.	c		e	f	g	a		i		j	k	l	m	n	b

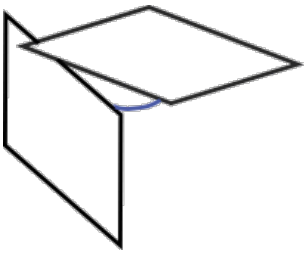
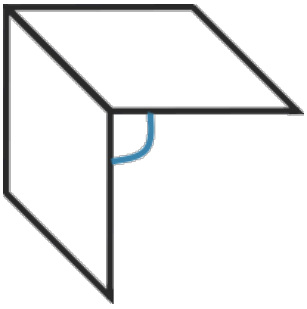
	M_SCORE_GRIP_BEST	Specifies to sort results by the best grip score. Note that this control value is only used if a grip label image that contains non-zero values was copied to the find surface 3D model finder context, using M3dmodCopy() with M_GRIP_LABEL_IMAGE; otherwise, found occurrences are returned in an arbitrary order.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE_TARGET	Specifies to sort results by the target score.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_DIRECTION		Sets whether results are sorted in ascending or descending order. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SORT_DOWN.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SORT_DOWN	Specifies to sort the results in descending order.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SORT_UP	Specifies to sort the results in ascending order.	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT		Sets the maximum amount of time for M3dmodFind() to complete the 3D model finder operation before generating a time-out error. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b

M_USE_COLOR	Sets whether to use the color information of the model and scene when searching for surface occurrences. This is useful when the model is similar in shape to other background objects in the scene, but has a different color distribution. When enabled, an occurrence will be returned only if its color score is greater than or equal to the color score acceptance level (M_ACCEPTANCE_COLOR). The color data is assumed to be in RGB format and stored in either the reflectance or intensity component of the respective point cloud container. The component must be a 3-band 8-bit unsigned buffer and have the same dimensions as the range component. The reflectance component is used if a valid reflectance component exists in the container. If a valid reflectance component does not exist and the container has a valid intensity component, the intensity component is used. Note that if gamma correction has been applied on the RGB source color data and you enable this control type, you must remove it, using M_CONVERSION_GAMMA set to M_ENABLE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to use the color information.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to use the color information.	c		e	f	g	a		i		j	k	l	m	n	b

The following ContextOrResult3dmodId, ControlType, and ControlValue parameter settings can be specified for a box, cylinder, rectangular plane, sphere, or surface 3D model when Index is set to 0.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling a box, cylinder, rectangular plane, sphere, or surface 3D model																
ContextOrResult3dmodId - Possible values returned	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ControlType																
ControlValue																
Find box 3D model finder context ID with a box model	Specifies a find box 3D model finder context, allocated using M3dmodAlloc() with M_FIND_BOX_CONTEXT, and used in M3dmodFind() operations. It must contain a box model.															

M_ACCEPTANCE	Sets the acceptance level for the score. An occurrence will be returned only if the match score between the target and the model is greater than or equal to this level. The score is a measure, as a percentage, of the model's surface that is covered by inlier points at the location of the occurrence. The score is defined relative to the maximum expected model coverage, such that an occurrence with a score of 100% has a model coverage equal to the maximum expected model coverage. Set the maximum expected coverage with M_COVERAGE_MAX. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 60.0%.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies an acceptable score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_FACE_PARALLELISM_THRESHOLD	Sets the maximum shear angle allowed between adjacent rectangles for them to be considered faces of the same box occurrence. INQ 	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 20.0.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 45.0	Specifies the maximum angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_FACE_PERPENDICULARITY_THRESHOLD	Sets the maximum deviation from 90 degrees allowed between adjacent rectangles for them to be considered faces of the same box occurrence. INQ 	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 10.0.	c		e	f	g	a		i		j	k	l	m	n	b

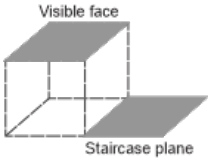
	0.0 < *Value* <= 90.0	Specifies the maximum angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_CERTAINTY		Sets the certainty level for the score, as a percentage. If the score is greater than or equal to the specified certainty level, the occurrence is considered a match, without searching the rest of the target for better matches (provided the specified number of occurrences has been found). The certainty level is defined relative to the maximum expected model coverage, such that setting M_CERTAINTY to 100% indicates that certain matches must have a total model coverage equal to the maximum expected model coverage. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 90.0%.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the certainty level for the score, as a percentage. If you set the certainty level too high (close to 100.0%), you slow down the search because you force the search algorithm to check the whole point cloud for the best possible match(es). A good certainty level is slightly lower than the expected score, so that the search can finish as soon as a match is found. However, if you set the certainty level too low, false matches might be found.	c		e	f	g	a		i		j	k	l	m	n	b
M_COMPLETION_ANGLE_TOLERANCE		Sets the completion angular tolerance, when only one box face (plane) is found. Aurora Imaging Library only extrudes the face if its normal is less than the specified number of degrees away from the reference direction (M_DIRECTION_REFERENCE_...). This control type is used only when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. Note that when only one box face is found, and the completion angular tolerance is not met, the occurrence is not returned as a match. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 90.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 90.0	Specifies the completion angular tolerance, when only one box face (plane) is found.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COMPLETION_SIZE_X	Sets the length along X to use to establish the missing dimension, when only one box face (plane) is found. Aurora Imaging Library compares the plane's two dimensions (length and width) to the specified completion sizes (M_COMPLETION_SIZE_...). The most dissimilar size determines the missing dimension. If M_COMPLETION_TO_USER_SIZE is enabled, the M_COMPLETION_SIZE_... of the missing dimension will be the length to which Aurora Imaging Library extrudes the visible face to complete the box, unless M_COMPLETION_TO_BACKGROUND and/or M_COMPLETION_TO_STAIRCASE are also enabled. If M_COMPLETION_TO_BACKGROUND and M_COMPLETION_TO_STAIRCASE are enabled, Aurora Imaging Library will attempt to extrude the visible face to the background plane and, if not possible, then to an edge of a staircase plane before using the completion size to complete the box. The completion methods are attempted in the following order: 1. M_COMPLETION_TO_BACKGROUND. 2. M_COMPLETION_TO_STAIRCASE. 3. M_COMPLETION_TO_USER_SIZE. Note that the completion sizes refer to the box model's dimensions and are not related to the axes of the working coordinate system. This control type is used only when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SAME_AS_NOMINAL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME_AS_NOMINAL	Specifies that the length along X, to use to establish the missing dimension, is the same as the nominal length in X of a nominal model. When searching for a range-type model, this corresponds to the middle of the specified size range (M_SIZE_X_MAX - M_SIZE_X_MIN).	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the length along X to use to establish the missing dimension.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COMPLETION_SIZE_Y	Sets the length along Y to use to establish the missing dimension, when only one box face (plane) is found. Aurora Imaging Library compares the plane's two dimensions (length and width) to the specified completion sizes (M_COMPLETION_SIZE_...). The most dissimilar size determines the missing dimension. If M_COMPLETION_TO_USER_SIZE is enabled, the M_COMPLETION_SIZE_... of the missing dimension will be the length to which Aurora Imaging Library extrudes the visible face to complete the box, unless M_COMPLETION_TO_BACKGROUND and/or M_COMPLETION_TO_STAIRCASE are also enabled. If M_COMPLETION_TO_BACKGROUND and M_COMPLETION_TO_STAIRCASE are enabled, Aurora Imaging Library will attempt to extrude the visible face to the background plane and, if not possible, then to an edge of a staircase plane before using the completion size to complete the box. The completion methods are attempted in the following order: 1. M_COMPLETION_TO_BACKGROUND. 2. M_COMPLETION_TO_STAIRCASE. 3. M_COMPLETION_TO_USER_SIZE. Note that the completion sizes refer to the box model's dimensions and are not related to the axes of the working coordinate system. This control type is used only when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SAME_AS_NOMINAL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME_AS_NOMINAL	Specifies that the length along Y, to use to establish the missing dimension, is the same as the nominal length in Y of a nominal model. When searching for a range-type model, this corresponds to the middle of the specified size range (M_SIZE_Y_MAX - M_SIZE_Y_MIN).	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the length along Y to use to establish the missing dimension.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COMPLETION_SIZE_Z	Sets the length along Z to use to establish the missing dimension, when only one box face (plane) is found. Aurora Imaging Library compares the plane's two dimensions (length and width) to the specified completion sizes (M_COMPLETION_SIZE_...). The most dissimilar size determines the missing dimension. If M_COMPLETION_TO_USER_SIZE is enabled, the M_COMPLETION_SIZE_... of the missing dimension will be the length to which Aurora Imaging Library extrudes the visible face to complete the box, unless M_COMPLETION_TO_BACKGROUND and/or M_COMPLETION_TO_STAIRCASE are also enabled. If M_COMPLETION_TO_BACKGROUND and M_COMPLETION_TO_STAIRCASE are enabled, Aurora Imaging Library will attempt to extrude the visible face to the background plane and, if not possible, then to an edge of a staircase plane before using the completion size to complete the box. The completion methods are attempted in the following order: 1. M_COMPLETION_TO_BACKGROUND. 2. M_COMPLETION_TO_STAIRCASE. 3. M_COMPLETION_TO_USER_SIZE. Note that the completion sizes refer to the box model's dimensions and are not related to the axes of the working coordinate system. This control type is used only when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SAME_AS_NOMINAL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME_AS_NOMINAL	Specifies that the length along Z, to use to establish the missing dimension, is the same as the nominal length in Z of a nominal model. When searching for a range-type model, this corresponds to the middle of the specified size range (M_SIZE_Z_MAX - M_SIZE_Z_MIN).	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the length along Z to use to establish the missing dimension.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COMPLETION_TO_BACKGROUND	Sets whether the visible face can be extruded to a background plane to complete the box, when only one box face (plane) is found. When enabled, Aurora Imaging Library will attempt to extrude the face to the closest background plane (includes the floor plane, if defined using M3dmodCopy() with M_FLOOR, and planes found in the scene) that intersects with the projection of the face. Note that the background plane must intersect with at least 80% of the projection of the face. If the closest background plane would yield an extrusion smaller than M_SIZE_..._MIN, the face cannot be extruded. In this case, no other completion methods will be attempted and the box is rejected. If no background plane intersects with the projection of the face, or the closest one would yield an extrusion larger than M_SIZE_..._MAX, Aurora Imaging Library will attempt to extrude the face using another completion method, if one is enabled. The completion methods are attempted in the following order: 1. M_COMPLETION_TO_BACKGROUND. 2. M_COMPLETION_TO_STAIRCASE. 3. M_COMPLETION_TO_USER_SIZE. This control type is used only when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to complete the box using a background plane.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to attempt to complete the box using a background plane.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COMPLETION_TO_STAIRCASE	Sets whether the visible face can be extruded to an edge of a staircase plane to complete the box, when only one box face (plane) is found. A staircase plane is any plane whose edge touches the extrusion of the face (if the box were extruded to this plane, the two would seem like a staircase).  When enabled, if M_COMPLETION_TO_BACKGROUND was disabled, or the closest background plane was too far to complete the box, Aurora Imaging Library will attempt to extrude the face to the farthest plane (includes the floor plane, if defined using M3dmodCopy() with M_FLOOR, and planes found in the scene) whose edge touches the projection of the face and falls inside the size range for the missing dimension (for example, for range-type models, M_SIZE_..._MIN to M_SIZE_..._MAX). If no plane meets these criteria, the face will be extruded to the completion size, if M_COMPLETION_TO_USER_SIZE is enabled. The completion methods are attempted in the following order: 1. M_COMPLETION_TO_BACKGROUND. 2. M_COMPLETION_TO_STAIRCASE. 3. M_COMPLETION_TO_USER_SIZE. This control type is used only when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to complete the box using a staircase plane.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to attempt to complete the box using a staircase plane.	c		e	f	g	a		i		j	k	l	m	n	b

M_COMPLETION_TO_USER_SIZE	Sets whether the visible face can be extruded to the specified completion size to complete the box, when only one box face (plane) is found. When enabled, if no other completion method was successful, Aurora Imaging Library will extrude the face to M_COMPLETION_SIZE_... for the missing dimension. The completion methods are attempted in the following order: 1. M_COMPLETION_TO_BACKGROUND. 2. M_COMPLETION_TO_STAIRCASE. 3. M_COMPLETION_TO_USER_SIZE. Note that the completion sizes refer to the box model's dimensions and are not related to the axes of the working coordinate system. This control type is used only when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to complete the box using the completion size.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to complete the box using the completion size.	c		e	f	g	a		i		j	k	l	m	n	b
M_COVERAGE_MAX	Sets the maximum expected model coverage. The model coverage is the percentage of the model's surface covered with inlier points found in the occurrence. An occurrence with a model coverage greater than or equal to the maximum expected model coverage has a score of 100%. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.0.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 100.0	Specifies the maximum expected model coverage, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELONGATION_MAX	Sets the maximum elongation of the box occurrence. The elongation is defined as the maximum side/minimum side. This is used to prevent lines from being incorrectly identified as boxes when a very large size range is used. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b

		M_INFINITE	Specifies that the box occurrence has no maximum elongation. It can be infinitely long.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 1.0	Specifies the box occurrence's maximum elongation, defined as maximum side / minimum side.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELONGATION_MIN		Sets the minimum elongation of the box occurrence. The elongation is defined as the maximum side/minimum side. This is used to prevent lines from being incorrectly identified as boxes when a very large size range is used. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 1.0	Specifies the box occurrence's minimum elongation, defined as maximum side / minimum side.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NORMAL_ANGLE_TOLERANCE		Sets the angular tolerance to use for M_NORMAL_CONDITION, when searching for box occurrences. Aurora Imaging Library only returns a box occurrence as a match if the normal of one of its faces meets the specified M_NORMAL_CONDITION, +/- the specified angular tolerance, relative to the vector M_NORMAL_.... INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 10.0.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 < *Value* <= 90.0	Specifies the normal angular tolerance, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

M_NORMAL_CONDITION	Sets whether to find only box occurrences that satisfy the specified condition, relative to the vector M_NORMAL_..., or to find any box occurrence. A box occurrence is only considered a match if the normal of one of its faces meets the specified parallel condition when compared to the vector specified with M_NORMAL_... (within the specified tolerance M_NORMAL_ANGLE_TOLERANCE). For example, you can set M_NORMAL_... to the normal of the floor plane and then set M_NORMAL_CONDITION to M_PARALLEL to find box occurrences parallel to the floor. If you know that your scene only contains parallel boxes, you should use this control to force the search algorithm to ignore non-parallel candidates. This can help to avoid false matches, particularly when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. It can also help to speed up the search, since candidates that do not meet the parallel condition will not be further examined. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_UNCONDITIONAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_PARALLEL	Specifies that the normal of one of the faces must be parallel to M_NORMAL_..., within the specified angular tolerance.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNCONDITIONAL	Specifies that there is no constraint on the normal.	c		e	f	g	a		i		j	k	l	m	n	b
M_VISIBLE_FACE_PARALLEL	Specifies that the normal of one of the visible faces must be parallel to M_NORMAL_..., within the specified angular tolerance. If the face whose normal is parallel is inferred, the box occurrence is not selected as a match.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_X	Sets the X-component of the vector against which to compare box occurrences, as specified using M_NORMAL_CONDITION. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_Y	Sets the Y-component of the vector against which to compare box occurrences, as specified using M_NORMAL_CONDITION. INQ	c		e	f	g	a		i		j	k	l	m	n	b

M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_Z	Sets the Z-component of the vector against which to compare box occurrences, as specified using M_NORMAL_CONDITION. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Z-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER	Sets the maximum number of occurrences for which to search. Once the required number of occurrences with scores greater than or equal to M_CERTAINTY have been found, the search will stop to avoid exhaustive checking of all possible candidates. If not all the required number of occurrences have been found, Aurora Imaging Library will continue to search for the required number of matches greater than or equal to the acceptance level and with the best score. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Specifies to find all occurrences. Note that this setting can increase the search time; always set M_NUMBER to a specific number whenever possible.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the number of occurrences for which to search.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS_MIN	Sets the minimum number of points per occurrence found. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 50.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the minimum number of points per occurrence found.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_VISIBLE_FACES_MAX	Sets the maximum number of visible faces (planes) required for a box occurrence to be accepted. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 6.	c		e	f	g	a		i		j	k	l	m	n	b
	1 <= *Value* <= 6	Specifies the maximum number of visible faces (planes) required for a box occurrence to be accepted.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_VISIBLE_FACES_MIN		Sets the minimum number of visible faces (planes) required for a box occurrence to be accepted. If set to 1, a box can be extruded from a single plane according to the completion constraints (M_COMPLETION_...) and the reference direction (M_DIRECTION_REFERENCE_...). The plane will be extruded if its normal is less than M_COMPLETION_ANGLE_TOLERANCE away from the reference direction. If these constraints are not met, the plane is not considered a box occurrence. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 2.	c		e	f	g	a		i		j	k	l	m	n	b
	1 <= *Value* <= 6	Specifies the maximum number of visible faces (planes) required for a box occurrence to be accepted.	c		e	f	g	a		i		j	k	l	m	n	b
M_PLANE_ACCEPTANCE		Sets the acceptance level used for individual faces of the box occurrence. A plane will only be considered a visible face of the box occurrence if its score is greater than or equal to this level. You can set M_PLANE_ACCEPTANCE to a low value to detect very occluded faces, while maintaining a high acceptance level for the whole box occurrence using M_ACCEPTANCE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 60.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the acceptance level used for individual faces of the box occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_PLANE_CERTAINTY		Sets the certainty level for individual faces of the box occurrence. If the score is greater than or equal to the specified certainty level, the plane is considered a visible face of the box, without searching for better faces. You can set M_PLANE_CERTAINTY to a low value to detect very occluded faces, while maintaining a high certainty level for the whole box occurrence using M_CERTAINTY. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 90.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the certainty value for individual faces of the box occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_PLANE_MAX_COVERAGE		Sets the maximum expected coverage for individual faces of the box occurrence. A plane with a coverage greater than or equal to the maximum expected coverage has a score of 100%. You can set M_PLANE_MAX_COVERAGE to a low value to detect very occluded faces, while maintaining a high maximum expected coverage for the whole box occurrence using M_COVERAGE_MAX. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 100.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 100.0	Specifies the maximum coverage for individual faces of the box occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_POLARITY		Sets whether the normals should point inside or outside the box occurrence. A box occurrence is only considered a match if its normals satisfy the specified condition. For example, you can set M_POLARITY to M_INSIDE to only accept box occurrences that the 3D sensor views from the inside. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SAME.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies that the normals can point inside or outside.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INSIDE	Specifies that the normals must point inside.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OUTSIDE	Specifies that the normals must point outside.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies that the normals can point either inside or outside, but the direction must be consistent for all faces.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X		Sets the size of a nominal box model (M_BOX) along its X-axis, changing its originally defined size. INQ	c		e	f	g	a		i		j	k	l	m	n	b

Value >= 0.0	Specifies the expected size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X_MAX	Sets the maximum size of a range-type box model (M_BOX_RANGE) along its X-axis, changing its originally defined maximum size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the maximum size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the maximum size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X_MIN	Sets the minimum size of a range-type box model (M_BOX_RANGE) along its X-axis, changing its originally defined minimum size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the minimum size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y	Sets the size of a nominal box model (M_BOX) along its Y-axis, changing its originally defined size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the expected size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y_MAX	Sets the maximum size of a range-type box model (M_BOX_RANGE) along its Y-axis, changing its originally defined maximum size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the maximum size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the maximum size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y_MIN	Sets the minimum size of a range-type box model (M_BOX_RANGE) along its Y-axis, changing its originally defined minimum size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the minimum size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Z	Sets the size of a nominal box model (M_BOX) along its Z-axis, changing its originally defined size. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	Value >= 0.0	Specifies the expected size along the Z-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Z_MAX		Sets the maximum size of a range-type box model (M_BOX_RANGE) along its Z-axis, changing its originally defined maximum size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no constraint on the maximum size along the Z-axis.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the maximum size along the Z-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Z_MIN		Sets the minimum size of a range-type box model (M_BOX_RANGE) along its Z-axis, changing its originally defined minimum size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the minimum size along the Z-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOLERANCE_X		Sets the tolerance for the size of a nominal box model (M_BOX) along its X-axis, changing its originally defined tolerance. Occurrences must have a size, along their X-axis, within the specified nominal size (M_SIZE_X) +/- the specified tolerance. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no constraint on the size along the X-axis. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, since the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the tolerance for the size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOLERANCE_Y		Sets the tolerance for the size of a nominal box model (M_BOX) along its Y-axis, changing its originally defined tolerance. Occurrences must have a size, along their Y-axis, within the specified nominal size (M_SIZE_Y) +/- the specified tolerance. INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_INFINITE	Specifies no constraint on the size along the Y-axis. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, since the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 0.0	Specifies the tolerance for the size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TOLERANCE_Z		Sets the tolerance for the size of a nominal box model (M_BOX) along its Z-axis, changing its originally defined tolerance. Occurrences must have a size, along their Z-axis, within the specified nominal size (M_SIZE_Z) +/- the specified tolerance. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_INFINITE	Specifies no constraint on the size along the Z-axis. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, since the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 0.0	Specifies the tolerance for the size along the Z-axis.	c		e	f	g	a		i		j	k	l	m	n	b
Find cylinder 3D model finder context ID with a cylinder model			Specifies a find cylinder 3D model finder context, allocated using M3dmodAlloc() with M_FIND_CYLINDER_CONTEXT, and used in M3dmodFind() operations. It must contain a cylinder model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCEPTANCE		Sets the acceptance level for the score. An occurrence will be returned only if the match score between the target and the model is greater than or equal to this level. The score is a measure, as a percentage, of the model's surface that is covered by inlier points at the location of the occurrence. The score is defined relative to the maximum expected model coverage, such that an occurrence with a score of 100% has a model coverage equal to the maximum expected model coverage. Set the maximum expected coverage with M_COVERAGE_MAX. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 60.0%.	c		e	f	g	a		i		j	k	l	m	n	b

0.0 <= *Value* <= 100.0	Specifies an acceptable score, as a percentage. A 100% score indicates that the total model coverage is equal to M_COVERAGE_MAX.	c		e	f	g	a		i		j	k	l	m	n	b
M_BASES	Sets whether the cylinder model includes bases. Setting this value reserves points corresponding to a found occurrence's bases. Note that points found on an occurrence's bases will not affect the fit regardless of this control type. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_WITH_BASES	Specifies that the cylinder model includes bases.	c		e	f	g	a		i		j	k	l	m	n	b
M_WITHOUT_BASES	Specifies that the cylinder model does not include bases.	c		e	f	g	a		i		j	k	l	m	n	b
M_CERTAINTY	Sets the certainty level for the score, as a percentage. If the score is greater than or equal to the specified certainty level, the occurrence is considered a match, without searching the rest of the target for better matches (provided the specified number of occurrences has been found). The certainty level is defined relative to the maximum expected model coverage, such that setting M_CERTAINTY to 100% indicates that certain matches must have a total model coverage equal to the maximum expected model coverage. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 90.0%.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the certainty level for the score, as a percentage. If you set the certainty level too high (close to 100.0%), you slow down the search because you force the search algorithm to check the whole point cloud for the best possible match(es). A good certainty level is slightly lower than the expected score, so that the search can finish as soon as a match is found. However, if you set the certainty level too low, false matches might be found.	c		e	f	g	a		i		j	k	l	m	n	b
M_COVERAGE_MAX	Sets the maximum expected model coverage. The model coverage is the percentage of the model's surface covered with inlier points found in the occurrence. An occurrence with a model coverage greater than or equal to the maximum expected model coverage has a score of 100%. INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Specifies the default value; the default value is 45.0%.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 < *Value* <= 100.0	Specifies the maximum expected model coverage, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
	M_LENGTH		Sets the length of a nominal cylinder model (M_CYLINDER), changing its originally defined length. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 0.0	Specifies the length.	c		e	f	g	a		i		j	k	l	m	n	b
	M_LENGTH_MAX		Sets the maximum length of a range-type cylinder model (M_CYLINDER_RANGE), changing its originally defined maximum length. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_INFINITE	Specifies no constraint on the maximum length.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the maximum length.	c		e	f	g	a		i		j	k	l	m	n	b
	M_LENGTH_MIN		Sets the minimum length of a range-type cylinder model (M_CYLINDER_RANGE), changing its originally defined minimum length. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 0.0	Specifies the minimum length.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIN_SEPARATION_DISTANCE		Sets the minimum gap distance along the length of a cylinder before considering the cylinder as two separate occurrences of the cylinder model. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to split cylinders.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the minimum gap distance.	c		e	f	g	a		i		j	k	l	m	n	b

M_NUMBER	Sets the maximum number of occurrences for which to search. Once the required number of occurrences with scores greater than or equal to M_CERTAINTY have been found, the search will stop to avoid exhaustive checking of all possible candidates. If not all the required number of occurrences have been found, Aurora Imaging Library will continue to search for the required number of matches greater than or equal to the acceptance level and with the best score. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Specifies to find all occurrences. Note that this setting can increase the search time; always set M_NUMBER to a specific number whenever possible.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the number of occurrences for which to search.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS_MIN	Sets the minimum number of points per occurrence found. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 50.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the minimum number of points per occurrence found.	c		e	f	g	a		i		j	k	l	m	n	b
M_RADIUS	Sets the radius of a nominal cylinder model (M_CYLINDER), changing its originally defined radius. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the radius of the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_RADIUS_MAX	Sets the maximum radius of a range-type cylinder model (M_CYLINDER_RANGE), changing its originally defined maximum radius. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the maximum radius.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the maximum radius.	c		e	f	g	a		i		j	k	l	m	n	b

M_RADIUS_MIN	Sets the minimum radius of a range-type cylinder model (M_CYLINDER_RANGE), changing its originally defined minimum radius. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the minimum radius.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESERVED_POINTS_DISTANCE	Sets the reserved area around the occurrence, defined as a percentage of the radius. Points found in this area are not considered in the fit, and cannot be considered for other occurrences. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 10.0.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies the reserved points distance for the model, as a percentage of its radius.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOLERANCE_LENGTH	Sets the tolerance for the length of a nominal cylinder model (M_CYLINDER), changing its originally defined length tolerance. Occurrences must have a length within the specified nominal length (M_LENGTH) +/- the specified tolerance. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the length. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, since the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the tolerance for the length.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOLERANCE_RADIUS	Sets the tolerance for the radius of a nominal cylinder model (M_CYLINDER), changing its originally defined radius tolerance. Occurrences must have a radius within the specified nominal radius (M_RADIUS) +/- the specified tolerance. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the radius. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, since the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the tolerance for the radius.	c		e	f	g	a		i		j	k	l	m	n	b

Find rectangular plane 3D model finder context ID with a rectangular plane model	Specifies a find rectangular plane 3D model finder context, allocated using M3dmodAlloc() with M_FIND_RECTANGULAR_PLANE_CONTEXT, and used in M3dmodFind() operations. It must contain a rectangular plane model.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE	Sets the acceptance level for the score. An occurrence will be returned only if the match score between the target and the model is greater than or equal to this level. The score is a measure, as a percentage, of the model's surface that is covered by inlier points at the location of the occurrence. The score is defined relative to the maximum expected model coverage, such that an occurrence with a score of 100% has a model coverage equal to the maximum expected model coverage. Set the maximum expected coverage with M_COVERAGE_MAX. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 60.0%.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies an acceptable score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_CERTAINTY	Sets the certainty level for the score, as a percentage. If the score is greater than or equal to the specified certainty level, the occurrence is considered a match, without searching the rest of the target for better matches (provided the specified number of occurrences has been found). The certainty level is defined relative to the maximum expected model coverage, such that setting M_CERTAINTY to 100% indicates that certain matches must have a total model coverage equal to the maximum expected model coverage. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 90.0%.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the certainty level for the score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_COVERAGE_MAX	Sets the maximum expected model coverage. The model coverage is the percentage of the model's surface covered with inlier points found in the occurrence. An occurrence with a model coverage greater than or equal to the maximum expected model coverage has a score of 100%. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 100.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 100.0	Specifies the maximum expected model coverage, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELONGATION_MAX		Sets the maximum elongation of the rectangular plane occurrence. The elongation is defined as the maximum side/minimum side. This is used to prevent lines from being incorrectly identified as rectangular planes when a very large size range is used. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies that the rectangular plane occurrence has no maximum elongation. It can be infinitely long.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 1.0	Specifies the rectangular plane occurrence's maximum elongation, defined as maximum side / minimum side.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELONGATION_MIN		Sets the minimum elongation of the rectangular plane occurrence. The elongation is defined as the maximum side/minimum side. This is used to prevent lines from being incorrectly identified as rectangular planes when a very large size range is used. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1.0	Specifies the rectangular plane occurrence's minimum elongation, defined as maximum side / minimum side.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_ANGLE_TOLERANCE		Sets the angular tolerance to use for M_NORMAL_CONDITION, when searching for rectangular plane occurrences. Aurora Imaging Library only returns the occurrence as a match if the normal of the rectangular plane meets the specified M_NORMAL_CONDITION, +/- the specified angular tolerance, relative to the vector M_NORMAL_.... INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 90.0	Specifies the normal angular tolerance, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

M_NORMAL_CONDITION		Sets whether to find only rectangular plane occurrences that satisfy the specified condition, relative to the vector M_NORMAL_..., or to find any rectangular plane occurrence. A rectangular plane occurrence is only considered a match if its normal meets the specified condition when compared to the vector specified with M_NORMAL_... (within the specified tolerance M_NORMAL_ANGLE_TOLERANCE). For example, you can set M_NORMAL_... to the normal of the floor plane and then set M_NORMAL_CONDITION to M_PARALLEL to find rectangular plane occurrences parallel to the floor. If you know that your scene only contains parallel and/or perpendicular rectangular planes, you should use this control to force the search algorithm to ignore all other candidates. This can help to avoid false matches and speed up the search, since candidates that do not meet the specified condition will not be further examined. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_UNCONDITIONAL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PARALLEL	Specifies that the normal of the rectangular plane must be parallel to the vector specified using M_NORMAL_..., within the specified angular tolerance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PARALLEL_OR_PERPENDICULAR	Specifies that the normal of the rectangular plane must be either parallel or perpendicular to the vector specified using M_NORMAL_..., within the specified angular tolerance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PERPENDICULAR	Specifies that the normal of the rectangular plane must be perpendicular to the vector specified using M_NORMAL_..., within the specified angular tolerance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_UNCONDITIONAL	Specifies that there is no constraint on the normal of the rectangular plane.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_X		Sets the X-component of the vector against which to compare rectangular plane occurrences, as specified using M_NORMAL_CONDITION. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the X-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b

M_NORMAL_Y		Sets the Y-component of the vector against which to compare rectangular plane occurrences, as specified using M_NORMAL_CONDITION. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_Z		Sets the Z-component of the vector against which to compare rectangular plane occurrences, as specified using M_NORMAL_CONDITION. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Z-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER		Sets the maximum number of occurrences for which to search. Once the required number of occurrences with scores greater than or equal to M_CERTAINTY have been found, the search will stop to avoid exhaustive checking of all possible candidates. If not all the required number of occurrences have been found, Aurora Imaging Library will continue to search for the required number of matches greater than or equal to the acceptance level and with the best score. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ALL	Specifies to find all occurrences. Note that this setting can increase the search time; always set M_NUMBER to a specific number whenever possible.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the number of occurrences for which to search.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS_MIN		Sets the minimum number of points per occurrence found. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the minimum number of points per occurrence found.	c		e	f	g	a		i		j	k	l	m	n	b

M_SIZE_X	Sets the size of a nominal rectangular plane model (M_RECTANGLE) along its X-axis, changing its originally defined size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the expected size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X_MAX	Sets the maximum size of a range-type rectangular plane model (M_RECTANGLE_RANGE) along its X-axis, changing its originally defined maximum size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the maximum size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the maximum size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X_MIN	Sets the minimum size of a range-type rectangular plane model (M_RECTANGLE_RANGE) along its X-axis, changing its originally defined minimum size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the minimum size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y	Sets the size of a nominal rectangular plane model (M_RECTANGLE) along its Y-axis, changing its originally defined size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the expected size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y_MAX	Sets the maximum size of a range-type rectangular plane model (M_RECTANGLE_RANGE) along its Y-axis, changing its originally defined maximum size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the maximum size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the maximum size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b

M_SIZE_Y_MIN	Sets the minimum size of a range-type rectangular plane model (M_RECTANGLE_RANGE) along its Y-axis, changing its originally defined minimum size.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the minimum size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOLERANCE_X	Sets the tolerance for the size of a nominal rectangular plane model (M_RECTANGLE) along its X-axis, changing its originally defined tolerance. Occurrences must have a size, along their X-axis, within the specified nominal size (M_SIZE_X) +/- the specified tolerance. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the size along the X-axis. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, since the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the tolerance for the size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOLERANCE_Y	Sets the tolerance for the size of a nominal rectangular plane model (M_RECTANGLE) along its Y-axis, changing its originally defined tolerance. Occurrences must have a size, along their Y-axis, within the specified nominal size (M_SIZE_Y) +/- the specified tolerance. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the size along the Y-axis. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, since the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the tolerance for the size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
Find sphere 3D model finder context ID with a sphere model	Specifies a find sphere 3D model finder context, allocated using M3dmodAlloc() with M_FIND_SPHERE_CONTEXT, and used in M3dmodFind() operations. It must contain a sphere model.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACCEPTANCE	Sets the acceptance level for the score. An occurrence will be returned only if the match score between the target and the model is greater than or equal to this level. The score is a measure, as a percentage, of the model's surface that is covered by inlier points at the location of the occurrence. The score is defined relative to the maximum expected model coverage, such that an occurrence with a score of 100% has a model coverage equal to the maximum expected model coverage. Set the maximum expected coverage with M_COVERAGE_MAX. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 60.0%.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies an acceptable score, as a percentage. A 100% score indicates that the total model coverage is equal to M_COVERAGE_MAX.	c		e	f	g	a		i		j	k	l	m	n	b
M_CERTAINTY	Sets the certainty level for the score, as a percentage. If the score is greater than or equal to the specified certainty level, the occurrence is considered a match, without searching the rest of the target for better matches (provided the specified number of occurrences has been found). The certainty level is defined relative to the maximum expected model coverage, such that setting M_CERTAINTY to 100% indicates that certain matches must have a total model coverage equal to the maximum expected model coverage. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 90.0%.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies the certainty level for the score, as a percentage. If you set the certainty level too high (close to 100.0%), you slow down the search because you force the search algorithm to check the whole point cloud for the best possible match(es). A good certainty level is slightly lower than the expected score, so that the search can finish as soon as a match is found. However, if you set the certainty level too low, false matches might be found.	c		e	f	g	a		i		j	k	l	m	n	b

M_COVERAGE_MAX	Sets the maximum expected model coverage. The model coverage is the percentage of the model's surface covered with inlier points found in the occurrence. An occurrence with a model coverage greater than or equal to the maximum expected model coverage has a score of 100%. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 45.0.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 100.0	Specifies the maximum expected model coverage, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER	Sets the maximum number of occurrences for which to search. Once the required number of occurrences with scores greater than or equal to M_CERTAINTY have been found, the search will stop to avoid exhaustive checking of all possible candidates. If not all the required number of occurrences have been found, Aurora Imaging Library will continue to search for the required number of matches greater than or equal to the acceptance level and with the best score. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Specifies to find all occurrences. Note that this setting can increase the search time; always set M_NUMBER to a specific number whenever possible.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the number of occurrences for which to search.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS_MIN	Sets the minimum number of points per occurrence found. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 50.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the minimum number of points per occurrence found.	c		e	f	g	a		i		j	k	l	m	n	b
M_RADIUS	Sets the radius of a nominal cylinder model (M_SPHERE), changing its originally defined radius. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the expected radius.	c		e	f	g	a		i		j	k	l	m	n	b

M_RADIUS_MAX	Sets the maximum radius of a range-type sphere model (M_SPHERE_RANGE), changing its originally defined maximum radius. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the maximum radius.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the maximum radius.	c		e	f	g	a		i		j	k	l	m	n	b
M_RADIUS_MIN	Sets the minimum radius for a range-type sphere model (M_SPHERE_RANGE), changing its originally defined minimum radius. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the minimum radius.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESERVED_POINTS_DISTANCE	Sets the reserved area around the occurrence, defined as a percentage of the radius. Points found in this area are not considered in the fit, and cannot be considered for other occurrences. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 10.0.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies the reserved points distance for the model, as a percentage of its radius.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOLERANCE_RADIUS	Sets the tolerance for the radius of a nominal sphere model (M_SPHERE), changing its originally defined radius tolerance. Occurrences must have a radius within the specified nominal radius (M_RADIUS) +/- the specified tolerance. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the radius. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, since the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the tolerance for the radius.	c		e	f	g	a		i		j	k	l	m	n	b
Find surface or planar surface 3D model finder context ID with a surface model	Specifies a find surface or planar surface 3D model finder context, allocated using M3dmodAlloc() with M_FIND_SURFACE_CONTEXT or M_FIND_PLANAR_SURFACE_CONTEXT, and used in M3dmodFind() operations. It must contain a surface model.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACCEPTANCE	Sets the acceptance level for the score. An occurrence will be returned only if the match score between the target and the model is greater than or equal to this level. The score is a measure, as a percentage, of the model's surface that is covered by inlier points at the location of the occurrence. The score is defined relative to the maximum expected model coverage, such that an occurrence with a score of 100% has a model coverage equal to the maximum expected model coverage. Set the maximum expected coverage with M_COVERAGE_MAX. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 60.0%.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies an acceptable score, as a percentage. A 100% score indicates that the total model coverage is equal to M_COVERAGE_MAX.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE_COLOR	Sets the acceptance level for the color score. An occurrence will be returned only if the color score between the target and the model is greater than or equal to this level. The color score is a measure, as a percentage, of the similarity between the color of the model and the color of the target. Note that this control type has no effect unless M_USE_COLOR is set to M_ENABLE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the acceptance level for the color score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE_TARGET	Sets the acceptance level for the target score. An occurrence will be returned only if the target score between the target and the model is greater than or equal to this level. The target score is a measure, as a percentage, of the total points found in the occurrence that are also found in the model. Points found in the occurrence that are not present in the model will reduce the target score. For example, a target score of 100% means that there are no points in the occurrence that do not fit the model. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b

0.0 <= *Value* < 100.0	Specifies the acceptance level for the target score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_CERTAINTY	Sets the certainty level for the score, as a percentage. If the score is greater than or equal to the specified certainty level, the occurrence is considered a match, without searching the rest of the target for better matches (provided the specified number of occurrences has been found). The certainty level is defined relative to the maximum expected model coverage, such that setting M_CERTAINTY to 100% indicates that certain matches must have a total model coverage equal to the maximum expected model coverage. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 90.0%.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies the certainty level for the score, as a percentage. If you set the certainty level too high (close to 100.0%), you slow down the search because you force the search algorithm to check the whole point cloud for the best possible match(es). A good certainty level is slightly lower than the expected score, so that the search can finish as soon as a match is found. However, if you set the certainty level too low, false matches might be found.	c		e	f	g	a		i		j	k	l	m	n	b
M_COVERAGE_MAX	Sets the maximum expected model coverage. The model coverage is the percentage of the model's surface covered with inlier points found in the occurrence. An occurrence with a model coverage greater than or equal to the maximum expected model coverage has a score of 100%. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.0.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 100.0	Specifies the maximum expected model coverage, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_SCORE_MIN	Sets the minimum expected occurrence fit score. Note that this control type has no effect unless M_SAVE_FIT_INFO is set to M_ENABLE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b

	0.0 <= *Value* <= 100.0	Specifies the minimum expected fit score.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_PLANES_OCCLUSION_MODE		Sets the occlusion level of the planar regions of the model in the scene. If less than 30% of a planar region's surface would be occluded at the position of an occurrence, set M_MODEL_PLANES_OCCLUSION_MODE to M_LOW. If the occlusion level is between 30% and 50%, set M_MODEL_PLANES_OCCLUSION_MODE to M_MEDIUM. If the occlusion level is between 50% and 80%, set M_MODEL_PLANES_OCCLUSION_MODE to M_HIGH. If the occlusion level is greater than 80%, set M_MODEL_PLANES_OCCLUSION_MODE to M_VERY_HIGH. If different planar regions have differing occlusion levels, use the occlusion level for the highest occluded planar region. Note that this control type is only available for find planar surface 3D model finder contexts. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_LOW.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Specifies a high model plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b
	M_LOW	Specifies a low model plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MEDIUM	Specifies a medium model plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b
	M_VERY_HIGH	Specifies a very high model plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER		Sets the maximum number of occurrences for which to search. Once the required number of occurrences with scores greater than or equal to M_CERTAINTY have been found, the search will stop to avoid exhaustive checking of all possible candidates. If not all the required number of occurrences have been found, Aurora Imaging Library will continue to search for the required number of matches greater than or equal to the acceptance level and with the best score. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ALL	Specifies to find all occurrences. Note that this setting can increase the search time; always set M_NUMBER to a specific number whenever possible.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the number of occurrences for which to search.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS_MIN		Sets the minimum number of points per occurrence found. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the minimum number of points per occurrence found.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_OUTLIERS		Sets whether to perform outlier removal on the model, changing the model's originally defined outlier removal setting. When enabled, Aurora Imaging Library removes outliers such as clusters of points around the perimeter of the model. Enable outlier removal if the model contains stray points, especially if M_REFINE_REGISTRATION is enabled. The default for this control type is set when adding the surface model to the context using M3dmodDefine() with M_SURFACE. Note that you should not enable outlier removal if your model was defined from a point cloud created from a CAD file (using MbufImport() / MbufRestore()) since parts of the model can be removed, which can cause the search to fail/return false positives. You can use M3dmodDraw3d() to draw the remaining points after preprocessing (M3dmodControlDraw() with M_DRAW_MODEL_PREPROCESSED). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to perform outlier removal on the model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to perform outlier removal on the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_OUTLIERS_SEARCH_MODE		Sets the search mode for finding outliers when M_REMOVE_OUTLIERS is enabled. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_AUTO.	c		e	f	g	a		i		j	k	l	m	n	b

		M_AUTO	Specifies to automatically set the search mode based on the point cloud's organization; either M_ORGANIZED if organized or M_TREE if unorganized.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ORGANIZED	Specifies to use the point cloud's organizational structure to determine the outliers. This option is supported only for an organized point cloud.	c		e	f	g	a		i		j	k	l	m	n	b
		M_TREE	Specifies to use a KD tree search mode to determine the outliers.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RESTING_PLANE_ANGLE_TOLERANCE		Sets the angular tolerance for the resting plane constraint. The resting plane provides contextual information about how the occurrence lies within the scene. Aurora Imaging Library will only return occurrences that rest on this plane, +/- the specified angular tolerance. Note that this does not affect any of the scores, but it filters occurrences that do not meet this constraint. It also does not decrease the search time because occurrences are filtered near the end of the search. You can copy the resting plane to or from the specified find surface or planar surface 3D model finder context using M3dmodCopy() with M_RESTING_PLANE. Note that to use a resting plane, you must also copy a floor plane (M_FLOOR) to the find surface or planar surface 3D model finder context. A small angular tolerance is typically required since the angle found is not always exact. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 10.0.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 < *Value* <= 180.0	Specifies the angular tolerance, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

M_SEARCH_POINT_RESOLUTION	Sets the resolution of the target point cloud (scene), where the resolution is expressed as the average distance between points. Very different resolutions between the scene and the model might cause missed occurrences. You should specify the resolution of the scene when it is incompatible with the resolution of the model. For example, if the distance between points in the scene is greater than that of the model, Aurora Imaging Library will subsample the model to match the resolution of the target. If the distance between points in the scene is lower than that of the model's, Aurora Imaging Library will upsample the model to match the resolution of the target, using its associated mesh. If there is no associated mesh in the find surface or planar surface 3D model finder context when Aurora Imaging Library attempts to upsample the model, an error is generated. After an M3dmodFind() operation, you can check whether the resolutions of the scene and model were compatible, using M3dmodGetResult() with M_STATUS_SCENE_RESOLUTION. You can inquire the resolution of the defined model (before preprocessing) using M_MODEL_RESOLUTION. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_SAME_AS_MODEL.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAME_AS_MODEL	Specifies that the target point cloud (scene) has the same resolution as the model.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the expected resolution of the target point cloud (scene), where the resolution is expressed as the average distance between points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SET_POSITION_X	Changes the X-coordinate of the model's reference axis origin. A surface occurrence's found position is determined by the model's reference axis; position results return the coordinates of the model's reference axis origin transformed at the model occurrence. Note that this does not transform the model, but moves the origin of the model coordinate system to this position.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b

M_SET_POSITION_Y	Changes the Y-coordinate of the model's reference axis origin. A surface occurrence's found position is determined by the model's reference axis; position results return the coordinates of the model's reference axis origin transformed at the model occurrence. Note that this does not transform the model, but moves the origin of the model coordinate system to this position.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_SET_POSITION_Z	Changes the Z-coordinate of the model's reference axis origin. A surface occurrence's found position is determined by the model's reference axis; position results return the coordinates of the model's reference axis origin transformed at the model occurrence. Note that this does not transform the model, but moves the origin of the model coordinate system to this position.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Z-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_TARGET_PLANES_OCCLUSION_MODE	Sets the target occlusion level when a planar region of the model is a portion of a planar region of an object in the scene. This sets how much of the object's planar region is not found in the model. If less than 30% of an object's planar region in the scene is not present in the model's surface, set M_TARGET_PLANES_OCCLUSION_MODE to M_LOW. If the occlusion level is between 30% and 50%, set M_TARGET_PLANES_OCCLUSION_MODE to M_MEDIUM. If the occlusion level is between 50% and 80%, set M_TARGET_PLANES_OCCLUSION_MODE to M_HIGH. If the occlusion level is greater than 80%, set M_TARGET_PLANES_OCCLUSION_MODE to M_VERY_HIGH. Note that this control type is only available for find planar surface 3D model finder contexts. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_LOW.	c		e	f	g	a		i		j	k	l	m	n	b
M_HIGH	Specifies a high target plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b
M_LOW	Specifies a low target plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b

		M_MEDIUM	Specifies a medium target plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b
		M_VERY_HIGH	Specifies a very high target plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRANSFORM_MODEL		Transforms the model (rotation and translation) to a new position and orientation. When the transformation occurs, the orientation and position of the model prior to the transformation is lost. Note that a surface occurrence's found position is determined by the model's reference axis; position results return the coordinates of the model's reference axis origin transformed at the model occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		Transformation matrix object ID	Specifies the identifier of a transformation matrix object. The transformation matrix object must have been previously allocated using M3dgeoAlloc() with M_TRANSFORMATION_MATRIX. The transformation matrix must be of type M_RIGID.	c		e	f	g	a		i		j	k	l	m	n	b

The following ContextOrResult3dmodId, ControlType, and ControlValue parameter settings can be specified for a 3D model finder result buffer when Index is set to M_GENERAL or M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For a 3D model finder result buffer																		
ContextOrResult3dmodId - Possible values returned			Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlType																	
		ControlValue																
3D model finder result buffer ID			Specifies a 3D model finder result buffer, allocated using M3dmodAllocResult().															
	M_RESET		Removes all results from the 3D model finder result buffer. This does not delete the result buffer, unlike M3dmodFree().	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STOP_FIND		Stops the current execution of M3dmodFind() from another thread. Any completed results from the 3D model finder operation will be available.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void M3dmodControlInt64 (AIL_ID ContextOrResult3dmodId, AIL_INT64 Index, AIL_INT64 ControlType, AIL_INT64 ControlValue)
```

Details

Parameters

ContextOrResult3dmodId

See ContextOrResult3dmodId of the main function for a description.

Index

See Index of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

```
void M3dmodControlDouble (AIL_ID ContextOrResult3dmodId, AIL_INT64 Index, AIL_INT64 ControlType, AIL_DOUBLE ControlValue)
```

Details

Parameters

ContextOrResult3dmodId

See ContextOrResult3dmodId of the main function for a description.

Index

See Index of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodControlDraw

Synopsis

Control a setting of a draw 3D model finder context.

Syntax

```
void M3dmodControlDraw(
    AIL_ID DrawContext3dmodId, //out
    AIL_INT64 Operation,       //in
    AIL_INT64 ControlType,     //in
    AIL_DOUBLE ControlValue    //in
)
```

Description

This function controls a specified setting of a draw 3D model finder context. These settings establish which features of a model or which results of found model occurrences to draw into the 3D graphics list (and how to draw them), when calling M3dmodDraw3d().

You can inquire about most of these settings using M3dmodInquireDraw().

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **M3dmodControlDraw()** internally calls the *AIL_DOUBLE* version of this function (M3dmodControlDrawDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (M3dmodControlDrawInt64()).

Parameters

DrawContext3dmodId

Specifies the identifier of the draw 3D model finder context to control. The draw 3D model finder context must have been previously allocated on the required system using M3dmodAlloc() with either M_DRAW_3D_GEOMETRIC_CONTEXT (to draw occurrences of geometric models) or M_DRAW_3D_SURFACE_CONTEXT (to draw surface models or occurrences of surface models).

Operation

Specifies the draw operation.

See the Parameter associations section for possible values that can be specified.

ControlType

Specifies the setting to change.

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the setting's new value.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The tables below list possible values for the **Operation**, **ControlType**, and **ControlValue** parameters.

- For a draw geometric or surface 3D model finder context
- For a draw surface 3D model finder context

The following Operation, ControlType, and ControlValue parameter settings can be specified for a draw geometric or surface 3D model finder context.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For a draw geometric or surface 3D model finder context																	
Operation / ControlType / ControlValue	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ALL	Applies the setting to all operations that support the control type.															
M_ACTIVE	Sets whether to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets the appearance of the graphic(s) to solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR).	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR	Sets the color with which to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_AXES			Sets whether and how to draw the axes of either the model or each specified occurrence when calling M3dmodDraw3d(), depending on whether a find 3D model finder context or result buffer is specified. The resulting graphic will be of type M_GRAPHIC_TYPE_AXIS.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE		Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Specifies the default value. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
	M_LENGTH		Sets the length of the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_DEFAULT_LENGTH.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT_LENGTH	Specifies that the length is the default size of the 3D model.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the length, in world units.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REFERENCE_X		Sets the X-position of the graphic(s) origin, relative to the model or occurrence. Note that M_REFERENCE_X is only available for a draw surface 3D model finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_CENTER.	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER	Specifies to draw the graphic(s) at the center of the model or occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX	Specifies to draw the graphic(s) at the model's or occurrence's maximum X-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN	Specifies to draw the graphic(s) at the model's or occurrence's minimum X-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REFERENCE_Y		Sets the Y-position of the graphic(s) origin, relative to the model or occurrence. Note that M_REFERENCE_Y is only available for a draw surface 3D model finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_CENTER.	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER	Specifies to draw the graphic(s) at the center of the model or occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX	Specifies to draw the graphic(s) at the model's or occurrence's maximum Y-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b

M_MIN	Specifies to draw the graphic(s) at the model's or occurrence's minimum Y-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_Z	Sets the Z-position of the graphic(s) origin, relative to the model or occurrence. Note that M_REFERENCE_Z is only available for a draw surface 3D model finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_CENTER.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER	Specifies to draw the graphic(s) at the center of the model or occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX	Specifies to draw the graphic(s) at the model's or occurrence's maximum Z-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN	Specifies to draw the graphic(s) at the model's or occurrence's minimum Z-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_BOX	Sets whether and how to draw the bounding box of each specified occurrence when calling M3dmodDraw3d(). The bounding box is the smallest-axis aligned box that contains the ideal geometric shape or surface of the model at the location of the occurrence, depending on the model type. If you enable M_DRAW_MODEL, the bounding box is the smallest axis-aligned box that contains that graphic. Note that it does not necessarily contain all inlier points. The resulting graphic will be of type M_GRAPHIC_TYPE_BOX.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b

M_APPEARANCE	Sets the appearance of the graphic(s) to solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_WIREFRAME.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The 3D graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_WHITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															

		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_INDEX	Sets whether and how to draw the index of each specified occurrence when calling M3dmodDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_TEXT.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_YELLOW.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															

			Green Specifies the green band, as a value between 0 and 255.																	
			Blue Specifies the blue band, as a value between 0 and 255.																	
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FONT_SIZE		Sets the size of the font with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the font size. This is the height of one line of text, in world units.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity of the 3D graphic(s) added to the 3D graphics list. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the graphic(s) . A graphic(s) with an opacity of 100.0 is completely opaque, while an opacity of 0.0 makes it completely transparent (invisible).	c		e	f	g	a		i		j	k	l	m	n	b
	M_REFERENCE_X		Sets the X-position of the graphic(s) origin, relative to the occurrence. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_CENTER.	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER	Specifies to draw the graphic(s) at the center of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX	Specifies to draw the graphic(s) at the occurrence's maximum X-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b

M_MIN	Specifies to draw the graphic(s) at the occurrence's minimum X-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_Y	Sets the Y-position of the graphic(s) origin, relative to the occurrence. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_CENTER.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER	Specifies to draw the graphic(s) at the center of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX	Specifies to draw the graphic(s) at the occurrence's maximum Y-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN	Specifies to draw the graphic(s) at the occurrence's minimum Y-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_Z	Sets the Z-position of the graphic(s) origin, relative to the occurrence. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_CENTER.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER	Specifies to draw the graphic(s) at the center of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX	Specifies to draw the graphic(s) at the occurrence's maximum Z-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN	Specifies to draw the graphic(s) at the occurrence's minimum Z-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_RENDER_LAYER	Sets the layer on which to draw the graphic(s). The graphics on higher layer are always drawn completely in front of graphics on lower layers even if those on higher layers are farther away. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 2.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <= 9	Specifies the layer on which the graphic will be rendered.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_DIRECTION	Sets the direction to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_SAME.	c		e	f	g	a		i		j	k	l	m	n	b

M_FLIP	Specifies to draw the graphic(s) such that it is mirrored across the occurrence's X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAME	Specifies to draw the graphic(s) in the direction of the occurrence's X-axis, and facing the direction of its Z-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_INLIER_POINTS	Sets whether and how to draw the inlier points, based on M_FIT_DISTANCE, of each specified occurrence when calling M3dmodDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_DOTS. Note that for a surface model, M_SAVE_FIT_INFO must be enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. The default value for draw geometric 3D model finder contexts is M_ENABLE. The default value for draw surface 3D model finder contexts is M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_PSEUDO_COLOR.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															

		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_MODEL	Sets whether and how to draw the defined model or each specified occurrence when calling M3dmodDraw3d(), depending on whether a find 3D model finder context or result buffer is specified. In the case of a geometric model occurrence, the resulting graphic will be of the same geometric type as the model. In the case of a surface model or surface model occurrence, the resulting graphic will be of type M_GRAPHIC_TYPE_POINT_CLOUD.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets the appearance of the graphic(s) to solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. The default value for draw geometric 3D model finder contexts is M_PSEUDO_COLOR. The default value for draw surface 3D model finder contexts is either M_AUTO_COLOR for a model in a find surface 3D model finder context or M_PSEUDO_COLOR for a result in a find surface 3D model finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_AUTO_COLOR	Specifies to color the graphic according to specific components in the source container. If the container is a point cloud, the graphic is colored based on the reflectance component. If the reflectance component does not exist, the intensity component is used. If there is no reflectance or intensity, use M_PSEUDO_COLOR.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY	Sets the opacity of the 3D graphic(s) added to the 3D graphics list. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 30.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the graphic(s) . A graphic(s) with an opacity of 100.0 is completely opaque, while an opacity of 0.0 makes it completely transparent (invisible).	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_RESERVED_POINTS	Sets whether and how to draw the reserved points of each specified occurrence when calling M3dmodDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_DOTS. Occurrences of all types of models have reserved points and you can draw them; only occurrences of cylinder and sphere models allow you to control the distance of reserved points (using M3dmodControl() with M_RESERVED_POINTS_DISTANCE).	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_PSEUDO_COLOR.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_GLOBAL_DRAW_SETTINGS			Sets a global draw 3D model finder context setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_COLOR_OFFSET		Sets the offset to apply to either the model's or the occurrence's index when drawing with M_PSEUDO_COLOR. You can use this control type to prevent the same colors from being assigned to different occurrences when drawing several results at once. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 0.	c		e	f	g	a		i		j	k	l	m	n	b

		Value >= 0	Specifies the offset to apply to the index.	c		e	f	g	a		i		j	k	l	m	n	b

The following Operation, ControlType, and ControlValue parameter settings can be specified for a draw surface 3D model finder context.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For a draw surface 3D model finder context																	
Operation	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ControlType																
ControlValue																
M_DRAW_AXES_POSITION	Sets whether and how to draw the axes at either the model's or each specified occurrence's position when calling M3dmodDraw3d(), depending on whether a find 3D model finder context or result buffer is specified. The axes are drawn at the model's reference axis origin/the model's reference axis origin transformed at each occurrence. The resulting graphic will be of type M_GRAPHIC_TYPE_AXIS.															
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_LENGTH	Sets the length of the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DEFAULT_LENGTH.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_LENGTH	Specifies that the length is the default size of the 3D model.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the length, in world units.	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b

M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_BACKGROUND_POINTS	Sets whether and how to draw the points that were considered part of the background during the search. Background points include those of objects that are too big or too small to be part of an occurrence of the model. The resulting graphic will be of type M_GRAPHIC_TYPE_DOTS. This drawing operation is only supported if M_REMOVE_BACKGROUND was enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_GRAY.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FLOOR_POINTS	Sets whether and how to draw the points that were considered part of the floor during the search. Floor points are determined using the floor plane, defined using M3dmodCopy() with M_FLOOR, and the floor removal direction and offset (M3dmodControl() with M_REMOVE_FLOOR_DIRECTION and M_REMOVE_FLOOR_OFFSET). The resulting graphic will be of type M_GRAPHIC_TYPE_DOTS. This drawing operation is only supported if M_REMOVE_FLOOR was enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_MAGENTA.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															

			Red Specifies the red band, as a value between 0 and 255.																
			Green Specifies the green band, as a value between 0 and 255.																
			Blue Specifies the blue band, as a value between 0 and 255.																
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_MODEL_PREPROCESSED			Sets whether and how to draw the preprocessed model that is in the specified find 3D model finder context. The resulting graphic will be of type M_GRAPHIC_TYPE_POINT_CLOUD. M3dmodPreprocess() must be called before calling M3dmodDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE		Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR		Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Same as M_COLOR_YELLOW.	c		e	f	g	a		i		j	k	l	m	n	b
		M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
			Parameters															
			Red Specifies the red band, as a value between 0 and 255.															
			Green Specifies the green band, as a value between 0 and 255.															
			Blue Specifies the blue band, as a value between 0 and 255.															
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity of the 3D graphic(s) added to the 3D graphics list. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 30.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the graphic(s) . A graphic(s) with an opacity of 100.0 is completely opaque, while an opacity of 0.0 makes it completely transparent (invisible).	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void M3dmodControlDrawInt64 (AIL_ID DrawContext3dmodId, AIL_INT64 Operation, AIL_INT64 ControlType, AIL_INT64 ControlValue)
```

Details

Parameters

DrawContext3dmodId

See DrawContext3dmodId of the main function for a description.

Operation

See Operation of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

```
void M3dmodControlDrawDouble (AIL_ID DrawContext3dmodId, AIL_INT64 Operation, AIL_INT64 ControlType, AIL_DOUBLE ControlValue)
```

Details

Parameters

DrawContext3dmodId

See DrawContext3dmodId of the main function for a description.

Operation

See Operation of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodCopy

Synopsis

Copy data to or from a 3D model finder context.

Syntax

```
void M3dmodCopy(
    AIL_ID SrcContext3dmodId, //in
    AIL_INT64 SrcIndex,       //in
    AIL_ID DstAilObjectId,    //out
    AIL_INT64 DstIndex,       //in
    AIL_INT64 CopyType,       //in
    AIL_INT64 ControlFlag     //in
)
```

Description

This function copies data to or from a 3D model finder context. For example, you can copy the floor plane (M_FLOOR) or the resting plane (M_RESTING_PLANE) to the 3D model finder context.

Parameters

SrcContext3dmodId

Specifies the identifier of the source find 3D model finder context, 3D geometry object, or image buffer, depending on the type of copy operation.

See the Parameter associations section for possible values that can be specified.

SrcIndex

Specifies the index of the model in the source find 3D model finder context, depending on the type of copy operation.

See the Parameter associations section for possible values that can be specified.

DstAilObjectId

Specifies the identifier of the destination find 3D model finder context, 3D registration context, 3D geometry object, container, or image buffer, depending on the type of copy operation.

See the Parameter associations section for possible values that can be returned.

DstIndex

Specifies the index of the model in the destination find 3D model finder context, depending on the type of copy operation.

See the Parameter associations section for possible values that can be specified.

CopyType

Specifies the type of copy operation.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion and must be set to M_DEFAULT.

Parameter associations

The table below lists possible values for the **SrcContext3dmodId**, **SrcIndex**, **DstIndex**, and **CopyType** parameters and possible values returned to the **DstAilObjectId** parameter.

- For specifying the copy type

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the copy type																	
CopyType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
SrcContext3dmodId																
SrcIndex																
DstAilObjectId - Possible values returned																
DstIndex																
M_FLOOR	Specifies to copy the plane that defines the floor of the scene to or from the specified find box, surface, or planar surface 3D model finder context. The floor plane provides contextual information about the position and orientation of objects within the scene. For a find surface or planar surface 3D model finder context, you must copy a floor plane to the find surface or planar surface 3D model finder context if you want to use a resting plane. The resting plane uses the floor plane as a reference when establishing the tilt angle of the occurrence. For a find box 3D model finder context, the floor plane is considered another possible background plane (M_COMPLETION_TO_BACKGROUND), so a single-faced box could be extruded to the floor plane in some situations. Refer to the Completion size when a single box face is found subsection of the Search settings specific to geometric models section of Chapter 37: 3D model finder for more information. Note that M_REMOVE_BACKGROUND does not use the specified plane to establish the background points to remove. You can remove floor points, using M_REMOVE_FLOOR.															
M_NULL	Specifies to remove the plane that defines the floor of the scene from the specified find box, surface, or planar surface 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b
SrcIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
DstAilObjectId	Specifies the identifier of a find box, surface, or planar surface 3D model finder context from which to remove the plane that defines the floor of the scene.	c		e	f	g	a		i		j	k	l	m	n	b

	DstIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
3D plane geometry object ID from which to copy		Specifies the identifier of a 3D plane geometry object from which to copy the floor plane. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY and defined as a plane.	c		e	f	g	a		i		j	k	l	m	n	b
	SrcIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
	DstAilObjectId	Specifies the identifier of a find box, surface, or planar surface 3D model finder context in which to copy the plane that defines the floor of the scene.	c		e	f	g	a		i		j	k	l	m	n	b
	DstIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
Find box 3D model finder context ID from which to copy		Specifies the identifier of a find box 3D model finder context from which to copy the floor plane.	c		e	f	g	a		i		j	k	l	m	n	b
	SrcIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
	DstAilObjectId	Specifies the identifier of the destination 3D geometry object in which to copy the floor plane. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY. Note that after calling **M3dmodCopy**(), the destination 3D geometry object will return type M_PLANE if inquired using M3dgeoInquire() with M_GEOMETRY_TYPE.	c		e	f	g	a		i		j	k	l	m	n	b
	DstIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
Find surface or planar surface 3D model finder context ID from which to copy		Specifies the identifier of a find surface or planar surface 3D model finder context from which to copy the floor plane.	c		e	f	g	a		i		j	k	l	m	n	b
	SrcIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b

		DstAilObjectId	Specifies the identifier of the destination 3D geometry object in which to copy the floor plane. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY. Note that after calling **M3dmodCopy**(), the destination 3D geometry object will return type M_PLANE if inquired using M3dgeoInquire() with M_GEOMETRY_TYPE.	c		e	f	g	a		i		j	k	l	m	n	b
		DstIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRIP_LABEL_IMAGE			Specifies to copy the grip label image to or from the specified model in the specified find surface 3D model finder context. In a grip label image, the gray value of each pixel is the grip label of the corresponding point in the model. Pixels with a value of zero represent unlabeled points. Note that the location of each pixel in the grip label image corresponds to the point at the same location in the range component of the model point cloud container. When an occurrence of a model with a grip label image is found, a score is calculated for each labeled surface. This is useful if you want to determine the best surface of the occurrence for gripping. You can retrieve the grip scores using M3dmodGetResult() with M_SCORE_GRIP().	c		e	f	g	a		i		j	k	l	m	n	b
	M_NULL		Specifies to remove the grip label image from the specified find surface 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b
		SrcIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
		DstAilObjectId	Specifies the identifier of a surface 3D model finder context from which to remove the grip label image.	c		e	f	g	a		i		j	k	l	m	n	b
		DstIndex	Specifies the index of the model in the destination context from which to remove the grip label image.	c		e	f	g	a		i		j	k	l	m	n	b
		0 <= *Value* <M_NUMBER_MODELS	Specifies the index of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	Find surface 3D model finder context ID from which to copy		Specifies the identifier of a find surface 3D model finder context from which to copy the grip label image.	c		e	f	g	a		i		j	k	l	m	n	b
		SrcIndex	Specifies the index of the model in the source context from which to copy the grip label image.	c		e	f	g	a		i		j	k	l	m	n	b

0 <= *Value* <M_NUMBER_MODELS	Specifies the index of the model.	c		e	f	g	a		i		j	k	l	m	n	b
DstAilObjectId	Specifies the identifier of the destination image buffer in which to copy the grip label image. The image buffer must be a 1-band, 8-bit unsigned buffer. The image buffer size must equal the size of the container (range component) used to define the model, in both X and Y. To retrieve the required size, use M3dmodInquire() with M_MODEL_RANGE_SIZE_X and M_MODEL_RANGE_SIZE_Y.	c		e	f	g	a		i		j	k	l	m	n	b
DstIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
Grip label image buffer ID from which to copy	Specifies the identifier of an image buffer from which to copy the grip label image. The image buffer must be a 1-band, 8-bit unsigned buffer. The image buffer size must equal the size of the container (range component) used to define the model, in both X and Y. To retrieve the required size, use M3dmodInquire() with M_MODEL_RANGE_SIZE_X and M_MODEL_RANGE_SIZE_Y.	c		e	f	g	a		i		j	k	l	m	n	b
SrcIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
DstAilObjectId	Specifies the identifier of a find surface 3D model finder context in which to copy the grip label image.	c		e	f	g	a		i		j	k	l	m	n	b
DstIndex	Specifies the index of the model in the destination context in which to copy the grip label image.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <M_NUMBER_MODELS	Specifies the index of the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL	Specifies to copy the point cloud of the model, as it was extracted before preprocessing, from the specified find surface or planar surface 3D model finder context to the specified container. The model must have been previously defined with M3dmodDefine().	c		e	f	g	a		i		j	k	l	m	n	b
Find surface or planar surface 3D model finder context ID from which to copy	Specifies the identifier of the find surface or planar surface 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b
SrcIndex	Specifies the index of the model to copy from the source context.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <M_NUMBER_MODELS	Specifies the index of the model to copy.	c		e	f	g	a		i		j	k	l	m	n	b

		DstAilObjectId	Specifies the identifier of the container in which to copy the point cloud of the model. The container must have been previously allocated using MbufAllocContainer() with M_PROC, and must not be a child container.	c		e	f	g	a		i		j	k	l	m	n	b
		DstIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_PREPROCESSED			Specifies to copy the point cloud of the preprocessed version of the model, from the specified find surface or planar surface 3D model finder context to the specified container. The model must have been previously defined using M3dmodPreprocess(), and must have been called before this call.	c		e	f	g	a		i		j	k	l	m	n	b
	Find surface or planar surface 3D model finder context ID from which to copy		Specifies the identifier of the find surface or planar surface 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b
		SrcIndex	Specifies the index of the model to copy from the source context.	c		e	f	g	a		i		j	k	l	m	n	b
		0 <= *Value* <M_NUMBER_MODELS	Specifies the index of the model to copy.	c		e	f	g	a		i		j	k	l	m	n	b
		DstAilObjectId	Specifies the identifier of the container in which to copy the point cloud of the model. The container must have been previously allocated using MbufAllocContainer() with M_PROC, and must not be a child container.	c		e	f	g	a		i		j	k	l	m	n	b
		DstIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_REFINE_REGISTRATION			Specifies to copy the refine registration settings available for use by the specified find surface or planar surface 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIND_SURFACE_CONTEXT_REFINEMENT_FAST		Specifies a predefined 3D registration context with settings that prioritize performance when fine-tuning results (M_REFINE_REGISTRATION with M_FIND_SURFACE_REFINEMENT_FAST).	c		e	f	g	a		i		j	k	l	m	n	b
		SrcIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
		DstAilObjectId	Specifies the identifier of the 3D registration context in which to copy the settings.	c		e	f	g	a		i		j	k	l	m	n	b
		3D registration context identifier	Specifies an Aurora Imaging Library 3D registration context, allocated using M3dregAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

			Internal 3D registration context ID	Specifies the internal 3D registration context used to perform the refine registration when M3dmodControl() with M_REFINE_REGISTRATION is set to M_USER_DEFINED_REGISTRATION. Retrieve this identifier by calling M3dmodInquire() with M_USER_DEFINED_REGISTRATION_CONTEXT_ID.	c		e	f	g	a		i		j	k	l	m	n	b
		DstIndex		This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIND_SURFACE_CONTEXT_REFINEMENT_PRECISE			Specifies a predefined 3D registration context with settings that prioritize accuracy when fine-tuning results (M_REFINE_REGISTRATION with M_FIND_SURFACE_REFINEMENT_PRECISE).	c		e	f	g	a		i		j	k	l	m	n	b
		SrcIndex		This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
		DstAilObjectId		Specifies the identifier of the 3D registration context in which to copy the settings.	c		e	f	g	a		i		j	k	l	m	n	b
			3D registration context identifier	Specifies an Aurora Imaging Library 3D registration context, allocated using M3dregAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
			Internal 3D registration context ID	Specifies the internal 3D registration context used to perform the refine registration when M3dmodControl() with M_REFINE_REGISTRATION is set to M_USER_DEFINED_REGISTRATION. Retrieve this identifier by calling M3dmodInquire() with M_USER_DEFINED_REGISTRATION_CONTEXT_ID.	c		e	f	g	a		i		j	k	l	m	n	b
		DstIndex		This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
	Find surface or planar surface 3D model finder context ID from which to copy			Specifies the identifier of the find surface or planar surface 3D model finder context from which to copy the current refine registration settings.	c		e	f	g	a		i		j	k	l	m	n	b
		SrcIndex		This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
		DstAilObjectId		Specifies the identifier of the 3D registration context in which to copy the settings.	c		e	f	g	a		i		j	k	l	m	n	b
			3D registration context identifier	Specifies an Aurora Imaging Library 3D registration context, allocated using M3dregAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

			Internal 3D registration context ID	Specifies the internal 3D registration context used to perform the refine registration when M3dmodControl() with M_REFINE_REGISTRATION is set to M_USER_DEFINED_REGISTRATION. Retrieve this identifier by calling M3dmodInquire() with M_USER_DEFINED_REGISTRATION_CONTEXT_ID.	c		e	f	g	a		i		j	k	l	m	n	b
		DstIndex		This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESTING_PLANE				Specifies to copy the resting plane to or from the specified find surface or planar surface 3D model finder context. The resting plane is used to provide contextual information about how the model occurrence lies within the scene. When matching, Aurora Imaging Library considers all poses of the occurrence unless a resting plane is specified. You can use M3dmodControl() with M_RESTING_PLANE_ANGLE_TOLERANCE to limit the acceptable angle of the model occurrence relative to the resting plane. This does not affect any of the scores, but it filters occurrences that do not sit on the resting plane, +/- the specified tolerance. Note that defining a resting plane does not decrease the search time because occurrences are filtered near the end of the search. Note that to use a resting plane, you must also copy a floor plane (M_FLOOR) to the find surface or planar surface 3D model finder context; the floor plane denotes the base of the scene. The resting plane uses the floor plane as a reference when establishing the tilt angle of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NULL			Specifies to remove the resting plane from the specified find surface or planar surface 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b
		SrcIndex		This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
		DstAilObjectId		Specifies the identifier of a find surface or planar surface 3D model finder context from which to remove the resting plane.	c		e	f	g	a		i		j	k	l	m	n	b
		DstIndex		Specifies the index of the model in the destination context, from which to remove the resting plane.	c		e	f	g	a		i		j	k	l	m	n	b
			0 <= *Value* <M_NUMBER_MODELS	Specifies the index of the model.	c		e	f	g	a		i		j	k	l	m	n	b

3D plane geometry object ID from which to copy	Specifies the identifier of a 3D plane geometry object from which to copy the resting plane. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY and defined as a plane.	c		e	f	g	a		i		j	k	l	m	n	b
SrcIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
DstAilObjectId	Specifies the identifier of a find surface or planar surface 3D model finder context in which to copy the resting plane.	c		e	f	g	a		i		j	k	l	m	n	b
DstIndex	Specifies the index of the model in the destination context in which to copy the plane.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <M_NUMBER_MODELS	Specifies the index in which to copy the plane.	c		e	f	g	a		i		j	k	l	m	n	b
Find surface or planar surface 3D model finder context ID from which to copy	Specifies the identifier of a find surface or planar surface 3D model finder context from which to copy the resting plane.	c		e	f	g	a		i		j	k	l	m	n	b
SrcIndex	Specifies the index of the model in the source context, from which to copy the resting plane.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <M_NUMBER_MODELS	Specifies the index of the model to copy.	c		e	f	g	a		i		j	k	l	m	n	b
DstAilObjectId	Specifies the identifier of the destination 3D geometry object in which to copy the resting plane. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY. Note that after calling **M3dmodCopy**(), the destination 3D geometry object will return type M_PLANE if inquired using M3dgeoInquire() with M_GEOMETRY_TYPE.	c		e	f	g	a		i		j	k	l	m	n	b
DstIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b

M_WEIGHT_IMAGE	Specifies to copy the weight image to or from the specified model in the specified find surface 3D model finder context. In a weight image, the gray value of each pixel is the weight of the corresponding point in the model. The weight image must consist of only non-zero pixels. Pixels with a value of zero will result in an error. Note that the location of each pixel in the weight image corresponds to the point at the same location in the range component of the model point cloud container. M3dmodFind() gives more importance to higher weighted points of the model when searching for occurrences. Model points that have a higher weight will also have a greater impact on the model score.	c		e	f	g	a		i		j	k	l	m	n	b
M_NULL	Specifies to remove the weight image from the specified find surface 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b
SrcIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
DstAilObjectId	Specifies the identifier of a surface 3D model finder context from which to remove the weight image.	c		e	f	g	a		i		j	k	l	m	n	b
DstIndex	Specifies the index of the model in the destination context from which to remove the weight image.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <M_NUMBER_MODELS	Specifies the index of the model.	c		e	f	g	a		i		j	k	l	m	n	b
Find surface 3D model finder context ID from which to copy	Specifies the identifier of a find surface 3D model finder context from which to copy the weight image.	c		e	f	g	a		i		j	k	l	m	n	b
SrcIndex	Specifies the index of the model in the source context from which to copy the weight image.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <M_NUMBER_MODELS	Specifies the index of the model.	c		e	f	g	a		i		j	k	l	m	n	b
DstAilObjectId	Specifies the identifier of the destination image buffer in which to copy the weight image. The image buffer must be a 1-band, 8-bit unsigned buffer. The image buffer size must equal the size of the container (range component) used to define the model, in both X and Y. To retrieve the required size, use M3dmodInquire() with M_MODEL_RANGE_SIZE_X and M_MODEL_RANGE_SIZE_Y.	c		e	f	g	a		i		j	k	l	m	n	b
DstIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b

	Weight image buffer ID from which to copy	Specifies the identifier of an image buffer from which to copy the weight image. The image buffer must be a 1-band, 8-bit unsigned buffer. The image buffer size must equal the size of the container (range component) used to define the model, in both X and Y. To retrieve the required size, use M3dmodInquire() with M_MODEL_RANGE_SIZE_X and M_MODEL_RANGE_SIZE_Y.	c		e	f	g	a		i		j	k	l	m	n	b
	SrcIndex	This parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
	DstAilObjectId	Specifies the identifier of a find surface 3D model finder context in which to copy the weight image.	c		e	f	g	a		i		j	k	l	m	n	b
	DstIndex	Specifies the index of the model in the destination context in which to copy the weight image.	c		e	f	g	a		i		j	k	l	m	n	b
	0 <= *Value* <M_NUMBER_MODELS	Specifies the index of the model.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodCopyResult

Synopsis

Copy a group of results from a 3D model finder result buffer into a 3D geometry object, an image buffer, a transformation matrix object, a model in a find surface or planar surface 3D model finder context, or a surface or planar surface 3D model finder result buffer.

Syntax

```
void M3dmodCopyResult(
    AIL_ID SrcResult3dmodId, //in
    AIL_INT64 SrcIndex,      //in
    AIL_ID DstAilObjectId,   //out
    AIL_INT64 DstIndex,      //in
    AIL_INT64 CopyType,      //in
    AIL_INT64 ControlFlag    //in
)
```

Description

This function copies a group of results from a 3D model finder result buffer into a 3D geometry object, an image buffer, a transformation matrix object, a model in a find surface or planar surface 3D model finder context, or a surface or planar surface 3D model finder result buffer, according to the specified copy operation.

Parameters

SrcResult3dmodId

Specifies the identifier of the 3D model finder result buffer from which to copy. The 3D model finder result buffer must have been allocated using M3dmodAllocResult() and must contain the results of a call to M3dmodFind().

SrcIndex

Specifies the index of the model occurrence in the result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the index of the model occurrence to copy																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Specifies to copy the result for all model occurrences.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Specifies to copy the result relating to the entire result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* < M_NUMBER	Specifies the index of the model occurrence for which to copy results.	c		e	f	g	a		i		j	k	l	m	n	b

DstAilObjectId

Specifies the identifier of the 3D geometry object, transformation matrix object, image buffer, find surface or planar surface 3D model finder context, or surface or planar surface 3D model finder result buffer in which to store the copied results.

See the Parameter associations section for possible values that can be returned.

DstIndex

Specifies the index of the element in the destination object in which to copy.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

CopyType

Specifies the type of copy operation to perform.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Parameter associations

The tables below list possible values for the **DstIndex** and **CopyType** parameters and possible values returned to the **DstAilObjectId** parameter.

- For copying general results from the source 3D model finder result buffer
- For copying the results of a specific occurrence or all occurrences from the source 3D model finder result buffer
- For copying the results of a specific occurrence from the source 3D model finder result buffer

To copy general results, set the CopyType parameter to the following value. In this case, you must set SrcIndex to M_GENERAL or M_DEFAULT.

Note that any unused parameters should be set to **M_DEFAULT**.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For copying general results from the source 3D model finder result buffer																		
CopyType			Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	DstAilObjectId - Possible values returned																	
		DstIndex																
M_BACKGROUND_MASK			Specifies to copy a non-zero value into an image buffer for each of the occurrence's background points and a 0 value elsewhere, creating a background mask image. Information about a point is set in the image buffer at the same location as the point's location in the range component of its point cloud container. Note that this is only available if M_REMOVE_BACKGROUND was enabled.															

Image buffer identifier in which to copy	Specifies the identifier of the image buffer in which to copy results. The image buffer must be a 1-band buffer. The image buffer size must equal the size of the container used during the 3D model finder operation, in both X and Y. To retrieve the size, use M3dmodGetResult() with M_RESULT_IMAGE_SIZE_X and M_RESULT_IMAGE_SIZE_Y. This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.	c		e	f	g	a		i		j	k	l	m	n	b
M_FLOOR_MASK	Specifies to copy a non-zero value into an image buffer for each of the floor points and a 0 value elsewhere, creating a floor mask image. Information about a point is set in the image buffer at the same location as the point's location in the range component of its point cloud container. Note that this is only available if M_REMOVE_FLOOR was enabled. Note that M_FLOOR_MASK is only available for surface and planar surface results.	c		e	f	g	a		i		j	k	l	m	n	b
Image buffer identifier in which to copy	Specifies the identifier of the image buffer in which to copy results. The image buffer must be a 1-band buffer. The image buffer size must equal the size of the container used during the 3D model finder operation, in both X and Y. To retrieve the size, use M3dmodGetResult() with M_RESULT_IMAGE_SIZE_X and M_RESULT_IMAGE_SIZE_Y. This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.	c		e	f	g	a		i		j	k	l	m	n	b

M_RESULT	Specifies to copy surface or planar surface results into another surface or planar surface 3D model finder result buffer, respectively, to reuse them in subsequent searches. You should use this copy type when you want to reuse the results as an optimization to specify the approximate location for multiple searches. For example, if you expect occurrences to be found in similar locations in multiple future searches, set up two result buffers: one for the initial search and another for use with subsequent searches. Before each subsequent search, copy into its result buffer the results from the initial search (so as not to change the original results), and use M3dmodControl() with M_REUSE_RESULT set to M_ENABLE to reuse them at the beginning of the search to speed up the operation. M3dmodFind() will search for occurrences at the stored positions before searching the rest of the target. You can use M3dmodModifyResult() to modify or delete results before performing the search. Note that if you only want to reuse the results once, you don't need to use this copy type; if no results are copied and M3dmodControl() with M_REUSE_RESULT is enabled, M3dmodFind() will reuse the most recent results in the specified result buffer. Note that M_RESULT is only available for surface and planar surface results.	c		e	f	g	a		i		j	k	l	m	n	b
Planar surface 3D model finder result buffer identifier in which to copy	Specifies the identifier of a planar surface 3D model finder result buffer in which to copy planar surface results. The result buffer must have been previously allocated using M3dmodAllocResult() with M_FIND_PLANAR_SURFACE_RESULT.	c		e	f	g	a		i		j	k	l	m	n	b
Surface 3D model finder result buffer identifier in which to copy	Specifies the identifier of a surface 3D model finder result buffer in which to copy surface results. The result buffer must have been previously allocated using M3dmodAllocResult() with M_FIND_SURFACE_RESULT.	c		e	f	g	a		i		j	k	l	m	n	b

To copy the results of a specific occurrence or all occurrences, set the CopyType parameter to one of the following values. In this case, you must set SrcIndex to M_ALL or the index of the occurrence.

Note that any unused parameters should be set to **M_DEFAULT**.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For copying the results of a specific occurrence or all occurrences from the source 3D model finder result buffer																	
CopyType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
DstAilObjectId - Possible values returned																
DstIndex																
M_INLIER_INDEX_IMAGE	Specifies to copy the model occurrence's index into an image buffer for each of the occurrence's inlier points, creating an inlier area index image. The inlier points of the occurrence are based on M_FIT_DISTANCE. Information about a point is set in the image buffer at the same location as the point's location in the range component of its point cloud container. Points that are not part of any occurrence are given the value of -1 in signed buffers, and are set to the maximum buffer value in unsigned buffers. Note that for surface and planar surface results, this is only available if M_SAVE_FIT_INFO was enabled.															
Image buffer identifier in which to copy	Specifies the identifier of the image buffer in which to copy results. The image buffer must be a 1-band buffer. The image buffer size must equal the size of the container used during the 3D model finder operation, in both X and Y. To retrieve the size, use M3dmodGetResult() with M_RESULT_IMAGE_SIZE_X and M_RESULT_IMAGE_SIZE_Y. This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.	c		e	f	g	a		i		j	k	l	m	n	b
M_INLIER_MASK	Specifies to copy a non-zero value into an image buffer for each of the occurrence's inlier points and a 0 value elsewhere, creating an inlier mask image. The non-zero value is 1 for floating-point image buffers and -1 for signed image buffers. For unsigned image buffers, the maximum buffer value is used. All other pixels in the buffer are set to 0. The inlier points of the occurrence are based on M_FIT_DISTANCE. Information about a point is set in the image buffer at the same location as the point's location in the range component of its point cloud container. Note that for surface and planar surface results, this is only available if M_SAVE_FIT_INFO was enabled.	c		e	f	g	a		i		j	k	l	m	n	b

	Image buffer identifier in which to copy	Specifies the identifier of the image buffer in which to copy results. The image buffer must be a 1-band buffer. The image buffer size must equal the size of the container used during the 3D model finder operation, in both X and Y. To retrieve the size, use M3dmodGetResult() with M_RESULT_IMAGE_SIZE_X and M_RESULT_IMAGE_SIZE_Y. This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESERVED_INDEX_IMAGE		Specifies to copy the model occurrence's index into the destination image buffer for each of the points in the reserved area of the occurrence, creating a reserved area index image. For cylinder and sphere results, the points in the reserved area of the occurrence are based on M_RESERVED_POINTS_DISTANCE. Information about a point is set in the image buffer at the same location as the point's location in the range component of its point cloud container. Points that are not part of any occurrence are given the value of -1 in signed buffers, and are set to the maximum buffer value in unsigned buffers.	c		e	f	g	a		i		j	k	l	m	n	b
	Image buffer identifier in which to copy	Specifies the identifier of the image buffer in which to copy results. The image buffer must be a 1-band buffer. The image buffer size must equal the size of the container used during the 3D model finder operation, in both X and Y. To retrieve the size, use M3dmodGetResult() with M_RESULT_IMAGE_SIZE_X and M_RESULT_IMAGE_SIZE_Y. This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.	c		e	f	g	a		i		j	k	l	m	n	b

		c		e	f	g	a		i		j	k	l	m	n	b
M_RESERVED_MASK	Specifies to copy a non-zero value into an image buffer for each of the occurrence's reserved points and a 0 value elsewhere, creating a reserved mask image. The non-zero value is 1 for floating-point image buffers and -1 for signed image buffers. For unsigned image buffers, the maximum buffer value is used. All other pixels in the buffer are set to 0. For cylinder and sphere results, the points in the reserved area of the occurrence are based on M_RESERVED_POINTS_DISTANCE. Information about a point is set in the image buffer at the same location as the point's location in the range component of its point cloud container. Note that this copy type is useful for creating a mask image to use, for example, with M3dimCrop().															
Image buffer identifier in which to copy	Specifies the identifier of the image buffer in which to copy results. The image buffer must be a 1-band buffer. The image buffer size must equal the size of the container used during the 3D model finder operation, in both X and Y. To retrieve the size, use M3dmodGetResult() with M_RESULT_IMAGE_SIZE_X and M_RESULT_IMAGE_SIZE_Y. This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.	c		e	f	g	a		i		j	k	l	m	n	b

To copy the results of a specific occurrence, set the CopyType parameter to one of the following values. In this case, you must set SrcIndex to the index of the occurrence.

Note that any unused parameters should be set to **M_DEFAULT**.

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For copying the results of a specific occurrence from the source 3D model finder result buffer																
CopyType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
DstAilObjectId - Possible values returned																
DstIndex																
M_BOUNDING_BOX	Specifies to copy the oriented bounding box of the found model occurrence into the specified 3D geometry object, establishing a 3D box geometry.															
3D geometry object identifier in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b

M_CENTER +	Specifies to copy, into a 3D geometry object, the center of the specified face of the box occurrence, establishing a point geometry. Note that M_CENTER is only available for box occurrences. You must specify a combination value from the following table: • To specify for which face of the box occurrence to copy results	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object identifier in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY. Note that after calling **M3dmodCopyResult**(), the destination 3D geometry object will return type M_POINT if inquired using M3dgeoInquire() with M_GEOMETRY_TYPE.	c		e	f	g	a		i		j	k	l	m	n	b
M_FACE +	Specifies to copy, into a 3D geometry object, the specified face of the box occurrence, establishing a plane geometry. Note that M_FACE is only available for box occurrences. You must specify a combination value from the following table: • To specify for which face of the box occurrence to copy results	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object identifier in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY. Note that after calling **M3dmodCopyResult**(), the destination 3D geometry object will return type M_PLANE if inquired using M3dgeoInquire() with M_GEOMETRY_TYPE.	c		e	f	g	a		i		j	k	l	m	n	b

M_FIXTURING_MATRIX +	Specifies to copy, into the transformation matrix, the transformation that can move the working coordinate system to the found model occurrence. For a box occurrence, the transformation moves the working coordinate system's origin to the box's center. The box's rotation is applied if there is one. You can add M_FACE_INDEX() to M_FIXTURING_MATRIX to copy the fixturing matrix for one face of a box occurrence. For a face, the transformation moves the working coordinate system's X- and Y- axis to the center of the face, and the normal becomes the Z-axis. For a cylinder occurrence, the transformation moves the working coordinate system's origin to the cylinder's start point, and the cylinder's central axis becomes the positive Z-axis. The rotation of the X- and Y-axis is determined by the minimal rotation that must be applied to align the positive Z-axis with the cylinder's central axis. For a rectangular plane occurrence, the transformation moves the working coordinate system's X- and Y- axis to the rectangular plane, and the normal becomes the Z-axis. For a sphere occurrence, the transformation moves the working coordinate system's origin to the sphere's center. No rotation is applied. For a surface occurrence, the transformation moves the working coordinate system such that it aligns with the model's reference axis transformed at the occurrence. You can specify a combination value from the following table: • To specify for which face of the box occurrence to copy results	c		e	f	g	a		i		j	k	l	m	n	b
Transformation matrix object identifier in which to copy	Specifies the identifier of the transformation matrix object in which to copy results. The transformation matrix object must have been previously allocated using M3dgeoAlloc() with M_TRANSFORMATION_MATRIX.	c		e	f	g	a		i		j	k	l	m	n	b

M_GEOMETRY	Specifies to copy the geometry of the found model occurrence, into the specified 3D geometry object, establishing a 3D geometry of the same type. Note that M_GEOMETRY is only available for sphere, cylinder, and box model results.	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object identifier in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b
M_OCCURRENCE_MATRIX	Specifies to copy, into the transformation matrix, the transformation that transforms the model to the same location as the found occurrence. Note that M_OCCURRENCE_MATRIX is only available for surface and planar surface results.	c		e	f	g	a		i		j	k	l	m	n	b
Transformation matrix object identifier in which to copy	Specifies the identifier of the transformation matrix object in which to copy results. The transformation matrix object must have been previously allocated using M3dgeoAlloc() with M_TRANSFORMATION_MATRIX.	c		e	f	g	a		i		j	k	l	m	n	b
M_PLANE	Specifies to copy into a 3D geometry object the infinite plane on which the rectangular plane occurrence is found. Note that M_PLANE is only available for rectangular plane results.	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object identifier in which to copy	Specifies the identifier of a 3D geometry object in which to copy results. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESTING_PLANE	Specifies to copy the resting plane on which the occurrence lies (set using M3dmodCopy() with M_FLOOR), to the model in the specified find surface or planar surface 3D model finder context or into the specified 3D geometry object. Note that M_RESTING_PLANE is only available for surface and planar surface results.	c		e	f	g	a		i		j	k	l	m	n	b

3D geometry object identifier in which to copy	Specifies the identifier of a 3D geometry object in which to copy the resting plane. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b
Find planar surface 3D model finder context identifier in which to copy	Specifies the identifier of a find planar surface 3D model finder context in which to copy the resting plane. The context must have been previously allocated using M3dmodAlloc() with M_FIND_PLANAR_SURFACE_CONTEXT.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* < M_NUMBER_MODELS	Specifies the index of the model in the find planar surface 3D model finder context in which to copy the resting plane.	c		e	f	g	a		i		j	k	l	m	n	b
Find surface 3D model finder context identifier in which to copy	Specifies the identifier of a find surface 3D model finder context in which to copy the resting plane. The context must have been previously allocated using M3dmodAlloc() with M_FIND_SURFACE_CONTEXT.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* < M_NUMBER_MODELS	Specifies the index of the model in the find surface 3D model finder context in which to copy the resting plane.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for M_CENTER; M_FACE; M_FIXTURING_MATRIX.

You must add the following value to the above-mentioned values to specify for which face of the box occurrence to copy results. Note that this combination is optional for some values. Refer to the individual values to determine if this is the case.

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For specifying the box face's index																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FACE_INDEX(AIL_INT Index)	Specifies to copy results for a specific face of a box occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Index Set this parameter to the following:															

			0 <= *Value* < 6	Specifies the index of the face for which to copy the result. Indices 0 and 5 correspond to the faces with the minimum XYZ and maximum XYZ coordinate values, respectively.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodDefine

Synopsis

Add a model to, or delete a model from, a 3D model finder context.

Syntax

```
AIL_INT64 M3dmodDefine(
    AIL_ID Context3dmodId, //out
    AIL_INT64 Operation,   //in
    AIL_INT64 ModelType,   //in
    AIL_DOUBLE Param1,     //in
    AIL_DOUBLE Param2,     //in
    AIL_DOUBLE Param3,     //in
    AIL_DOUBLE Param4,     //in
    AIL_DOUBLE Param5,     //in
    AIL_DOUBLE Param6,     //in
    AIL_INT64 ControlFlag  //in
)
```

Description

This function allows you to add a model to, or remove a model from, a 3D model finder context. Note, when a model is added or deleted, the 3D model finder context must be preprocessed again, using M3dmodPreprocess(). You can define one model per model finder context.

A geometric model can be defined either nominally or using a range of accepted values. Nominal models should be used when the expected occurrences are all approximately the same size; define models as ranges when you want to accept a variety of occurrences at different scales. Note that for range-type models, the proportions of the occurrences don't have to be the same as that of the model, as long as they are in the valid range.

Models are defined from specified numeric constraints, from one or two specified 3D geometries, or from a point cloud.

The search is performed according to the general search settings specified in the 3D model finder context (M_DEFAULT), as well as the individual model search settings. Both the context and individual model search settings can be specified using M3dmodControl().

Parameters

Context3dmodId

Specifies the 3D model finder context to which to add, or from which to delete, the model. The 3D model finder context must have been previously allocated using M3dmodAlloc() with M_FIND_..._CONTEXT.

Operation

Specifies whether to add or remove a model.

See the Parameter associations section for possible values that can be specified.

ModelType

Specifies the type of model to define when adding models to the context. Note that the type of model specified must match the specified 3D model finder context.

See the Parameter associations section for possible values that can be specified.

Param1

Specifies the first parameter. Its definition is dependent on the model type chosen.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

Param2

Specifies the second parameter. Its definition is dependent on the model type chosen.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

Param3

Specifies the third parameter. Its definition is dependent on the model type chosen.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

Param4

Specifies the fourth parameter. Its definition is dependent on the model type chosen.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

Param5

Specifies the fifth parameter. Its definition is dependent on the model type chosen.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

Param6

Specifies the sixth parameter. Its definition is dependent on the model type chosen.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Parameter associations

The table below lists possible values for the **Operation**, **ModelType**, **Param1**, **Param2**, **Param3**, **Param4**, **Param5**, and **Param6** parameters.

- For adding or removing a model from the context

To add a model to the 3D model finder context, the Operation, ModelType, Param1, Param2, Param3, Param4, Param5, and Param6 parameters can be set to the following values. Note that any unused parameters should be set to **M_DEFAULT**.

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For adding or removing a model from the context

Operation	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ModelType																
Param1																
Param2																
Param3																
Param4																
Param5																
Param6																
M_ADD	Adds a model to the 3D model finder context.															
M_BOX	Specifies to add a nominal box as the model.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the nominal length of the box model along its first (X) axis.	c		e	f	g	a		i		j	k	l	m	n	b
Param2	Specifies the nominal length of the box model along its second (Y) axis.	c		e	f	g	a		i		j	k	l	m	n	b
Param3	Specifies the nominal length of the box model along its third (Z) axis.	c		e	f	g	a		i		j	k	l	m	n	b
Param4	Specifies the tolerance for the length of the occurrences along their first (X) axis. Occurrences must have a length along their first axis within the specified nominal length (Param1) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
Param5	Specifies the tolerance for the length of the occurrences along their second (Y) axis. Occurrences must have a length along their second axis within the specified nominal length (Param2) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b

	Param6	Specifies the tolerance for the length of the occurrences along their third (Z) axis. Occurrences must have a length along their third axis within the specified nominal length (Param3) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_RANGE		Specifies to add a box, defined with a range of possible values, as the model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1	Specifies the minimum length of the first (X) axis of the box model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2	Specifies the minimum length of the second (Y) axis of the box model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param3	Specifies the minimum length of the third (Z) axis of the box model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param4	Specifies the maximum length of the first (X) axis of the box model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param5	Specifies the maximum length of the second (Y) axis of the box model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param6	Specifies the maximum length of the third (Z) axis of the box model.	c		e	f	g	a		i		j	k	l	m	n	b
M_CYLINDER		Specifies to add a nominal cylinder as the model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1	Specifies the nominal radius of the cylinder model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2	Specifies the nominal length of the cylinder model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param3	Specifies the tolerance for the radius of the occurrences. Occurrences must have a radius within the specified nominal radius (Param1) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b

		Param4	Specifies the tolerance for the length of occurrences. Occurrences must have a length within the specified nominal length (Param2) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
		Param5	Specifies whether the cylinder model includes bases. Setting this value reserves points corresponding to a found occurrence's bases. > Note that points found on an occurrence's bases will not affect the fit regardless of this setting.	c		e	f	g	a		i		j	k	l	m	n	b
			M_DEFAULT	Same as M_WITH_BASES.	c		e	f	g	a		i		j	k	l	m	n	b
			M_WITH_BASES	Specifies that the cylinder model includes bases.	c		e	f	g	a		i		j	k	l	m	n	b
			M_WITHOUT_BASES	Specifies that the cylinder model does not include bases.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CYLINDER_RANGE			Specifies to add a cylinder, defined with a range of possible values, as the model.	c		e	f	g	a		i		j	k	l	m	n	b
		Param1		Specifies the minimum radius of the cylinder model.	c		e	f	g	a		i		j	k	l	m	n	b
		Param2		Specifies the minimum length of the cylinder model.	c		e	f	g	a		i		j	k	l	m	n	b
		Param3		Specifies the maximum radius of the cylinder model.	c		e	f	g	a		i		j	k	l	m	n	b
		Param4		Specifies the maximum length of the cylinder model.	c		e	f	g	a		i		j	k	l	m	n	b
		Param5		Specifies whether the cylinder model includes bases. Setting this value reserves points corresponding to a found occurrence's bases. > Note that points found on an occurrence's bases will not affect the fit regardless of this setting.	c		e	f	g	a		i		j	k	l	m	n	b
			M_DEFAULT	Same as M_WITH_BASES.	c		e	f	g	a		i		j	k	l	m	n	b

M_WITH_BASES	Specifies that the cylinder model includes bases.	c		e	f	g	a		i		j	k	l	m	n	b
M_WITHOUT_BASES	Specifies that the cylinder model does not include bases.	c		e	f	g	a		i		j	k	l	m	n	b
M_RECTANGLE	Specifies to add a nominal rectangular plane as the model.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the nominal length of the rectangular plane model along its first (X) axis.	c		e	f	g	a		i		j	k	l	m	n	b
Param2	Specifies the nominal length of the rectangular plane model along its second (Y) axis.	c		e	f	g	a		i		j	k	l	m	n	b
Param3	Specifies the tolerance for the length of the occurrences along their first (X) axis. Occurrences must have a length along their first axis within the specified nominal length (Param1) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
Param4	Specifies the tolerance for the length of the occurrences along their second (Y) axis. Occurrences must have a length along their second axis within the specified nominal length (Param2) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
M_RECTANGLE_RANGE	Specifies to add a rectangular plane, defined with a range of possible values, as the model.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the minimum length of the first (X) axis of the rectangular plane model.	c		e	f	g	a		i		j	k	l	m	n	b
Param2	Specifies the minimum length of the second (Y) axis of the rectangular plane model.	c		e	f	g	a		i		j	k	l	m	n	b
Param3	Specifies the maximum length of the first (X) axis of the rectangular plane model.	c		e	f	g	a		i		j	k	l	m	n	b

Param4	Specifies the maximum length of the second (Y) axis of the rectangular plane model.	c		e	f	g	a		i		j	k	l	m	n	b
M_SPHERE	Specifies to add a nominal sphere as the model.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the nominal radius of the sphere model.	c		e	f	g	a		i		j	k	l	m	n	b
Param2	Specifies the tolerance for the radius of the occurrences. Occurrences must have a radius within the specified nominal radius (Param1) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
M_SPHERE_RANGE	Specifies to add a sphere, defined with a range of possible values, as the model.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the minimum radius of the sphere model.	c		e	f	g	a		i		j	k	l	m	n	b
Param2	Specifies the maximum radius of the sphere model.	c		e	f	g	a		i		j	k	l	m	n	b
M_ADD_FROM_GEOMETRY	Adds a model to the 3D model finder context using predefined 3D geometries.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX	Specifies to add a nominal box as the model, using a predefined 3D geometry.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the identifier of a 3D box geometry object, previously allocated using M3dgeoAlloc() with M_GEOMETRY, and successfully defined as a box.	c		e	f	g	a		i		j	k	l	m	n	b
Param2	Specifies the tolerance for the length of the occurrences along their first (X) axis. Occurrences must have a length along their first axis within the specified 3D geometry's length (Param1) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b

		Param3	Specifies the tolerance for the length of the occurrences along their second (Y) axis. Occurrences must have a length along their second axis within the specified 3D geometry's length (Param1) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
		Param4	Specifies the tolerance for the length of the occurrences along their third (Z) axis. Occurrences must have a length along their third axis within the specified 3D geometry's length (Param3) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOX_RANGE		Specifies to add a box, defined with a range of possible sizes, as the model, using predefined 3D geometries.	c		e	f	g	a		i		j	k	l	m	n	b
		Param1	Specifies the identifier of a 3D box geometry object that defines the minimum lengths along the three axes (X, Y, Z). The geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY, and successfully defined as a box.	c		e	f	g	a		i		j	k	l	m	n	b
		Param2	Specifies the identifier of a 3D box geometry object that defines the maximum lengths along the three axes (X, Y, Z). The geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY, and successfully defined as a box.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CYLINDER		Specifies to add a nominal cylinder as the model, using a predefined 3D geometry.	c		e	f	g	a		i		j	k	l	m	n	b
		Param1	Specifies the identifier of a 3D cylinder geometry object, previously allocated using M3dgeoAlloc() with M_GEOMETRY, and successfully defined as a cylinder.	c		e	f	g	a		i		j	k	l	m	n	b

	Param2	Specifies the tolerance for the radius of occurrences. Occurrences must have a radius within the specified 3D geometry's radius (Param1) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
	Param3	Specifies the tolerance for the length of occurrences. Occurrences must have a length within the specified 3D geometry's length (Param1) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b
M_CYLINDER_RANGE		Specifies to add a cylinder, defined with a range of possible sizes, as the model, using predefined 3D geometries.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1	Specifies the identifier of a 3D cylinder geometry object that defines the minimum radius and length. The geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY, and successfully defined as a cylinder.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2	Specifies the identifier of a 3D cylinder geometry object that defines the maximum radius and length. The geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY, and successfully defined as a cylinder.	c		e	f	g	a		i		j	k	l	m	n	b
M_SPHERE		Specifies to add a nominal sphere as the model, using predefined 3D geometries.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1	Specifies the identifier of a 3D sphere geometry object, previously allocated using M3dgeoAlloc() with M_GEOMETRY, and successfully defined as a sphere.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2	Specifies the tolerance for the radius of the occurrences. Occurrences must have a radius within the specified 3D geometry's radius (Param1) +/- the specified tolerance. Note that if you expect occurrences to have a significantly different size than the model, you should use a range-type model instead, as the algorithm used for this type of model is better suited.	c		e	f	g	a		i		j	k	l	m	n	b

M_SPHERE_RANGE	Specifies to add a sphere, defined with a range of possible sizes, as the model, using predefined 3D geometries.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the identifier of a 3D sphere geometry object that defines the minimum radius. The geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY, and successfully defined as a sphere.	c		e	f	g	a		i		j	k	l	m	n	b
Param2	Specifies the identifier of a 3D sphere geometry object that defines the maximum radius. The geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY, and successfully defined as a sphere.	c		e	f	g	a		i		j	k	l	m	n	b
M_ADD_FROM_POINT_CLOUD	Adds a model to the 3D model finder context using the points in a point cloud.	c		e	f	g	a		i		j	k	l	m	n	b
M_SURFACE	Specifies to add an arbitrary surface, defined by a point cloud, as the model.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the identifier of a container containing a point cloud of a surface. The container must be 3D-processable (that is, if you call MbufInquireContainer() with M_3D_PROCESSABLE, the function returns M_PROCESSABLE).	c		e	f	g	a		i		j	k	l	m	n	b
Param2	Specifies whether to remove outliers from the model. When enabled, Aurora Imaging Library removes outliers such as clusters of points around the perimeter of the model. Enable outlier removal if the model contains stray points, especially if M_REFINE_REGISTRATION will be enabled. You can change this setting later, using M3dmodControl() with M_REMOVE_OUTLIERS. Note that you should not enable outlier removal if your model is defined from a point cloud created from a CAD file (using MbufImport() / MbufRestore()) since parts of the model can be removed, which can cause the search to fail/return false positives.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to perform outlier removal on the model.	c		e	f	g	a		i		j	k	l	m	n	b

			M_ENABLE	Specifies to perform outlier removal on the model.	c		e	f	g	a		i		j	k	l	m	n	b
		Param3		Specifies the position of the model's reference axis origin. A surface occurrence's found position is determined by the model's reference axis; position results return the coordinates of the model's reference axis origin transformed at the model occurrence. By default, the reference axis origin is at the origin of the working coordinate system of the original point cloud from which the model is defined. This does not necessarily correspond to the center of the model. You can use this parameter to move the origin of the working coordinate system to the required position.	c		e	f	g	a		i		j	k	l	m	n	b
			M_DEFAULT	Specifies the default behavior; the origin of the model's reference axis is at the origin of the working coordinate system of the original point cloud from which the model is defined.	c		e	f	g	a		i		j	k	l	m	n	b
			M_BOX_CENTER	Specifies to move the origin of the working coordinate system to the center of the point cloud's bounding box, setting the model's reference axis origin to this position.	c		e	f	g	a		i		j	k	l	m	n	b
			M_CENTROID	Specifies to move the origin of the working coordinate system to the point cloud's centroid (center of mass), setting the model's reference axis origin to this position.	c		e	f	g	a		i		j	k	l	m	n	b
			M_GEOMETRY_CENTER	Specifies to move the origin of the working coordinate system to the center of the specified 3D geometry, setting the model's reference axis origin to this position.	c		e	f	g	a		i		j	k	l	m	n	b
			M_TRANSFORMATION_MATRIX	Specifies to transform the point cloud according to the specified rigid transformation matrix.	c		e	f	g	a		i		j	k	l	m	n	b
		Param4		Specifies the identifier of a transformation matrix object or a 3D geometry object, depending on Param3. You must set this parameter to M_DEFAULT unless Param3 is set to M_GEOMETRY_CENTER or M_TRANSFORMATION_MATRIX.	c		e	f	g	a		i		j	k	l	m	n	b
			M_DEFAULT	Specifies the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b

			3D geometry object ID	Specifies the identifier of a 3D geometry object. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY, and must have been successfully defined as a box or a sphere.	c		e	f	g	a		i		j	k	l	m	n	b
			Transformation matrix object ID	Specifies the identifier of a transformation matrix object. The transformation object must have been previously allocated using M3dgeoAlloc() with M_TRANSFORMATION_MATRIX. The transformation matrix must be of type M_RIGID.	c		e	f	g	a		i		j	k	l	m	n	b
M_DELETE				Deletes a model from the 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT			Specifies the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b
		Param1		Specifies the index of the model to delete.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

Reserved for future expansion.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodDraw3d

Synopsis

Draw features of models, or results of found model occurrences, into a 3D graphics list.

Syntax

```
AIL_INT64 M3dmodDraw3d(
    AIL_ID OperationDraw3dContext3dmodId, //in
    AIL_ID SrcContextOrResult3dmodId,     //in
    AIL_INT64 SrcIndex,                   //in
    AIL_ID DstList3dgraId,                //out
    AIL_INT64 DstParentLabel,             //in
    AIL_INT64 ControlFlag                 //in
)
```

Description

This function draws the specified feature of a surface model, or result of any found model occurrence(s), stored in the specified find surface or planar surface 3D model finder context or find 3D model finder result buffer, into a 3D graphics list. Set the draw operations and options for the draw using M3dmodControlDraw().

> Note that for a 3D model finder context, the model must be defined using M3dmodDefine() and, if M_DRAW_MODEL_PREPROCESSED is active, the context must be preprocessed using M3dmodPreprocess(), before calling this function. For a 3D model finder result buffer, M3dmodFind() must be called before calling this function.

Parameters

OperationDraw3dContext3dmodId

Specifies the identifier of the draw 3D model finder context that specifies the annotations to draw and how to draw them. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the draw 3D model finder context																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies a predefined draw 3D model finder context with all draw operations (M3dmodControlDraw()) set to use default settings.	c		e	f	g	a		i		j	k	l	m	n	b
Draw 3D model finder context identifier	Specifies a valid draw 3D model finder context identifier, which you have allocated using M3dmodAlloc() with either M_DRAW_3D_GEOMETRIC_CONTEXT (to draw occurrences of geometric models) or M_DRAW_3D_SURFACE_CONTEXT (to draw occurrences of surface models or to draw features of surface models).	c		e	f	g	a		i		j	k	l	m	n	b

SrcContextOrResult3dmodId

Specifies the identifier of the 3D model finder context, previously allocated using M3dmodAlloc() with

M_FIND_SURFACE_CONTEXT or M_FIND_PLANAR_SURFACE_CONTEXT, or the identifier of the 3D model finder result buffer, previously allocated using M3dmodAllocResult() with M_FIND_BOX_RESULT, M_FIND_CYLINDER_RESULT, M_FIND_RECTANGULAR_PLANE_RESULT, M_FIND_SPHERE_RESULT, M_FIND_SURFACE_RESULT, or M_FIND_PLANAR_SURFACE_RESULT.

SrcIndex

Specifies the index of the model in the 3D model finder context, or of the occurrence in the 3D model finder result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the index of the model or occurrence to draw																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_ALL.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Draws the specified feature(s) of all models or all occurrences. A sub-parent label for each model or occurrence will be created under the parent label returned by this function.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* < M_NUMBER	Specifies the index of the model occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* < M_NUMBER_MODELS	Specifies the index of the model.	c		e	f	g	a		i		j	k	l	m	n	b

DstList3dgraId

Specifies the identifier of the 3D graphics list in which to draw. You can specify a 3D graphics list that you have previously allocated using M3dgraAlloc(), or you can specify the identifier of the 3D display's internal graphics list (inquired using M3ddispInquire() with M_3D_GRAPHIC_LIST_ID).

DstParentLabel

Specifies the label of the 3D graphic in the 3D graphics list to use as the model annotation's parent.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the parent label																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_ROOT_NODE.	c		e	f	g	a		i		j	k	l	m	n	b
M_ROOT_NODE	Specifies the top-most node of the 3D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the label of the 3D graphic in the 3D graphics list. Label 0 is the 3D graphics list's root node.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Return value

Returns the parent label of all 3D graphics that the function added to the 3D graphics list. If the specified 3D model

finder result buffer contains no occurrences, a node 3D graphic is added to the graphics list and the node's label is returned.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodFind

Synopsis

Search for the model in the specified 3D model finder context in the specified point cloud.

Syntax

```
void M3dmodFind(
    AIL_ID Context3dmodId, //in
    AIL_ID ContainerBufId, //in
    AIL_ID Result3dmodId,  //out
    AIL_INT64 ControlFlag  //in
)
```

Description

This function searches for the model in the specified 3D model finder context, in the specified point cloud. Results are stored in the specified 3D model finder result buffer. You can retrieve results using M3dmodGetResult().

To consider the color information of the surface model and target point cloud, set M3dmodControl() with M_USE_COLOR to M_ENABLE.

If you expect occurrences to be found in similar locations to the occurrences found in the previous search, you can use M3dmodControl() with M_REUSE_RESULT set to M_ENABLE to reuse the previous results to speed up the current search. **M3dmodFind**() will search the target point cloud at the positions of the occurrences found during the previous search, before searching for occurrences in the rest of the point cloud. To reuse the same results for multiple searches, set up two result buffers: one for the initial search and another for use with subsequent searches. Before each subsequent search, copy into its result buffer the results from the initial search, using M3dmodCopyResult() with M_RESULT, and use this result buffer for the search. If there is a known movement between scans or an occurrence is not always present, you can use M3dmodModifyResult() to transform or delete results for an occurrence.

A normals component (M_COMPONENT_NORMALS_AIL) must exist in the specified point cloud container. Otherwise, the **M3dmodFind**() operation will not complete successfully and M3dmodGetResult() with M_STATUS will return M_MISSING_COMPONENT_NORMALS_AIL. You can use M3dimNormals() to generate a normals component.

If an occurrence has a score greater than or equal to its model's certainty level (specified using M3dmodControl() with M_CERTAINTY), it is automatically considered an occurrence (default 90%); the remaining occurrences will be the best of those greater than or equal to the model's acceptance level (specified using M3dmodControl() with M_ACCEPTANCE).

The maximum number of occurrences found depends on the maximum number specified for the context (specified using M3dmodControl() with M_NUMBER).

The 3D model finder context must be preprocessed with M3dmodPreprocess() before calling this function.

Parameters

Context3dmodId

Specifies the identifier of the 3D model finder context to use for the search, and other settings related to the 3D model finder operation. The context must have been previously allocated using M3dmodAlloc() with M_FIND_BOX_CONTEXT, M_FIND_CYLINDER_CONTEXT, M_FIND_RECTANGULAR_PLANE_CONTEXT, M_FIND_SPHERE_CONTEXT, M_FIND_SURFACE_CONTEXT, or M_FIND_PLANAR_SURFACE_CONTEXT and a corresponding model must have been previously defined using M3dmodDefine().

ContainerBufId

Specifies the identifier of the 3D-processable point cloud container in which to search for the model.

The container must be 3D-processable (that is, if you call MbufInquireContainer() with M_3D_PROCESSABLE, the function returns M_PROCESSABLE).

Result3dmodId

Specifies the 3D model finder result buffer identifier in which to store results. The result buffer must have been previously allocated using M3dmodAllocResult().

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodFree

Synopsis

Free a 3D model finder context or a 3D model finder result buffer.

Syntax

```
void M3dmodFree(
    AIL_ID ContextOrResult3dmodId //in
)
```

Description

This function deletes the specified 3D model finder context or 3D model finder result buffer, and releases any memory associated with it.

All 3D model finder contexts and 3D model finder result buffers allocated on a particular system must be freed before the system can be freed, unless M_UNIQUE_ID was specified during allocation.

Parameter

ContextOrResult3dmodId

Specifies the identifier of the 3D model finder context or 3D model finder result buffer to free. These must have been successfully allocated (using M3dmodAlloc() or M3dmodAllocResult()) prior to calling this function.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodGetResult

Synopsis

Get the specified type of result(s) from a 3D model finder result buffer.

Syntax

```
AIL_DOUBLE M3dmodGetResult(
    AIL_ID Result3dmodId,  //in
    AIL_INT64 ResultIndex, //in
    AIL_INT64 ResultType,  //in
    void *ResultArrayPtr   //out
)
```

Description

This function retrieves the result(s) of the specified type from a 3D model finder result buffer.

Parameters

Result3dmodId

Specifies the identifier of the 3D model finder result buffer from which to retrieve results. The result buffer must have been previously allocated on the system using M3dmodAllocResult() with M_FIND_..._RESULT.

See the Parameter associations section for possible values that can be specified.

ResultIndex

Specifies where to get results. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the index of the 3D model																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Specifies to retrieve results for all model occurrences in the 3D model finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Specifies to retrieve general results from the 3D model finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* < M_NUMBER	Specifies the index of the model occurrence for which to retrieve results.	c		e	f	g	a		i		j	k	l	m	n	b

ResultType

Specifies the type of result to retrieve.

See the Parameter associations section for possible values that can be specified.

ResultArrayPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]
- array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]
- array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]
- array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]
- AIL_DOUBLE
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address of the array in which to write results. Since **M3dmodGetResult**() also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **Result3dmodId** and **ResultType** parameters and possible values returned to the **ResultArrayPtr** parameter.

- For retrieving general results from the 3D model finder result buffer
- For retrieving a result of a single occurrence or all occurrences from the 3D model finder result buffer

To retrieve a general result, the ResultType parameter should be set to one of the following values. In this case, you must set ResultIndex to M_GENERAL or M_DEFAULT. The following results are only available for retrieval after a 3D model finder operation, unless otherwise specified.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving general results from the 3D model finder result buffer																	
Result3dmodId	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultType																
ResultArrayPtr - Possible values returned																
3D model finder result buffer ID	Specifies a box, cylinder, rectangular plane, sphere, surface, or planar surface 3D model finder result buffer, allocated using M3dmodAllocResult() with M_FIND_..._RESULT, and used to store M3dmodFind() results.															
M_FIT_DISTANCE_USED +	Retrieves the fit distance used during the 3D model finder operation. For a surface model, this result is only available if M_SAVE_FIT_INFO was enabled.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER +	Retrieves the number of model occurrences found. This result is always available.	c		e	f	g	a		i		j	k	l	m	n	b

	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_FLOOR_OFFSET_USED +		Retrieves the offset, relative to the defined floor plane, within which points are removed. This result is only available if M_REMOVE_FLOOR was enabled. This result is only available for surface and planar surface 3D model finder result buffers.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_RESULT_IMAGE_SIZE_X +		Retrieves the size in X of the range component of the target point cloud, in pixels. Note that you can use this value to allocate an image buffer of an appropriate size for M3dmodCopyResult() to create an occurrence index image or mask image.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_RESULT_IMAGE_SIZE_Y +		Retrieves the size in Y of the range component of the target point cloud, in pixels. Note that you can use this value to allocate an image buffer of an appropriate size for M3dmodCopyResult() to create an occurrence index image or mask image.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_STATUS +		Retrieves the global status of the last M3dmodFind() operation. This result is always available.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ALL_POINTS_REPLICATED	Specifies that all points of the scene were replicated. Note that this result is only available for surface and planar surface 3D model finder result buffers.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COMPLETE	Specifies that the 3D model finder operation completed successfully. 3D model finder results are available.	c		e	f	g	a		i		j	k	l	m	n	b

		M_CURRENTLY_CALCULATING	Specifies that the 3D model finder operation is ongoing.	c		e	f	g	a		i		j	k	l	m	n	b
		M_INTERNAL_ERROR	Specifies that an unexpected internal error occurred during the 3D model finder operation.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MISSING_COMPONENT_NORMALS_AIL	Specifies that the source point cloud container does not have the M_COMPONENT_NORMALS_AIL component, so the 3D model finder operation was unable to be completed. You can use M3dimNormals() to generate a normals component.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NOT_ENOUGH_FEATURES	Specifies that the surface 3D model finder operation was not completed because of an insufficient number of unique features.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NOT_ENOUGH_MEMORY	Specifies that the 3D model finder operation was not completed because of a lack of available memory.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NOT_ENOUGH_VALID_DATA	Specifies that the 3D model finder operation failed because there were not enough valid points.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NOT_INITIALIZED	Specifies that the 3D model finder result buffer was not used in a call to M3dmodFind(), and contains no results.	c		e	f	g	a		i		j	k	l	m	n	b
		M_STOPPED_BY_REQUEST	Specifies that the 3D model finder operation was stopped from another thread using M3dmodControl() with M_STOP_FIND.	c		e	f	g	a		i		j	k	l	m	n	b
		M_TIMEOUT_REACHED	Specifies that the 3D model finder operation took longer than the allowed value, specified using M3dmodControl() with M_TIMEOUT, and stopped before completion.	c		e	f	g	a		i		j	k	l	m	n	b
		M_TRANSFORMED	Specifies that the surface or planar surface 3D model finder result buffer contains results that were transformed using M3dmodModifyResult() with M_TRANSFORM.	c		e	f	g	a		i		j	k	l	m	n	b

M_STATUS_REFINE_REGISTRATION +	Retrieves the global status of the refine registration applied to the different occurrences. This result is only available if M_REFINE_REGISTRATION was enabled. This result is only available for surface and planar surface 3D model finder result buffers.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_APPLIED	Specifies that refine registration was applied successfully to all occurrences.	c		e	f	g	a		i		j	k	l	m	n	b
M_APPLIED_AND_REJECTED	Specifies that even though M_REFINE_REGISTRATION was enabled, at least one registration was not applied successfully to an occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_NOT_APPLIED	Specifies that refine registration was not enabled, so it was not applied to the occurrences.	c		e	f	g	a		i		j	k	l	m	n	b
M_STATUS_SCENE_RESOLUTION +	Retrieves whether the resolution of the surface model was compatible with the resolution of the scene. If the resolution of the model was much lower or higher than that of the scene, this result type will return M_INCOMPATIBLE. You can use M3dmodInquire() with M_MODEL_RESOLUTION to inquire the resolution of the model. You can use M3dmodControl() with M_SEARCH_POINT_RESOLUTION to specify the resolution of the scene; Aurora Imaging Library will attempt to increase or decrease the model's resolution accordingly. This result is only available for surface and planar surface 3D model finder result buffers.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_COMPATIBLE	Specifies that the resolution was compatible.	c		e	f	g	a		i		j	k	l	m	n	b
M_INCOMPATIBLE	Specifies that the resolution was not compatible.	c		e	f	g	a		i		j	k	l	m	n	b

To retrieve a result for a single occurrence or for all occurrences, the ResultType parameter should be set to one of the following values. In this case, you must set ResultIndex to the index of the occurrence or M_ALL. The following

results are only available for retrieval after a 3D model finder operation.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving a result of a single occurrence or all occurrences from the 3D model finder result buffer																	
Result3dmodId ResultType ResultArrayPtr - Possible values returned	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Box 3D model finder result buffer ID	Specifies a box 3D model finder result buffer, allocated using M3dmodAllocResult() with M_FIND_BOX_RESULT, and used to store M3dmodFind() results.															
M_AREA +	Retrieves the surface area of the model occurrence. The surface area is calculated using the following formula: $SA = 2(l * w + l * h + w * h)$.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_X +	Retrieves the X-coordinate of the center point of the box occurrence or its specified face (depending on whether you add M_FACE_INDEX() to M_CENTER_X), expressed in the working coordinate system. For a box occurrence, the X-coordinate of the box's center point is retrieved. For a box occurrence's face, the X-coordinate of the face's center point is retrieved. You can specify a combination value from the following table: • To specify for which face of the box to retrieve results	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_CENTER_Y +	Retrieves the Y-coordinate of the center point of the box occurrence or its specified face (depending on whether you add M_FACE_INDEX() to M_CENTER_Y), expressed in the working coordinate system. For a box occurrence, the Y-coordinate of the box's center point is retrieved. For a box occurrence's face, the Y-coordinate of the face's center point is retrieved. You can specify a combination value from the following table: • To specify for which face of the box to retrieve results	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Z +	Retrieves the Z-coordinate of the center point of the box occurrence or its specified face (depending on whether you add M_FACE_INDEX() to M_CENTER_Z), expressed in the working coordinate system. For a box occurrence, the Z-coordinate of the box's center point is retrieved. For a box occurrence's face, the Z-coordinate of the face's center point is retrieved. You can specify a combination value from the following table: • To specify for which face of the box to retrieve results	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_COMPLETION_STATUS +	Retrieves which method was used to complete the box.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_COMPLETION_TO_BACKGROUND	Specifies that the face was extruded to a background plane.	c		e	f	g	a		i		j	k	l	m	n	b
M_COMPLETION_TO_STAIRCASE	Specifies that the face was extruded to an edge of a staircase plane.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COMPLETION_TO_USER_SIZE	Specifies that the face was extruded to the completion size.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that no completion method was used because an occurrence with more than 1 face was found.	c		e	f	g	a		i		j	k	l	m	n	b
M_COVERAGE +		Retrieves the plane coverage of the box occurrence's specified face. The plane coverage is the percentage of a plane's surface found in the face and covered by inlier points. You must specify a combination value from the following table: • To specify for which face of the box to retrieve results	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the plane coverage, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_X +		Retrieves the X-component of the normal unit vector of the box occurrence's specified face. You must specify a combination value from the following table: • To specify for which face of the box to retrieve results	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	-1.0 <= *Value* <= 1.0	Specifies the X-component of the box face's normal unit vector, expressed in the working coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_Y +		Retrieves the Y-component of the normal unit vector of the box occurrence's specified face. You must specify a combination value from the following table: • To specify for which face of the box to retrieve results	c		e	f	g	a		i		j	k	l	m	n	b

	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	-1.0 <= *Value* <= 1.0	Specifies the Y-component of the box face's normal unit vector, expressed in the working coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_Z +		Retrieves the Z-component of the normal unit vector of the box occurrence's specified face. You must specify a combination value from the following table: • To specify for which face of the box to retrieve results	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	-1.0 <= *Value* <= 1.0	Specifies the Z-component of the box face's normal unit vector, expressed in the working coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS +		Retrieves the number of points in the box occurrence or its specified face, depending on whether you add M_FACE_INDEX() to M_NUMBER_OF_POINTS. You can specify a combination value from the following table: • To specify for which face of the box to retrieve results	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the number of points.	c		e	f	g	a		i		j	k	l	m	n	b

M_NUMBER_OF_POINTS_RESERVED +	Retrieves the number of reserved points in the box occurrence or its specified face, depending on whether you add M_FACE_INDEX() to M_NUMBER_OF_POINTS_RESERVED. These are the points in the reserved area around the occurrence. They are not considered in the fit, are not included in the calculation of the score, and are not considered for other occurrences. You can specify a combination value from the following table: • To specify for which face of the box to retrieve results	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the number of points.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_VISIBLE_FACES +	Retrieves the number of visible faces (planes) used to fit the box occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the number of visible faces (planes).	c		e	f	g	a		i		j	k	l	m	n	b
M_RMS_ERROR +	Retrieves the root-mean-square (RMS) error between the points found in the box occurrence or its specified face (depending on whether you add M_FACE_INDEX() to M_RMS_ERROR) and the source model. You can specify a combination value from the following table: • To specify for which face of the box to retrieve results	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_SCORE +	Retrieves the score of the occurrence, where the score is based on the occurrence's model coverage. The model coverage is the percentage of the model's surface found in the occurrence and covered by inlier points. The score is defined relative to the maximum expected model coverage, such that an occurrence with a score of 100% has a model coverage equal to the maximum expected model coverage. You can set the maximum expected model coverage, using M3dmodControl() with M_COVERAGE_MAX. You can add M_FACE_INDEX() to M_SCORE to retrieve this result for one or all of the box occurrence's faces. The score of a box face is defined relative to the maximum expected plane coverage. In this case, a face with a score of 100% has a plane coverage equal to the maximum expected plane coverage. You can set the maximum expected plane coverage, using M3dmodControl() with M_PLANE_MAX_COVERAGE. You can specify a combination value from the following table: • To specify for which face of the box to retrieve results	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the score of the occurrence, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X +	Retrieves the length of the box occurrence along its X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y +	Retrieves the length of the box occurrence along its Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Z +	Retrieves the length of the box occurrence along its Z-axis.	c		e	f	g	a		i		j	k	l	m	n	b

	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_UNROTATED_MAX_X +		Retrieves the box occurrence's maximum X-coordinate, ignoring the box's rotation. The coordinate is returned as if the box is axis-aligned.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_UNROTATED_MAX_Y +		Retrieves the box occurrence's maximum Y-coordinate, ignoring the box's rotation. The coordinate is returned as if the box is axis-aligned.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_UNROTATED_MAX_Z +		Retrieves the box occurrence's maximum Z-coordinate, ignoring the box's rotation. The coordinate is returned as if the box is axis-aligned.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_UNROTATED_MIN_X +		Retrieves the box occurrence's minimum X-coordinate, ignoring the box's rotation. The coordinate is returned as if the box is axis-aligned.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_UNROTATED_MIN_Y +		Retrieves the box occurrence's minimum Y-coordinate, ignoring the box's rotation. The coordinate is returned as if the box is axis-aligned.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_UNROTATED_MIN_Z +		Retrieves the box occurrence's minimum Z-coordinate, ignoring the box's rotation. The coordinate is returned as if the box is axis-aligned.	c	e	f	g	a	i	j	k	l	m	n	b

ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_VISIBLE +	Retrieves whether the box occurrence's specified face was visible in the target. A face is only considered a visible face of the box occurrence if its score is greater than or equal to M_PLANE_ACCEPTANCE. You must specify a combination value from the following table: • To specify for which face of the box to retrieve results	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the face was not visible.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the face was visible.	c		e	f	g	a		i		j	k	l	m	n	b
M_VOLUME +	Retrieves the volume of the model occurrence. The volume is calculated using the following formula: $V = l * w * h$.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
Cylinder 3D model finder result buffer ID	Specifies a cylinder 3D model finder result buffer, allocated using M3dmodAllocResult() with M_FIND_CYLINDER_RESULT, and used to store M3dmodFind() results.	c		e	f	g	a		i		j	k	l	m	n	b
M_AREA +	Retrieves the surface area of the model occurrence. The surface area of a cylinder occurrence with no bases is calculated using the following formula: $SA = 2\pi r\ h$. The surface area of a cylinder occurrence with bases is calculated using the following formula: $SA = 2\pi r\ h + 2\pi r^2$.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_AXIS_X +	Retrieves the X-component of the cylinder occurrence's central axis unit vector. This vector does not reflect the cylinder's length.	c		e	f	g	a		i		j	k	l	m	n	b

	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	-1.0 >= *Value* >= 1.0	Specifies the X-component of the unit vector, expressed in the working coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_AXIS_Y +		Retrieves the Y-component of the cylinder occurrence's central axis unit vector. This vector does not reflect the cylinder's length.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	-1.0 >= *Value* >= 1.0	Specifies the Y-component of the unit vector, expressed in the working coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_AXIS_Z +		Retrieves the Z-component of the cylinder occurrence's central axis unit vector. This vector does not reflect the cylinder's length.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	-1.0 >= *Value* >= 1.0	Specifies the Z-component of the unit vector, expressed in the working coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_X +		Retrieves the X-coordinate of the center point of the model occurrence, expressed in the working coordinate system. For a cylinder model occurrence, the X-coordinate of the center point on the cylinder's central axis is retrieved.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Y +		Retrieves the Y-coordinate of the center point of the model occurrence, expressed in the working coordinate system. For a cylinder model occurrence, the Y-coordinate of the center point on the cylinder's central axis is retrieved.	c		e	f	g	a		i		j	k	l	m	n	b

	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_CENTER_Z +		Retrieves the Z-coordinate of the center point of the model occurrence, expressed in the working coordinate system. For a cylinder model occurrence, the Z-coordinate of the center point on the cylinder's central axis is retrieved.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_END_POINT_X +		Retrieves the X-coordinate of the cylinder occurrence's end point, expressed in the working coordinate system. The end position is located at the center of the cylinder's second circular base.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_END_POINT_Y +		Retrieves the Y-coordinate of the cylinder occurrence's end point, expressed in the working coordinate system. The end position is located at the center of the cylinder's second circular base.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_END_POINT_Z +		Retrieves the Z-coordinate of the cylinder occurrence's end point, expressed in the working coordinate system. The end position is located at the center of the cylinder's second circular base.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_LENGTH +		Retrieves the length of the cylinder occurrence in world units.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_NUMBER_OF_POINTS +		Retrieves the number of points in the model occurrence.	c	e	f	g	a	i	j	k	l	m	n	b

	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c	e	f	g	a	i	j	k	l	m	n	b
	Value > 0	Specifies the number of points.	c	e	f	g	a	i	j	k	l	m	n	b
M_NUMBER_OF_POINTS_RESERVED +		Retrieves the number of reserved points in the model occurrence. These are the points in the reserved area around the occurrence (established using M3dmodControl() with M_RESERVED_POINTS_DISTANCE). They are not considered in the fit, are not included in the calculation of the score, and are not considered for other occurrences.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c	e	f	g	a	i	j	k	l	m	n	b
	Value > 0	Specifies the number of points.	c	e	f	g	a	i	j	k	l	m	n	b
M_RADIUS +		Retrieves the radius of the cylinder occurrence in world units.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_RMS_ERROR +		Retrieves the root-mean-square (RMS) error between the points found in the occurrence and the source model.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_SCORE +		Retrieves the score of the occurrence, where the score is based on the occurrence's model coverage. The model coverage is the percentage of the model's surface found in the occurrence and covered by inlier points. The score is defined relative to the maximum expected model coverage, such that an occurrence with a score of 100% has a model coverage equal to the maximum expected model coverage. You can set the maximum expected model coverage, using M3dmodControl() with M_COVERAGE_MAX.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b

0.0 <= *Value* < 100.0	Specifies the score of the occurrence, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_START_POINT_X +	Retrieves the X-coordinate of the cylinder occurrence's start point, expressed in the working coordinate system. The start position is located at the center of the cylinder's first circular base.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_START_POINT_Y +	Retrieves the Y-coordinate of the cylinder occurrence's start point, expressed in the working coordinate system. The start position is located at the center of the cylinder's first circular base.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_START_POINT_Z +	Retrieves the Z-coordinate of the cylinder's occurrence's start point, expressed in the working coordinate system. The start position is located at the center of the cylinder's first circular base.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_VOLUME +	Retrieves the volume of the model occurrence. The volume is calculated using the following formula: $V = \pi r^2 h$.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
Rectangular plane 3D model finder result buffer ID	Specifies a rectangular plane 3D model finder result buffer, allocated using M3dmodAllocResult() with M_FIND_RECTANGULAR_PLANE_RESULT, and used to store M3dmodFind() results.	c		e	f	g	a		i		j	k	l	m	n	b
M_AREA +	Retrieves the surface area of the model occurrence. The surface area is calculated using the following formula: $SA = l * w$.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

	M_CENTER_X +	Retrieves the X-coordinate of the center point of the model occurrence, expressed in the working coordinate system. For a rectangular plane model occurrence, the X-coordinate of the center point of the rectangular plane's bounding box is retrieved.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_Y +	Retrieves the Y-coordinate of the center point of the model occurrence, expressed in the working coordinate system. For a rectangular plane model occurrence, the Y-coordinate of the center point of the rectangular plane's bounding box is retrieved.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_Z +	Retrieves the Z-coordinate of the center point of the model occurrence, expressed in the working coordinate system. For a rectangular plane model occurrence, the Z-coordinate of the center point of the rectangular plane's bounding box is retrieved.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_CLOSEST_TO_ORIGIN_X +	Retrieves the X-coordinate of the point on the rectangular plane occurrence closest to the origin of the working coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_CLOSEST_TO_ORIGIN_Y +	Retrieves the Y-coordinate of the point on the rectangular plane occurrence closest to the origin of the working coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_CLOSEST_TO_ORIGIN_Z +	Retrieves the Z-coordinate of the point on the rectangular plane occurrence closest to the origin of the working coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_COEFFICIENT_A +	Retrieves the coefficient A of the plane equation, Ax + By + Cz + D = 0.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
-1.0 >= *Value* >= 1.0	Specifies the coefficient A of the plane equation.	c		e	f	g	a		i		j	k	l	m	n	b
M_COEFFICIENT_B +	Retrieves the coefficient B of the plane equation, Ax + By + Cz + D = 0.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
-1.0 >= *Value* >= 1.0	Specifies the coefficient B of the plane equation.	c		e	f	g	a		i		j	k	l	m	n	b
M_COEFFICIENT_C +	Retrieves the coefficient C of the plane equation, Ax + By + Cz + D = 0.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
-1.0 >= *Value* >= 1.0	Specifies the coefficient C of the plane equation.	c		e	f	g	a		i		j	k	l	m	n	b
M_COEFFICIENT_D +	Retrieves the coefficient D of the plane equation, Ax + By + Cz + D = 0.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the coefficient D of the plane equation.	c		e	f	g	a		i		j	k	l	m	n	b

M_NORMAL_X +	Retrieves the X-component of the rectangular plane occurrence's normal unit vector.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
-1.0 >= *Value* >= 1.0	Specifies the X-component of the rectangular plane occurrence's normal unit vector, expressed in the working coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_Y +	Retrieves the Y-component of the rectangular plane occurrence's normal unit vector.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
-1.0 >= *Value* >= 1.0	Specifies the Y-component of the rectangular plane occurrence's normal unit vector, expressed in the working coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_Z +	Retrieves the Z-component of the rectangular plane occurrence's normal unit vector.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
-1.0 >= *Value* >= 1.0	Specifies the Z-component of the rectangular plane occurrence's normal unit vector, expressed in the working coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS +	Retrieves the number of points in the model occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the number of points.	c		e	f	g	a		i		j	k	l	m	n	b

M_NUMBER_OF_POINTS_RESERVED +	Retrieves the number of reserved points in the model occurrence. These are the points in the reserved area around the occurrence. They are not considered in the fit, are not included in the calculation of the score, and are not considered for other occurrences.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the number of points.	c		e	f	g	a		i		j	k	l	m	n	b
M_RMS_ERROR +	Retrieves the root-mean-square (RMS) error between the points found in the occurrence and the source model.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE +	Retrieves the score of the occurrence, where the score is based on the occurrence's model coverage. The model coverage is the percentage of the model's surface found in the occurrence and covered by inlier points. The score is defined relative to the maximum expected model coverage, such that an occurrence with a score of 100% has a model coverage equal to the maximum expected model coverage. You can set the maximum expected model coverage, using M3dmodControl() with M_COVERAGE_MAX.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies the score of the occurrence, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X +	Retrieves the length of the rectangular plane occurrence along its X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y +	Retrieves the length of the rectangular plane occurrence along its Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b

ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
Sphere 3D model finder result buffer ID	Specifies a sphere 3D model finder result buffer, allocated using M3dmodAllocResult() with M_FIND_SPHERE_RESULT, and used to store M3dmodFind() results.	c		e	f	g	a		i		j	k	l	m	n	b
M_AREA +	Retrieves the surface area of the model occurrence. The surface area is calculated using the following formula: $SA = 4\pi r^2$.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_X +	Retrieves the X-coordinate of the center point of the model occurrence, expressed in the working coordinate system. For a sphere model occurrence, the X-coordinate of the sphere's center point is retrieved.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Y +	Retrieves the Y-coordinate of the center point of the model occurrence, expressed in the working coordinate system. For a sphere model occurrence, the Y-coordinate of the sphere's center point is retrieved.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Z +	Retrieves the Z-coordinate of the center point of the model occurrence, expressed in the working coordinate system. For a sphere model occurrence, the Z-coordinate of the sphere's center point is retrieved.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS +	Retrieves the number of points in the model occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c	e	f	g	a	i	j	k	l	m	n	b
	Value > 0	Specifies the number of points.	c	e	f	g	a	i	j	k	l	m	n	b
M_NUMBER_OF_POINTS_RESERVED +		Retrieves the number of reserved points in the model occurrence. These are the points in the reserved area around the occurrence (established using M3dmodControl() with M_RESERVED_POINTS_DISTANCE). They are not considered in the fit, are not included in the calculation of the score, and are not considered for other occurrences.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c	e	f	g	a	i	j	k	l	m	n	b
	Value > 0	Specifies the number of points.	c	e	f	g	a	i	j	k	l	m	n	b
M_RADIUS +		Retrieves the radius of the sphere occurrence in world units.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_RMS_ERROR +		Retrieves the root-mean-square (RMS) error between the points found in the occurrence and the source model.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
M_SCORE +		Retrieves the score of the occurrence, where the score is based on the occurrence's model coverage. The model coverage is the percentage of the model's surface found in the occurrence and covered by inlier points. The score is defined relative to the maximum expected model coverage, such that an occurrence with a score of 100% has a model coverage equal to the maximum expected model coverage. You can set the maximum expected model coverage, using M3dmodControl() with M_COVERAGE_MAX.	c	e	f	g	a	i	j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b

0.0 <= *Value* < 100.0	Specifies the score of the occurrence, as a percentage.	c	e	f	g	a	i	j	k	l	m	n	b
M_VOLUME +	Retrieves the volume of the model occurrence. The volume is calculated using the following formula: $V = (4/3)\pi r^3$.	c	e	f	g	a	i	j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c	e	f	g	a	i	j	k	l	m	n	b
Surface or planar surface 3D model finder result buffer ID	Specifies a surface or planar surface 3D model finder result buffer, allocated using M3dmodAllocResult() with M_FIND_SURFACE_RESULT or M_FIND_PLANAR_SURFACE_RESULT, and used to store M3dmodFind() results. Note that M_SCORE_GRIP(), M_SCORE_GRIP_BEST, and M_SCORE_GRIP_BEST_LABEL are not available for planar surface 3D model finder result buffers.	c	e	f	g	a	i	j	k	l	m	n	b
M_SCORE_GRIP(AIL_INT Value) +	Retrieves the grip score for the specified label. The grip score is the percentage of the model's labeled surface found in the occurrence and covered by inlier points. To retrieve the highest grip score among all labels, use M_SCORE_GRIP_BEST. This result is only available for surface 3D model finder result buffers and only if a grip label image that contains non-zero values was copied to the find surface 3D model finder context, using M3dmodCopy() with M_GRIP_LABEL_IMAGE.	c	e	f	g	a	i	j	k	l	m	n	b
	Parameters												
	Value Specifies the label.												
M_ALL	Specifies all labels.	c	e	f	g	a	i	j	k	l	m	n	b

		0 < *Value* <= 255 Specifies the valid label. The label must exist in the grip label image.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* < 100.0	Specifies the grip score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_X +		Retrieves the X-coordinate of the center point of the model occurrence, expressed in the working coordinate system. For a surface model occurrence, the X-coordinate of the center of the surface's bounding box is retrieved.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Y +		Retrieves the Y-coordinate of the center point of the model occurrence, expressed in the working coordinate system. For a surface model occurrence, the Y-coordinate of the center of the surface's bounding box is retrieved.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Z +		Retrieves the Z-coordinate of the center point of the model occurrence, expressed in the working coordinate system. For a surface model occurrence, the Z-coordinate of the center of the surface's bounding box is retrieved.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS +		Retrieves the number of points in the model occurrence. This result is only available if M_SAVE_FIT_INFO was enabled.	c		e	f	g	a		i		j	k	l	m	n	b

	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the number of points.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS_RESERVED +		Retrieves the number of reserved points in the model occurrence. These are the points in the reserved area around the occurrence. They are not considered in the fit, are not included in the calculation of the score, and are not considered for other occurrences.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the number of points.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_X +		Retrieves the X-coordinate of the surface occurrence. This is the X-position of the model's reference axis transformed at the occurrence. Note, the origin of the model's reference axis is at the origin of the working coordinate system of the original point cloud from which the surface model was defined.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_Y +		Retrieves the Y-coordinate of the surface occurrence. This is the Y-position of the model's reference axis transformed at the occurrence. Note, the origin of the model's reference axis is at the origin of the working coordinate system of the original point cloud from which the surface model was defined.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_POSITION_Z +	Retrieves the Z-coordinate of the surface occurrence. This is the Z-position of the model's reference axis transformed at the occurrence. Note, the origin of the model's reference axis is at the origin of the working coordinate system of the original point cloud from which the surface model was defined.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_RMS_ERROR +	Retrieves the root-mean-square (RMS) error between the points found in the occurrence and the source model. For a surface model, this result is only available if M_SAVE_FIT_INFO was enabled. To improve fit results, you should also enable refine registration (M3dmodControl() with M_REFINE_REGISTRATION) prior to performing the search. By default, Aurora Imaging Library uses the features of the model and target to find occurrences, but it doesn't do any fitting. To get more accurate score and pose information, you should enable refine registration.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE +	Retrieves the score of the occurrence, where the score is based on the occurrence's model coverage. The model coverage is the percentage of the model's surface found in the occurrence and covered by inlier points. The score is defined relative to the maximum expected model coverage, such that an occurrence with a score of 100% has a model coverage equal to the maximum expected model coverage. You can set the maximum expected model coverage, using M3dmodControl() with M_COVERAGE_MAX.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies the score of the occurrence, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b

M_SCORE_COLOR +	Retrieves the color score of the occurrence, where the color score indicates the similarity between the color of the model and the color of the occurrence. This result is only available if M_USE_COLOR was enabled.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies the color score of the occurrence, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE_FIT +	Retrieves the fit score of the occurrence, where the fit score is based on the Euclidean distance between the occurrence's points and the model. The score is calculated as follows: Fit Score = 1 - Normalized RMS error This result is only available if M_SAVE_FIT_INFO was enabled. To improve fit results, you should also enable refine registration (M3dmodControl() with M_REFINE_REGISTRATION) prior to performing the search. By default, Aurora Imaging Library uses the features of the model and target to find occurrences, but it doesn't do any fitting. To get more accurate score and pose information, you should enable refine registration.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies the fit score of the occurrence, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE_GRIP_BEST +	Retrieves the highest grip score among all labels. The grip score is the percentage of the model's labeled surface found in the occurrence and covered by inlier points. This result is only available for surface 3D model finder result buffers and only if a grip label image that contains non-zero values was copied to the find surface 3D model finder context, using M3dmodCopy() with M_GRIP_LABEL_IMAGE.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

	0.0 <= *Value* < 100.0	Specifies the best grip score.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE_GRIP_BEST_LABEL +		Retrieves the label with the highest grip score. The grip score is the percentage of the model's labeled surface found in the occurrence and covered by inlier points. This result is only available for surface 3D model finder result buffers and only if a grip label image that contains non-zero values was copied to the find surface 3D model finder context, using M3dmodCopy() with M_GRIP_LABEL_IMAGE.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
	0 < *Value* <= 255	Specifies the label.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE_TARGET +		Retrieves the target score of the occurrence, where the target score is based on the target coverage. Target coverage is a measure of the points found in the occurrence that don't have matching points in the original model (that is, extra points), weighted by the deviation in position of the common points. Points found in the occurrence that are not present in the model will reduce the target score. For example, a target score of 100% means that there are no points in the occurrence that do not fit the model.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* < 100.0	Specifies the target score of the occurrence, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_STATUS_REFINE_REGISTRATION +		Retrieves the status of the refine registration applied to the occurrence. This result is only available for surface model occurrences and only if M_REFINE_REGISTRATION was enabled.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_APPLIED	Specifies that refine registration was applied successfully to the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

		M_APPLIED_AND_REJECTED	Specifies that even though M_REFINE_REGISTRATION was enabled, at least one registration was not applied successfully to the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NOT_APPLIED	Specifies that refine registration was not enabled, so it was not applied to the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for [when Result3dmodId=Box 3D model finder result buffer ID: M_CENTER_X; M_CENTER_Y; M_CENTER_Z; M_COVERAGE; M_NORMAL_X; M_NORMAL_Y; M_NORMAL_Z; M_NUMBER_OF_POINTS; M_NUMBER_OF_POINTS_RESERVED; M_RMS_ERROR; M_SCORE; M_VISIBLE].

You must add the following value to the above-mentioned values to specify for which face of the box to retrieve results. Note that this combination is optional for some values. Refer to the individual values to determine if this is the case.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the box face's index																		
Combination value	Description					(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FACE_INDEX(AIL_INT Index)	Specifies to retrieve results for a specific face of a box occurrence.					c		e	f	g	a		i		j	k	l	m	n	b
		Parameters																		
		Index Set this parameter to one of the following:																		
			M_ALL	Specifies to get the result for all 6 faces.		c		e	f	g	a		i		j	k	l	m	n	b

			0 <= *Value* < 6 Specifies the index of the face for which to get the result. Indices 0 and 5 correspond to the faces with the minimum XYZ and maximum XYZ coordinate values, respectively.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for the values of the ResultType parameter listed in For retrieving a result of a single occurrence or all occurrences from the 3D model finder result buffer.

You can add the following value to the above-mentioned values to determine the required array size (number of elements) to store the returned values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining the required array size (number of elements) to store the returned values																	
ResultType combination value		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ResultArrayPtr - Possible values returned																
M_NB_ELEMENTS		Retrieves the required array size (number of elements) to store the returned values.															
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values of the ResultType parameter listed in For retrieving general results from the 3D model finder result buffer, For retrieving a result of a single occurrence or all occurrences from the 3D model finder result buffer.

You can add one of the following values to the above-mentioned values to cast the requested information to a required data type.

> Note: In C++, when using a standard vector (std::vector) instead of an array, the values listed in this combination table must not be used. If you need to use a specific data type, instantiate the vector with the necessary data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested information to an *AIL_DOUBLE*.															
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* **|** *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]* **|** *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]* **|** *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]* **|** *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_DOUBLE*. If the requested information does not fit into an *AIL_DOUBLE*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodInquire

Synopsis

Inquire about a setting of a find 3D model finder context.

Syntax

```
AIL_INT64 M3dmodInquire(
    AIL_ID Context3dmodId, //in
    AIL_INT64 Index,       //in
    AIL_INT64 InquireType, //in
    void *UserVarPtr       //out
)
```

Description

This function inquires about a specified setting of a find 3D model finder context.

To inquire about draw 3D model finder context settings, use M3dmodInquireDraw() instead.

Parameters

Context3dmodId

Specifies the identifier of the find 3D model finder context about which to inquire information. The find 3D model finder context must have been previously allocated on the required system using M3dmodAlloc() with M_FIND_..._CONTEXT.

See the Parameter associations section for possible values that can be specified.

Index

Specifies what to inquire. Set this parameter to one of the following values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a general context or individual model																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_CONTEXT.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTEXT	Specifies to inquire information about the find 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b
0	Specifies to inquire the 3D model defined in the context. Note that the model must have been added to the context, using M3dmodDefine(), prior to calling M3dmodControl() with this value.	c		e	f	g	a		i		j	k	l	m	n	b

InquireType

Specifies the type of setting about which to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_DOUBLE

- AIL_FLOAT
- AIL_ID
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write the requested information. Since the **M3dmodInquire**() function also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **Context3dmodId** and **InquireType** parameters and possible values returned to the **UserVarPtr** parameter.

- For inquiring a find box, cylinder, rectangular plane, sphere, surface, or planar surface 3D model finder context
- For inquiring the type of model in a find 3D model finder context
- For inquiring a box, cylinder, rectangular plane, sphere, or surface 3D model

The following Context3dmodId and InquireType parameter settings can be specified for a 3D model finder context when Index is set to M_CONTEXT or M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring a find box, cylinder, rectangular plane, sphere, surface, or planar surface 3D model finder context																	
Context3dmodId InquireType UserVarPtr - Possible values returned	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Find box 3D model finder context ID	Specifies a find box 3D model finder context, allocated using M3dmodAlloc() with M_FIND_BOX_CONTEXT, and used in M3dmodFind() operations.															
M_DIRECTION_MODE +	Inquires how to interpret the reference direction (M_DIRECTION_REFERENCE_...). For a find box 3D model finder context, this is only used when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_AWAY_FROM_POSITION	Specifies to interpret the reference direction as away from the position specified using M_DIRECTION_REFERENCE_....	c		e	f	g	a		i		j	k	l	m	n	b
M_TOWARDS_DIRECTION	Specifies to interpret the reference direction as towards the direction specified using M_DIRECTION_REFERENCE_....	c		e	f	g	a		i		j	k	l	m	n	b

	M_TOWARDS_POSITION	Specifies to interpret the reference direction as towards the position specified using M_DIRECTION_REFERENCE_....	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION_REFERENCE_X +		Inquires the X-coordinate of the position, or the X-component of the vector, that determines the reference direction, depending on M_DIRECTION_MODE. For a find box 3D model finder context, this is only used when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the X-coordinate of the position or the X-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION_REFERENCE_Y +		Inquires the Y-coordinate of the position, or the Y-component of the vector, that determines the reference direction, depending on M_DIRECTION_MODE. For a find box 3D model finder context, this is only used when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-coordinate of the position or the Y-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION_REFERENCE_Z +		Inquires the Z-coordinate of the position, or the Z-component of the vector, that determines the reference direction, depending on M_DIRECTION_MODE. For a find box 3D model finder context, this is only used when M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Z-coordinate of the position or the Z-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE +		Inquires the fit distance when M_FIT_DISTANCE_MODE is set to M_USER_DEFINED. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0.0	Specifies the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE_MODE +		Inquires how Aurora Imaging Library establishes the fit distance, which is used when determining whether points belong to a certain occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO	Specifies to automatically calculate the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_USER_DEFINED	Specifies to use the value set with M_FIT_DISTANCE as the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FLOOR_DEFINED +		Inquires whether a floor plane has been defined using M3dmodCopy() with M_FLOOR.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that a floor plane has not been defined.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that a floor plane has been defined.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_MODELS +		Inquires the number of models in the context.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the number of models.	c		e	f	g	a		i		j	k	l	m	n	b
M_PREPROCESSED +		Inquires whether the find 3D model finder context is preprocessed. The context must be preprocessed (using M3dmodPreprocess()) before calling M3dmodFind(). After certain settings of the context are changed using M3dmodControl(), a model is added or removed using M3dmodDefine(), or a floor plane is copied to the context using M3dmodCopy(), this inquire type will indicate that the context is no longer in its preprocessed state.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the context is not preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b

		M_TRUE	Specifies that the context is preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SORT +		Inquires the sorting key for result retrieval. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_AREA	Specifies to sort results by the area of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER_X	Specifies to sort the results by the X-coordinate of the occurrence's center point.	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER_Y	Specifies to sort the results by the Y-coordinate of the occurrence's center point.	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER_Z	Specifies to sort the results by the Z-coordinate of the occurrence's center point.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX_X	Specifies to sort the results by the maximum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX_Y	Specifies to sort the results by the maximum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX_Z	Specifies to sort the results by the maximum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN_X	Specifies to sort the results by the minimum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN_Y	Specifies to sort the results by the minimum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN_Z	Specifies to sort the results by the minimum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NUMBER_OF_POINTS	Specifies to sort the results by the number of points in the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE	Specifies to sort the results by the score.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_TARGET	Specifies to sort results by the target score.	c		e	f	g	a		i		j	k	l	m	n	b

		M_VOLUME	Specifies to sort results by the volume of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SORT_DIRECTION +		Inquires whether results are sorted in ascending or descending order. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_SORT_DOWN	Specifies to sort the results in descending order.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SORT_UP	Specifies to sort the results in ascending order.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TIMEOUT +		Inquires the maximum amount of time for M3dmodFind() to complete the 3D model finder operation before generating a time-out error. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b
Find cylinder 3D model finder context ID			Specifies a find cylinder 3D model finder context, allocated using M3dmodAlloc() with M_FIND_CYLINDER_CONTEXT, and used in M3dmodFind() operations.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIT_DISTANCE +		Inquires the fit distance when M_FIT_DISTANCE_MODE is set to M_USER_DEFINED. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIT_DISTANCE_MODE +		Inquires how Aurora Imaging Library establishes the fit distance, which is used when determining whether points belong to a certain occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_AUTO	Specifies to automatically calculate the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b

	M_USER_DEFINED	Specifies to use the value set with M_FIT_DISTANCE as the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_NORMALS_DISTANCE +		Inquires the acceptable deviation between a given point's normal vector and the normal vector of the model at the same point, to decide which points are used when fitting. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 90.0	Specifies the acceptable deviation from the normal vector of the model at the same point, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_MODELS +		Inquires the number of models in the context.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the number of models.	c		e	f	g	a		i		j	k	l	m	n	b
M_PERSEVERANCE +		Inquires the algorithm's search perseverance when searching for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the algorithm's search perseverance, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_PREPROCESSED +		Inquires whether the find 3D model finder context is preprocessed. The context must be preprocessed (using M3dmodPreprocess()) before calling M3dmodFind(). After certain settings of the context are changed using M3dmodControl(), or a model is added or removed using M3dmodDefine(), this inquire type will indicate that the context is no longer in its preprocessed state.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the context is not preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the context is preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT +		Inquires the sorting key for result retrieval. SET	c		e	f	g	a		i		j	k	l	m	n	b

		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_AREA	Specifies to sort results by the area of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER_X	Specifies to sort the results by the X-coordinate of the occurrence's center point. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER_Y	Specifies to sort the results by the Y-coordinate of the occurrence's center point. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER_Z	Specifies to sort the results by the Z-coordinate of the occurrence's center point. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX_X	Specifies to sort the results by the maximum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX_Y	Specifies to sort the results by the maximum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX_Z	Specifies to sort the results by the maximum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN_X	Specifies to sort the results by the minimum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN_Y	Specifies to sort the results by the minimum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN_Z	Specifies to sort the results by the minimum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NUMBER_OF_POINTS	Specifies to sort the results by the number of points in the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE	Specifies to sort the results by the score.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_TARGET	Specifies to sort results by the target score.	c		e	f	g	a		i		j	k	l	m	n	b
		M_VOLUME	Specifies to sort results by the volume of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SORT_DIRECTION +		Inquires whether results are sorted in ascending or descending order. SET	c		e	f	g	a		i		j	k	l	m	n	b

		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_SORT_DOWN	Specifies to sort the results in descending order.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SORT_UP	Specifies to sort the results in ascending order.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TIMEOUT +		Inquires the maximum amount of time for M3dmodFind() to complete the 3D model finder operation before generating a time-out error. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b
Find rectangular plane 3D model finder context ID			Specifies a find rectangular plane 3D model finder context, allocated using M3dmodAlloc() with M_FIND_RECTANGULAR_PLANE_CONTEXT, and used in M3dmodFind() operations.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIT_DISTANCE +		Inquires the fit distance when M_FIT_DISTANCE_MODE is set to M_USER_DEFINED. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIT_DISTANCE_MODE +		Inquires how Aurora Imaging Library establishes the fit distance, which is used when determining whether points belong to a certain occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_AUTO	Specifies to automatically calculate the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
		M_USER_DEFINED	Specifies to use the value set with M_FIT_DISTANCE as the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NUMBER_MODELS +		Inquires the number of models in the context.	c		e	f	g	a		i		j	k	l	m	n	b

		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the number of models.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PREPROCESSED +		Inquires whether the find 3D model finder context is preprocessed. The context must be preprocessed (using M3dmodPreprocess()) before calling M3dmodFind(). After certain settings of the context are changed using M3dmodControl(), or a model is added or removed using M3dmodDefine(), this inquire type will indicate that the context is no longer in its preprocessed state.	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_FALSE	Specifies that the context is not preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b
		M_TRUE	Specifies that the context is preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SORT +		Inquires the sorting key for result retrieval. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_AREA	Specifies to sort results by the area of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER_X	Specifies to sort the results by the X-coordinate of the occurrence's center point. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER_Y	Specifies to sort the results by the Y-coordinate of the occurrence's center point. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER_Z	Specifies to sort the results by the Z-coordinate of the occurrence's center point. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX_X	Specifies to sort the results by the maximum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX_Y	Specifies to sort the results by the maximum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX_Z	Specifies to sort the results by the maximum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

M_MIN_X	Specifies to sort the results by the minimum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Y	Specifies to sort the results by the minimum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Z	Specifies to sort the results by the minimum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS	Specifies to sort the results by the number of points in the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE	Specifies to sort the results by the score.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE_TARGET	Specifies to sort results by the target score.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_DIRECTION +	Inquires whether results are sorted in ascending or descending order. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_DOWN	Specifies to sort the results in descending order.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT_UP	Specifies to sort the results in ascending order.	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT +	Inquires the maximum amount of time for M3dmodFind() to complete the 3D model finder operation before generating a time-out error. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b
Find sphere 3D model finder context ID	Specifies a find sphere 3D model finder context, allocated using M3dmodAlloc() with M_FIND_SPHERE_CONTEXT, and used in M3dmodFind() operations.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE +	Inquires the fit distance when M_FIT_DISTANCE_MODE is set to M_USER_DEFINED. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0.0	Specifies the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE_MODE +		Inquires how Aurora Imaging Library establishes the fit distance, which is used when determining whether points belong to a certain occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO	Specifies to automatically calculate the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_USER_DEFINED	Specifies to use the value set with M_FIT_DISTANCE as the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_ITERATIONS_MAX +		Inquires the maximum number of fit iterations to perform, when finding a sphere occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1	Specifies the maximum number of sphere fit iterations.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_NORMALS_DISTANCE +		Inquires the acceptable deviation between a given point's normal vector and the normal vector of the model at the same point, to decide which points are used when fitting. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 90.0	Specifies the acceptable deviation from the normal vector of the model at the same point, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_MODELS +		Inquires the number of models in the context.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the number of models.	c		e	f	g	a		i		j	k	l	m	n	b
M_PERSEVERANCE +		Inquires the algorithm's search perseverance when searching for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	0.0 <= *Value* <= 100.0	Specifies the algorithm's search perseverance, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_PREPROCESSED +		Inquires whether the find 3D model finder context is preprocessed. The context must be preprocessed (using M3dmodPreprocess()) before calling M3dmodFind(). After certain settings of the context are changed using M3dmodControl(), or a model is added or removed using M3dmodDefine(), this inquire type will indicate that the context is no longer in its preprocessed state.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the context is not preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the context is preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT +		Inquires the sorting key for result retrieval. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_AREA	Specifies to sort results by the area of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_X	Specifies to sort the results by the X-coordinate of the occurrence's center point.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_Y	Specifies to sort the results by the Y-coordinate of the occurrence's center point.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_Z	Specifies to sort the results by the Z-coordinate of the occurrence's center point.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_X	Specifies to sort the results by the maximum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_Y	Specifies to sort the results by the maximum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_Z	Specifies to sort the results by the maximum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

		M_MIN_X	Specifies to sort the results by the minimum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN_Y	Specifies to sort the results by the minimum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN_Z	Specifies to sort the results by the minimum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NUMBER_OF_POINTS	Specifies to sort the results by the number of points in the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE	Specifies to sort the results by the score.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_TARGET	Specifies to sort results by the target score.	c		e	f	g	a		i		j	k	l	m	n	b
		M_VOLUME	Specifies to sort results by the volume of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SORT_DIRECTION +		Inquires whether results are sorted in ascending or descending order. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_SORT_DOWN	Specifies to sort the results in descending order.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SORT_UP	Specifies to sort the results in ascending order.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TIMEOUT +		Inquires the maximum amount of time for M3dmodFind() to complete the 3D model finder operation before generating a time-out error. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b
Find surface or planar surface 3D model finder context ID			Specifies a find surface or planar surface 3D model finder context, allocated using M3dmodAlloc() with M_FIND_SURFACE_CONTEXT or M_FIND_PLANAR_SURFACE_CONTEXT, and used in M3dmodFind() operations.	c		e	f	g	a		i		j	k	l	m	n	b

M_CONVERSION_GAMMA +	Inquires whether to remove gamma correction before the match. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to remove gamma correction. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to remove gamma correction. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION_MODE +	Inquires how to interpret the reference direction (M_DIRECTION_REFERENCE_...). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TOWARDS_DIRECTION	Specifies to interpret the reference direction as towards the direction specified using M_DIRECTION_REFERENCE_....	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION_REFERENCE_X +	Inquires the X-component of the vector that determines the reference direction, depending on M_DIRECTION_MODE. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION_REFERENCE_Y +	Inquires the Y-component of the vector that determines the reference direction, depending on M_DIRECTION_MODE. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_DIRECTION_REFERENCE_Z +	Inquires the Z-component of the vector that determines the reference direction, depending on M_DIRECTION_MODE. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Z-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b

M_EXHAUSTIVE_SEARCH +	Inquires whether to perform very robust but slow exhaustive searching. Note that this inquire type is not available for find planar surface 3D model finder contexts. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies to disable exhaustive searching.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to enable exhaustive searching.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE +	Inquires the fit distance when M_FIT_DISTANCE_MODE is set to M_USER_DEFINED. Note that for a find surface or planar surface 3D model finder context, this is only used when M_FIT_DISTANCE_MODE is set to M_USER_DEFINED and M_SAVE_FIT_INFO is enabled. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE_MODE +	Inquires how Aurora Imaging Library establishes the fit distance, which is used when determining whether points belong to a certain occurrence. Note that for a find surface or planar surface 3D model finder context, this is only used when M_SAVE_FIT_INFO is enabled. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO	Specifies to automatically calculate the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_DEFINED	Specifies to use the value set with M_FIT_DISTANCE as the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FLOOR_DEFINED +	Inquires whether a floor plane has been defined using M3dmodCopy() with M_FLOOR.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that a floor plane has not been defined.	c		e	f	g	a		i		j	k	l	m	n	b

	M_TRUE	Specifies that a floor plane has been defined.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_NORMAL_SEARCH_MODE +		Inquires the search mode for calculating model normals when the defined model has no normals component (M_COMPONENT_NORMALS_AIL). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO	Specifies to automatically set the search mode based on the point cloud's organization; either M_ORGANIZED if organized or M_TREE if unorganized.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ORGANIZED	Specifies to use the point cloud's organizational structure to determine the model normals. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_TREE	Specifies to use a KD tree search mode to determine the model normals.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_MODELS +		Inquires the number of models in the context.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the number of models.	c		e	f	g	a		i		j	k	l	m	n	b
M_PERSEVERANCE +		Inquires the algorithm's search perseverance when searching for occurrences. Note that this inquire type is not available for find planar surface 3D model finder contexts. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the algorithm's search perseverance, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_PREPROCESSED +		Inquires whether the find 3D model finder context is preprocessed. The context must be preprocessed (using M3dmodPreprocess()) before calling M3dmodFind(). After certain settings of the context are changed using M3dmodControl(), a model is added or removed using M3dmodDefine(), or a floor plane or resting plane is copied to the context using M3dmodCopy(), this inquire type will indicate that the context is no longer in its preprocessed state.	c		e	f	g	a		i		j	k	l	m	n	b

		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_FALSE	Specifies that the context is not preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b
		M_TRUE	Specifies that the context is preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REFINE_REGISTRATION +		Inquires whether and how to perform refine registration of each occurrence found. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies to disable refine registration.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FIND_SURFACE_REFINEMENT_FAST	Specifies to use a predefined registration context that prioritizes performance to fine-tune results.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FIND_SURFACE_REFINEMENT_PRECISE	Specifies to use a predefined registration context that prioritizes accuracy to fine-tune results.	c		e	f	g	a		i		j	k	l	m	n	b
		M_USER_DEFINED_REGISTRATION	Specifies to use a custom registration context. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_REMOVE_BACKGROUND +		Inquires whether to remove background points from the target point cloud before performing the match. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to remove background points.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to remove background points.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REMOVE_BACKGROUND_SEARCH_MODE +		Inquires the search mode for finding the background points when M_REMOVE_BACKGROUND is enabled. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_AUTO	Specifies to automatically set the search mode based on the organization of the point clouds; either M_ORGANIZED if organized or M_TREE if unorganized.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ORGANIZED	Specifies to use the organizational structure of the point clouds to determine the background points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_TREE	Specifies to use a KD tree search mode to determine the background points.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_FLOOR +		Inquires whether to remove floor points from the target point cloud before performing the match. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to remove floor points.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to remove floor points.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_FLOOR_DIRECTION +		Inquires the direction, relative to the defined floor plane, in which to remove points when M_REMOVE_FLOOR is enabled. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ABOVE	Specifies to remove points above the floor plane.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO	Specifies to remove points in the direction with the least number of valid points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_BELOW	Specifies to remove points below the floor plane.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_FLOOR_EXPECTED_PERCENTAGE +		Inquires the percentage of total points in the target point cloud that are expected to be floor points when M_REMOVE_FLOOR is enabled and M_REMOVE_FLOOR_OFFSET is set to M_AUTO_VALUE. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	5.0 <= *Value* < 100.0	Specifies the expected floor points, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_FLOOR_OFFSET +		Inquires the offset, relative to the defined floor plane, within which to remove points when M_REMOVE_FLOOR is enabled. SET	c		e	f	g	a		i		j	k	l	m	n	b

		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_AUTO_VALUE	Specifies to automatically determine the offset within which to remove points.	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 0.0	Specifies the offset within which to remove points.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REUSE_RESULT +		Inquires whether to reuse the result of the previous search at the beginning of the current search. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to reuse the previous result.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to reuse the previous result.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAVE_FIT_INFO +		Inquires whether to enable the calculation of the root-mean square (RMS) error and the fit score. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies to disable the calculation of the root-mean square (RMS) error and the fit score.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to enable the calculation of the root-mean square (RMS) error and the fit score.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCENE_COMPLEXITY +		Inquires the complexity of the scene. Note that this inquire type is not available for find planar surface 3D model finder contexts. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_HIGH	Specifies that the scene has a high complexity.	c		e	f	g	a		i		j	k	l	m	n	b
		M_LOW	Specifies that the scene has a low complexity.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MEDIUM	Specifies that the scene has a medium complexity.	c		e	f	g	a		i		j	k	l	m	n	b

M_SCENE_PROJECTION +	Inquires whether to apply occlusion handling to the model, which discards points of the model that are not visible to the 3D sensor at the position of the occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies to disable occlusion handling.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to enable occlusion handling.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE_MODE +	Inquires the mode with which to associate model and target points for score calculations. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FLEXIBLE	Specifies to use the flexible mode when calculating the scores.	c		e	f	g	a		i		j	k	l	m	n	b
M_STRICT	Specifies to use the strict mode when calculating the scores.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT +	Inquires the sorting key for result retrieval. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_X	Specifies to sort the results by the X-coordinate of the occurrence's center point. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Y	Specifies to sort the results by the Y-coordinate of the occurrence's center point. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Z	Specifies to sort the results by the Z-coordinate of the occurrence's center point. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_X	Specifies to sort the results by the maximum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_Y	Specifies to sort the results by the maximum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_Z	Specifies to sort the results by the maximum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

		M_MIN_X	Specifies to sort the results by the minimum X-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN_Y	Specifies to sort the results by the minimum Y-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN_Z	Specifies to sort the results by the minimum Z-coordinate of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NO_SORT	Specifies not to sort the results; found occurrences are returned in an arbitrary order.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NUMBER_OF_POINTS	Specifies to sort the results by the number of points in the occurrence. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE	Specifies to sort the results by the score.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_COLOR	Specifies to sort results by the color score. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_FIT	Specifies to sort results by the fit score. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_GRIP_BEST	Specifies to sort results by the best grip score. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_TARGET	Specifies to sort results by the target score.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SORT_DIRECTION +		Inquires whether results are sorted in ascending or descending order. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_SORT_DOWN	Specifies to sort the results in descending order.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SORT_UP	Specifies to sort the results in ascending order.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TIMEOUT +		Inquires the maximum amount of time for M3dmodFind() to complete the 3D model finder operation before generating a time-out error. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b
M_USE_COLOR +	Inquires whether to use the color information of the model and scene when searching for surface occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to use the color information.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to use the color information.	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_DEFINED_REGISTRATION_CONTEXT_ID +	Inquires the identifier of the internal registration context to use for fine tuning when M_REFINE_REGISTRATION is set to M_USER_DEFINED_REGISTRATION.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b

The following Context3dmodId and InquireType parameter settings can be specified for a model in a 3D model finder context when Index is set to 0.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring the type of model in a find 3D model finder context																	
Context3dmodId	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
InquireType																	
UserVarPtr - Possible values returned																	
Find 3D model finder context ID with a model	Specifies a find box, cylinder, rectangular plane, sphere, surface, or planar surface 3D model finder context, allocated using M3dmodAlloc() with M_FIND_..._CONTEXT, and used in M3dmodFind() operations. It must contain a model.																
M_MODEL_TYPE +	Inquires the type of model in the find 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX	Specifies that the model is a nominal box model.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_RANGE	Specifies that the model is a range-type box model.	c		e	f	g	a		i		j	k	l	m	n	b

		M_CYLINDER	Specifies that the model is a nominal cylinder model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_CYLINDER_RANGE	Specifies that the model is a range-type cylinder model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_RECTANGLE	Specifies that the model is a nominal rectangular plane model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_RECTANGLE_RANGE	Specifies that the model is a range-type rectangular-plane model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SPHERE	Specifies that the model is a nominal sphere model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SPHERE_RANGE	Specifies that the model is a range-type sphere model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SURFACE	Specifies that the model is a surface model, defined by a point cloud.	c		e	f	g	a		i		j	k	l	m	n	b

The following Context3dmodId and InquireType parameter settings can be specified for a box, cylinder, rectangular plane, sphere, or surface 3D model when Index is set to 0.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring a box, cylinder, rectangular plane, sphere, or surface 3D model																	
Context3dmodId InquireType UserVarPtr - Possible values returned	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Find box 3D model finder context ID with a box model	Specifies a find box 3D model finder context, allocated using M3dmodAlloc() with M_FIND_BOX_CONTEXT, and used in M3dmodFind() operations. It must contain a box model.															
M_ACCEPTANCE +	Inquires the acceptance level for the score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies an acceptable score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b

M_BOX_FACE_PARALLELISM_THRESHOLD +	Inquires the maximum shear angle allowed between adjacent rectangles for them to count as the same box. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 45.0	Specifies the maximum angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_FACE_PERPENDICULARITY_THRESHOLD +	Inquires the maximum deviation from 90 degrees allowed between adjacent rectangles for them to count as the same box. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 90.0	Specifies the maximum angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_CERTAINTY +	Inquires the certainty level for the score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the certainty level for the score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COMPLETION_ANGLE_TOLERANCE +	Inquires the completion angular tolerance, when only one box face (plane) is found. Note that the corresponding control type has no effect unless M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 90.0	Specifies the completion angular tolerance, when only one box face (plane) is found.	c		e	f	g	a		i		j	k	l	m	n	b
M_COMPLETION_SIZE_X +	Inquires the length along X to use to establish the missing dimension, when only one box face (plane) is found. Note that the corresponding control type has no effect unless M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

		M_SAME_AS_NOMINAL	Specifies that the length along X, to use to establish the missing dimension, is the same as the nominal length in X of a nominal model. When searching for a range-type model, this corresponds to the middle of the specified size range (M_SIZE_X_MAX - M_SIZE_X_MIN).	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the length along X to use to establish the missing dimension.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COMPLETION_SIZE_Y +		Inquires the length along Y to use to establish the missing dimension, when only one box face (plane) is found. Note that the corresponding control type has no effect unless M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_SAME_AS_NOMINAL	Specifies that the length along Y, to use to establish the missing dimension, is the same as the nominal length in Y of a nominal model. When searching for a range-type model, this corresponds to the middle of the specified size range (M_SIZE_Y_MAX - M_SIZE_Y_MIN).	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the length along Y to use to establish the missing dimension.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COMPLETION_SIZE_Z +		Inquires the length along Z to use to establish the missing dimension, when only one box face (plane) is found. Note that the corresponding control type has no effect unless M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_SAME_AS_NOMINAL	Specifies that the length along Z, to use to establish the missing dimension, is the same as the nominal length in Z of a nominal model. When searching for a range-type model, this corresponds to the middle of the specified size range (M_SIZE_Z_MAX - M_SIZE_Z_MIN).	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the length along Z to use to establish the missing dimension.	c		e	f	g	a		i		j	k	l	m	n	b

M_COMPLETION_TO_BACKGROUND +	Inquires whether the visible face could be extruded to a background plane to complete the box, when only one box face (plane) is found. Note that the corresponding control type has no effect unless M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to complete the box using a background plane.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to attempt to complete the box using a background plane.	c		e	f	g	a		i		j	k	l	m	n	b
M_COMPLETION_TO_STAIRCASE +	Inquires whether the visible face could be extruded to an edge of a staircase plane to complete the box, when only one box face (plane) is found. Note that the corresponding control type has no effect unless M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to complete the box using a staircase plane.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to attempt to complete the box using a staircase plane.	c		e	f	g	a		i		j	k	l	m	n	b
M_COMPLETION_TO_USER_SIZE +	Inquires whether the visible face could be extruded to the specified completion size to complete the box, when only one box face (plane) is found. Note that the corresponding control type has no effect unless M_NUMBER_OF_VISIBLE_FACES_MIN is set to 1. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to complete the box using the completion size.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to complete the box using the completion size.	c		e	f	g	a		i		j	k	l	m	n	b

M_COVERAGE_MAX +	Inquires the maximum expected model coverage. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 100.0	Specifies the maximum expected model coverage, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELONGATION_MAX +	Inquires the maximum elongation of the box occurrence. The elongation is defined as the maximum side/minimum side. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies that the box occurrence has no maximum elongation. INFO	c		e	f	g	a		i		j	k	l	m	n	b
Value > 1.0	Specifies the box occurrence's maximum elongation, defined as maximum side / minimum side.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELONGATION_MIN +	Inquires the minimum elongation of the box occurrence. The elongation is defined as the maximum side/minimum side. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1.0	Specifies the box occurrence's minimum elongation, defined as maximum side / minimum side.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_ANGLE_TOLERANCE +	Inquires the angular tolerance to use for M_NORMAL_CONDITION, when searching for box occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 90.0	Specifies the normal angular tolerance, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_CONDITION +	Inquires whether to find only box occurrences that satisfy the specified condition, relative to the vector M_NORMAL_..., or to find any box occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_PARALLEL	Specifies that the normal of one of the faces must be parallel to M_NORMAL_..., within the specified angular tolerance.	c		e	f	g	a		i		j	k	l	m	n	b

		M_UNCONDITIONAL	Specifies that there is no constraint on the normal.	c		e	f	g	a		i		j	k	l	m	n	b
		M_VISIBLE_FACE_PARALLEL	Specifies that the normal of one of the visible faces must be parallel to M_NORMAL_..., within the specified angular tolerance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NORMAL_X +		Inquires the X-component of the vector against which to compare box occurrences, as specified using M_NORMAL_CONDITION. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the X-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NORMAL_Y +		Inquires the Y-component of the vector against which to compare box occurrences, as specified using M_NORMAL_CONDITION. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the Y-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NORMAL_Z +		Inquires the Z-component of the vector against which to compare box occurrences, as specified using M_NORMAL_CONDITION. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the Z-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NUMBER +		Inquires the maximum number of occurrences for which to search. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_ALL	Specifies to find all occurrences. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0	Specifies the number of occurrences for which to search.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NUMBER_OF_POINTS_MIN +		Inquires the minimum number of points per occurrence found. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0	Specifies the minimum number of points per occurrence found.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_VISIBLE_FACES_MAX +		Inquires the maximum number of visible faces (planes) required for a box occurrence to be accepted. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	1 <= *Value* <= 6	Specifies the maximum number of visible faces (planes) required for a box occurrence to be accepted.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_VISIBLE_FACES_MIN +		Inquires the minimum number of visible faces (planes) required for a box occurrence to be accepted. If set to 1, a box can be extruded from a single plane according to the completion constraints (M_COMPLETION_...) and the reference direction (M_DIRECTION_REFERENCE_...). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	1 <= *Value* <= 6	Specifies the maximum number of visible faces (planes) required for a box occurrence to be accepted.	c		e	f	g	a		i		j	k	l	m	n	b
M_PLANE_ACCEPTANCE +		Inquires the acceptance level used for individual faces of the box occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the acceptance level used for individual faces of the box occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_PLANE_CERTAINTY +		Inquires the certainty level for individual faces of the box occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the certainty value for individual faces of the box occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_PLANE_MAX_COVERAGE +		Inquires the maximum expected coverage for individual faces of the box occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	0.0 < *Value* <= 100.0	Specifies the maximum coverage for individual faces of the box occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_POLARITY +		Inquires whether the normals should point inside or outside the box occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies that the normals can point inside or outside.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INSIDE	Specifies that the normals must point inside.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OUTSIDE	Specifies that the normals must point outside.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies that the normals can point either inside or outside, but the direction must be consistent for all faces.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X +		Inquires the size of a nominal box model (M_BOX) along its X-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the expected size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X_MAX +		Inquires the maximum size of a range-type box model (M_BOX_RANGE) along its X-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no constraint on the maximum size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the maximum size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X_MIN +		Inquires the minimum size of a range-type box model (M_BOX_RANGE) along its X-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the minimum size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b

M_SIZE_Y +		Inquires the size of a nominal box model (M_BOX) along its Y-axis, changing its originally defined size. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the expected size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y_MAX +		Inquires the maximum size of a range-type box model (M_BOX_RANGE) along its Y-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no constraint on the maximum size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the maximum size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y_MIN +		Inquires the minimum size of a range-type box model (M_BOX_RANGE) along its Y-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the minimum size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Z +		Inquires the size of a nominal box model (M_BOX) along its Z-axis, changing its originally defined size. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the expected size along the Z-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Z_MAX +		Inquires the maximum size of a range-type box model (M_BOX_RANGE) along its Z-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no constraint on the maximum size along the Z-axis.	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0.0	Specifies the maximum size along the Z-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Z_MIN +		Inquires the minimum size of a range-type box model (M_BOX_RANGE) along its Z-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the minimum size along the Z-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOLERANCE_X +		Inquires the tolerance for the size of a nominal box model (M_BOX) along its X-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no constraint on the size along the X-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the tolerance for the size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOLERANCE_Y +		Inquires the tolerance for the size of a nominal box model (M_BOX) along its Y-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no constraint on the size along the Y-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the tolerance for the size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOLERANCE_Z +		Inquires the tolerance for the size of a nominal box model (M_BOX) along its Z-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no constraint on the size along the Z-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the tolerance for the size along the Z-axis.	c		e	f	g	a		i		j	k	l	m	n	b

Find cylinder 3D model finder context ID with a cylinder model	Specifies a find cylinder 3D model finder context, allocated using M3dmodAlloc() with M_FIND_CYLINDER_CONTEXT, and used in M3dmodFind() operations. It must contain a cylinder model.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE +	Inquires the acceptance level for the score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies an acceptable score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_BASES +	Inquires whether the cylinder model includes bases. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_WITH_BASES	Specifies that the cylinder model includes bases.	c		e	f	g	a		i		j	k	l	m	n	b
M_WITHOUT_BASES	Specifies that the cylinder model does not include bases.	c		e	f	g	a		i		j	k	l	m	n	b
M_CERTAINTY +	Inquires the certainty level for the score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the certainty level for the score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COVERAGE_MAX +	Inquires the maximum expected model coverage. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 100.0	Specifies the maximum expected model coverage, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_LENGTH +	Inquires the length of a nominal cylinder model (M_CYLINDER). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the length.	c		e	f	g	a		i		j	k	l	m	n	b

M_LENGTH_MAX +	Inquires the maximum length of a range-type cylinder model (M_CYLINDER_RANGE). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the maximum length.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the maximum length.	c		e	f	g	a		i		j	k	l	m	n	b
M_LENGTH_MIN +	Inquires the minimum length of a range-type cylinder model (M_CYLINDER_RANGE). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the minimum length.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_SEPARATION_DISTANCE +	Inquires the minimum gap distance along the length of a cylinder before considering the cylinder as two separate occurrences of the cylinder model. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to split cylinders.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the minimum gap distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER +	Inquires the maximum number of occurrences for which to search. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Specifies to find all occurrences. INFO	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the number of occurrences for which to search.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS_MIN +	Inquires the minimum number of points per occurrence found. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the minimum number of points per occurrence found.	c		e	f	g	a		i		j	k	l	m	n	b

M_RADIUS +	Inquires the radius of a nominal cylinder model (M_CYLINDER). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the radius of the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_RADIUS_MAX +	Inquires the maximum radius of a range-type cylinder model (M_CYLINDER_RANGE). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the maximum radius.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the maximum radius.	c		e	f	g	a		i		j	k	l	m	n	b
M_RADIUS_MIN +	Inquires the minimum radius of a range-type cylinder model (M_CYLINDER_RANGE). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the minimum radius.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESERVED_POINTS_DISTANCE +	Inquires the reserved distance around the occurrence, defined as a percentage of the radius. Points found in this area are not considered in the fit, and cannot be considered for other occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies the reserved points distance for the model, as a percentage of its radius.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOLERANCE_LENGTH +	Inquires the tolerance for the length of a nominal cylinder model (M_CYLINDER). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the length. INFO	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the tolerance for the length.	c		e	f	g	a		i		j	k	l	m	n	b

M_TOLERANCE_RADIUS +	Inquires the tolerance for the radius of a nominal cylinder model (M_CYLINDER). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the radius. INFO	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the tolerance for the radius.	c		e	f	g	a		i		j	k	l	m	n	b
Find rectangular plane 3D model finder context ID with a rectangular plane model	Specifies a find rectangular plane 3D model finder context, allocated using M3dmodAlloc() with M_FIND_RECTANGULAR_PLANE_CONTEXT, and used in M3dmodFind() operations. It must contain a rectangular plane model.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE +	Inquires the acceptance level for the score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies an acceptable score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_CERTAINTY +	Inquires the certainty level for the score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the certainty level for the score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_COVERAGE_MAX +	Inquires the maximum expected model coverage. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 100.0	Specifies the maximum expected model coverage, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELONGATION_MAX +	Inquires the maximum elongation of the rectangular plane occurrence. The elongation is defined as the maximum side/minimum side. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

		M_INFINITE	Specifies that the rectangular plane occurrence has no maximum elongation. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 1.0	Specifies the rectangular plane occurrence's maximum elongation, defined as maximum side / minimum side.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELONGATION_MIN +		Inquires the minimum elongation of the rectangular plane occurrence. The elongation is defined as the maximum side/minimum side. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 1.0	Specifies the rectangular plane occurrence's minimum elongation, defined as maximum side / minimum side.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NORMAL_ANGLE_TOLERANCE +		Inquires the angular tolerance to use for M_NORMAL_CONDITION, when searching for rectangular plane occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 < *Value* <= 90.0	Specifies the normal angular tolerance, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NORMAL_CONDITION +		Inquires whether to find only rectangular plane occurrences that satisfy the specified condition, relative to the vector M_NORMAL_..., or to find any rectangular plane occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_PARALLEL	Specifies that the normal of the rectangular plane must be parallel to the vector specified using M_NORMAL_..., within the specified angular tolerance.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PARALLEL_OR_PERPENDICULAR	Specifies that the normal of the rectangular plane must be either parallel or perpendicular to the vector specified using M_NORMAL_..., within the specified angular tolerance.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PERPENDICULAR	Specifies that the normal of the rectangular plane must be perpendicular to the vector specified using M_NORMAL_..., within the specified angular tolerance.	c		e	f	g	a		i		j	k	l	m	n	b

	M_UNCONDITIONAL	Specifies that there is no constraint on the normal of the rectangular plane.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_X +		Inquires the X-component of the vector against which to compare rectangular plane occurrences, as specified using M_NORMAL_CONDITION. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the X-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_Y +		Inquires the Y-component of the vector against which to compare rectangular plane occurrences, as specified using M_NORMAL_CONDITION. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_Z +		Inquires the Z-component of the vector against which to compare rectangular plane occurrences, as specified using M_NORMAL_CONDITION. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Z-component of the vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER +		Inquires the maximum number of occurrences for which to search. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ALL	Specifies to find all occurrences. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the number of occurrences for which to search.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS_MIN +		Inquires the minimum number of points per occurrence found. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the minimum number of points per occurrence found.	c		e	f	g	a		i		j	k	l	m	n	b

M_SIZE_X +		Inquires the size of a nominal rectangular plane model (M_RECTANGLE) along its X-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the expected size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X_MAX +		Inquires the maximum size of a range-type rectangular plane model (M_RECTANGLE_RANGE) along its X-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no constraint on the maximum size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the maximum size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X_MIN +		Inquires the minimum size of a range-type rectangular plane model (M_RECTANGLE_RANGE) along its X-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the minimum size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y +		Inquires the size of a nominal rectangular plane model (M_RECTANGLE) along its Y-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the expected size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y_MAX +		Inquires the maximum size of a range-type rectangular plane model (M_RECTANGLE_RANGE) along its Y-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no constraint on the maximum size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b

Value > 0.0	Specifies the maximum size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y_MIN +	Inquires the minimum size of a range-type rectangular plane model (M_RECTANGLE_RANGE) along its Y-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the minimum size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOLERANCE_X +	Inquires the tolerance for the size of a nominal rectangular plane model (M_RECTANGLE) along its X-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the size along the X-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the tolerance for the size along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOLERANCE_Y +	Inquires the tolerance for the size of a nominal rectangular plane model (M_RECTANGLE) along its Y-axis. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies no constraint on the size along the Y-axis. INFO	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the tolerance for the size along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b
Find sphere 3D model finder context ID with a sphere model	Specifies a find sphere 3D model finder context, allocated using M3dmodAlloc() with M_FIND_SPHERE_CONTEXT, and used in M3dmodFind() operations. It must contain a sphere model.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE +	Inquires the acceptance level for the score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies an acceptable score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_CERTAINTY +	Inquires the certainty level for the score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies the certainty level for the score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COVERAGE_MAX +	Inquires the maximum expected model coverage. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 100.0	Specifies the maximum expected model coverage, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER +	Inquires the maximum number of occurrences for which to search. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Specifies to find all occurrences. INFO	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the number of occurrences for which to search.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS_MIN +	Inquires the minimum number of points per occurrence found. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the minimum number of points per occurrence found.	c		e	f	g	a		i		j	k	l	m	n	b
M_RADIUS +	Inquires the radius of a nominal sphere model (M_SPHERE). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the expected radius.	c		e	f	g	a		i		j	k	l	m	n	b
M_RADIUS_MAX +	Inquires the maximum radius of a range-type sphere model (M_SPHERE_RANGE). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

		M_INFINITE	Specifies no constraint on the maximum radius.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the maximum radius.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RADIUS_MIN +		Inquires the minimum radius of a range-type sphere model (M_SPHERE_RANGE). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 0.0	Specifies the minimum radius.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RESERVED_POINTS_DISTANCE +		Inquires the reserved distance around the occurrence, defined as a percentage of the radius. Points found in this area are not considered in the fit, and cannot be considered for other occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* < 100.0	Specifies the reserved points distance for the model, as a percentage of its radius.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TOLERANCE_RADIUS +		Inquires the tolerance for the radius of a nominal sphere model (M_SPHERE). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_INFINITE	Specifies no constraint on the radius. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 0.0	Specifies the tolerance for the radius.	c		e	f	g	a		i		j	k	l	m	n	b
Find surface or planar surface 3D model finder context ID with a surface model			Specifies a find surface or planar surface 3D model finder context, allocated using M3dmodAlloc() with M_FIND_SURFACE_CONTEXT or M_FIND_PLANAR_SURFACE_CONTEXT, and used in M3dmodFind() operations. It must contain a surface model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCEPTANCE +		Inquires the acceptance level for the score. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* < 100.0	Specifies an acceptable score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_ACCEPTANCE_COLOR +	Inquires the acceptance level for the color score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the acceptance level for the color score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE_TARGET +	Inquires the acceptance level for the target score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies the acceptance level for the target score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_CERTAINTY +	Inquires the certainty level for the score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 100.0	Specifies the certainty level for the score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COVERAGE_MAX +	Inquires the maximum expected model coverage. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 100.0	Specifies the maximum expected model coverage, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_SCORE_MIN +	Inquires the minimum expected occurrence fit score. Note that the corresponding control type has no effect unless M_SAVE_FIT_INFO is set to M_ENABLE. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the minimum expected fit score.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_PLANES_OCCLUSION_MODE +	Inquires the occlusion level of the planar regions of the model in the scene. Note that this inquire type is only available for find planar surface 3D model finder contexts. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

		M_HIGH	Specifies a high model plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b
		M_LOW	Specifies a low model plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MEDIUM	Specifies a medium model plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b
		M_VERY_HIGH	Specifies a very high model plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MODEL_RANGE_SIZE_X +		Inquires the size in X of the range component of the model point cloud. Note that you can use this value to allocate an image buffer of an appropriate size to pass to M3dmodCopy() with M_GRIP_LABEL_IMAGE or M_WEIGHT_IMAGE. Note that this inquire type is not available for find planar surface 3D model finder contexts.	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0	Specifies the X-size, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MODEL_RANGE_SIZE_Y +		Inquires the size in Y of the range component of the model point cloud. Note that you can use this value to allocate an image buffer of an appropriate size to pass to M3dmodCopy() with M_GRIP_LABEL_IMAGE or M_WEIGHT_IMAGE. Note that this inquire type is not available for find planar surface 3D model finder contexts.	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0	Specifies the Y-size, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MODEL_RESOLUTION +		Inquires the resolution of the model when it was originally defined (before preprocessing); the resolution is expressed as the average distance between points.	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0.0	Specifies the resolution of the defined model, where the resolution is expressed as the average distance between points.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER +		Inquires the maximum number of occurrences for which to search. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ALL	Specifies to find all occurrences. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the number of occurrences for which to search.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_POINTS_MIN +		Inquires the minimum number of points per occurrence found. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the minimum number of points per occurrence found.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_RESTING_PLANE +		Inquires the number of resting planes defined using M3dmodCopy() with M_RESTING_PLANE.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	0 <= *Value* <= 1	Specifies the number of resting planes.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_OUTLIERS +		Inquires whether to perform outlier removal on the model. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to perform outlier removal on the model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to perform outlier removal on the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_OUTLIERS_SEARCH_MODE +		Inquires the search mode for finding outliers when M_REMOVE_OUTLIERS is enabled. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

		M_AUTO	Specifies to automatically set the search mode based on the point cloud's organization; either M_ORGANIZED if organized or M_TREE if unorganized.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ORGANIZED	Specifies to use the point cloud's organizational structure to determine the outliers. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_TREE	Specifies to use a KD tree search mode to determine the outliers.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RESTING_PLANE_ANGLE_TOLERANCE +		Inquires the angular tolerance for the resting plane constraint. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 < *Value* <= 180.0	Specifies the angular tolerance, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SEARCH_POINT_RESOLUTION +		Inquires the resolution of the target point cloud (scene), where the resolution is expressed as the average distance between points. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_SAME_AS_MODEL	Specifies that the target point cloud (scene) has the same resolution as the model.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0	Specifies the expected resolution of the target point cloud (scene), where the resolution is expressed as the average distance between points.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TARGET_PLANES_OCCLUSION_MODE +		Inquires the target occlusion level when a planar region of the model is a portion of a planar region of an object in the scene. Note that this inquire type is only available for find planar surface 3D model finder contexts. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_HIGH	Specifies a high target plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b
		M_LOW	Specifies a low target plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b

		M_MEDIUM	Specifies a medium target plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b
		M_VERY_HIGH	Specifies a very high target plane occlusion.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For inquiring a find box, cylinder, rectangular plane, sphere, surface, or planar surface 3D model finder context
- For inquiring the type of model in a find 3D model finder context
- For inquiring a box, cylinder, rectangular plane, sphere, or surface 3D model

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to get information about the default value of an inquire type, regardless of the current value of the inquire type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the default value																		
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																	
M_DEFAULT		Inquires the default value of the specified inquire type.																
	UserVarPtr extra info	Data type info: Data type: *defined elsewhere*	c		e	f	g	a		i		j	k	l	m	n	b
M_IS_SET_TO_DEFAULT		Inquires whether the specified inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type is not set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For inquiring a find box, cylinder, rectangular plane, sphere, surface, or planar surface 3D model finder context
- For inquiring the type of model in a find 3D model finder context
- For inquiring a box, cylinder, rectangular plane, sphere, or surface 3D model

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to determine whether an inquire type has a default value or whether it is supported.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring whether an inquire type has a default value or whether it is supported																		
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

UserVarPtr - Possible values returned																
M_HAS_DEFAULT	Inquires whether the specified inquire type has a default value.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the inquire type does not have a default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the inquire type has a default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_SUPPORTED	Inquires whether the specified inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the inquire type is not supported.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for

Tables:

- For inquiring a find box, cylinder, rectangular plane, sphere, surface, or planar surface 3D model finder context
- For inquiring the type of model in a find 3D model finder context
- For inquiring a box, cylinder, rectangular plane, sphere, or surface 3D model

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to cast the requested information to the required data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested information to an *AIL_DOUBLE*.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_FLOAT	Casts the requested information to an *AIL_FLOAT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_FLOAT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_INT64*. If the requested information does not fit into an *AIL_INT64*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodInquireDraw

Synopsis

Inquire about a setting of a draw 3D model finder context.

Syntax

```
AIL_INT64 M3dmodInquireDraw(
    AIL_ID DrawContext3dmodId, //in
    AIL_INT64 Operation,       //in
    AIL_INT64 InquireType,     //in
    void *UserVarPtr           //out
)
```

Description

This function inquires about a specified setting of a draw 3D model finder context. These settings establish which features of a model or which results of found model occurrences to draw into the 3D graphics list when calling M3dmodDraw3d() and how to draw them.

You can control the draw 3D model finder context settings using M3dmodControlDraw().

Parameters

DrawContext3dmodId

Specifies the identifier of the draw 3D model finder context to inquire. The draw 3D model finder context must have been previously allocated on the required system using M3dmodAlloc() with either M_DRAW_3D_GEOMETRIC_CONTEXT (to draw occurrences of geometric models) or M_DRAW_3D_SURFACE_CONTEXT (to draw surface models or occurrences of surface models).

Operation

Specifies the draw operation about which to inquire.

See the Parameter associations section for possible values that can be specified.

InquireType

Specifies the type of setting about which to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_DOUBLE
- AIL_FLOAT
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write the requested information. Since the **M3dmodInquireDraw**() function also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **Operation** and **InquireType** parameters and possible values returned to the **UserVarPtr** parameter.

- For inquiring a draw geometric or surface 3D model finder context
- For inquiring a draw surface 3D model finder context

The following Operation and InquireType parameter settings can be specified for a draw 3D model finder context.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring a draw geometric or surface 3D model finder context																
Operation InquireType UserVarPtr - Possible values returned	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_AXES	Inquires whether and how to draw the axes of either the model or each specified occurrence when calling M3dmodDraw3d(), depending on whether a find 3D model finder context or result buffer is specified.															
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_LENGTH +	Inquires the length of the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_LENGTH	Specifies that the length is the default size of the 3D model.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the length, in world units.	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_X +	Inquires the X-position of the graphic(s) origin, relative to the model or occurrence. Note that M_REFERENCE_X is only available for a draw surface 3D model finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER	Specifies to draw the graphic(s) at the center of the model or occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX	Specifies to draw the graphic(s) at the model's or occurrence's maximum X-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b

		M_MIN	Specifies to draw the graphic(s) at the model's or occurrence's minimum X-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REFERENCE_Y +		Inquires the Y-position of the graphic(s) origin, relative to the model or occurrence. Note that M_REFERENCE_Y is only available for a draw surface 3D model finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER	Specifies to draw the graphic(s) at the center of the model or occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX	Specifies to draw the graphic(s) at the model's or occurrence's maximum Y-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN	Specifies to draw the graphic(s) at the model's or occurrence's minimum Y-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REFERENCE_Z +		Inquires the Z-position of the graphic(s) origin, relative to the model or occurrence. Note that M_REFERENCE_Z is only available for a draw surface 3D model finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER	Specifies to draw the graphic(s) at the center of the model or occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MAX	Specifies to draw the graphic(s) at the model's or occurrence's maximum Z-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MIN	Specifies to draw the graphic(s) at the model's or occurrence's minimum Z-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_BOX	Inquires whether and how to draw the bounding box of each specified occurrence when calling M3dmodDraw3d(). The bounding box is the smallest-axis aligned box that contains the ideal geometric shape or surface of the model at the location of the occurrence, depending on the model type. If you enable M_DRAW_MODEL, the bounding box is the smallest axis-aligned box that contains that graphic. Note that it does not necessarily contain all inlier points.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +	Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b

Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_INDEX	Inquires whether and how to draw the index of each specified occurrence when calling M3dmodDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FONT_SIZE +		Inquires the size of the font with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the font size. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the graphic(s) . INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_X +		Inquires the X-position of the graphic(s) origin, relative to the occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER	Specifies to draw the graphic(s) at the center of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX	Specifies to draw the graphic(s) at the occurrence's maximum X-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIN	Specifies to draw the graphic(s) at the occurrence's minimum X-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_Y +		Inquires the Y-position of the graphic(s) origin, relative to the occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER	Specifies to draw the graphic(s) at the center of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX	Specifies to draw the graphic(s) at the occurrence's maximum Y-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIN	Specifies to draw the graphic(s) at the occurrence's minimum Y-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_Z +		Inquires the Z-position of the graphic(s) origin, relative to the occurrence. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER	Specifies to draw the graphic(s) at the center of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

M_MAX	Specifies to draw the graphic(s) at the occurrence's maximum Z-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN	Specifies to draw the graphic(s) at the occurrence's minimum Z-coordinate.	c		e	f	g	a		i		j	k	l	m	n	b
M_RENDER_LAYER +	Inquires the layer on which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <= 9	Specifies the layer on which the graphic will be rendered.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_DIRECTION +	Inquires the direction to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FLIP	Specifies to draw the graphic(s) such that it is mirrored across the occurrence's X-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAME	Specifies to draw the graphic(s) in the direction of the occurrence's X-axis, and facing the direction of its Z-axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_INLIER_POINTS	Inquires whether and how to draw the inlier points, based on M_FIT_DISTANCE, of each specified occurrence when calling M3dmodDraw3d(). Note that for a surface model, M_SAVE_FIT_INFO must be enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b

		Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_MODEL			Inquires whether and how to draw either the defined model or each specified occurrence when calling M3dmodDraw3d(), depending on whether a find 3D model finder context or result buffer is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_APPEARANCE +		Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO_COLOR	Specifies to color the graphic according to specific components in the source container. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the graphic(s) . INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_RESERVED_POINTS			Inquires whether and how to draw the reserved points of each specified occurrence when calling M3dmodDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b

		M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_GLOBAL_DRAW_SETTINGS	Inquires a global draw 3D model finder context setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_COLOR_OFFSET +	Inquires the offset to apply to either the model's or the occurrence's index when drawing with M_PSEUDO_COLOR. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the offset to apply to the index.	c		e	f	g	a		i		j	k	l	m	n	b

The following Operation and InquireType parameter settings can be specified for a draw surface 3D model finder context.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring a draw surface 3D model finder context																
Operation	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
InquireType																
UserVarPtr - Possible values returned																

M_DRAW_AXES_POSITION	Inquires whether and how to draw the axes at either the model's or each specified occurrence's position when calling M3dmodDraw3d(), depending on whether a find 3D model finder context or result buffer is specified. The axes are drawn at the model's reference axis origin/the model's reference axis origin transformed at each occurrence.															
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_LENGTH +	Inquires the length of the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_LENGTH	Specifies that the length is the default size of the 3D model.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the length, in world units.	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_BACKGROUND_POINTS	Inquires whether and how to draw the points that were considered part of the background during the search. Background points include those of objects that are too big or too small to be part of an occurrence of the model. The corresponding drawing operation is only supported if M_REMOVE_BACKGROUND was enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b

		M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FLOOR_POINTS			Inquires whether and how to draw the points that were considered part of the floor during the search. Floor points are determined using the floor plane, defined using M3dmodCopy() with M_FLOOR, and the floor removal direction and offset (M3dmodControl() with M_REMOVE_FLOOR_DIRECTION and M_REMOVE_FLOOR_OFFSET). The corresponding drawing operation is only supported if M_REMOVE_FLOOR was enabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b

		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_MODEL_PREPROCESSED			Inquires whether and how to draw the preprocessed model that is in the specified find 3D model finder context. M3dmodPreprocess() must be called before calling M3dmodDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each occurrence with a different color, according to the mapping between each occurrence's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b

	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the graphic(s) . INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values of the InquireType parameter listed in For inquiring a draw geometric or surface 3D model finder context, For inquiring a draw surface 3D model finder context.

You can add one of the following values to the above-mentioned values to get information about the default value of an inquire type, regardless of the current value of the inquire type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the default value																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_DEFAULT		Inquires the default value of the specified inquire type.															
	UserVarPtr extra info	Data type info: Data type: *defined elsewhere*	c		e	f	g	a		i		j	k	l	m	n	b
M_IS_SET_TO_DEFAULT		Inquires whether the specified inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type is not set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b

	M_TRUE	Specifies that the inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values of the InquireType parameter listed in For inquiring a draw geometric or surface 3D model finder context, For inquiring a draw surface 3D model finder context.

You can add one of the following values to the above-mentioned values to determine whether an inquire type has a default value or whether it is supported.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring whether an inquire type has a default value or whether it is supported																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_HAS_DEFAULT		Inquires whether the specified inquire type has a default value.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type does not have a default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type has a default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_SUPPORTED		Inquires whether the specified inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type is not supported.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values of the InquireType parameter listed in For inquiring a draw geometric or surface 3D model finder context, For inquiring a draw surface 3D model finder context.

You can add one of the following values to the above-mentioned values to cast the requested information to the required data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_TYPE_AIL_DOUBLE		Casts the requested information to an *AIL_DOUBLE*.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

M_TYPE_AIL_FLOAT	Casts the requested information to an *AIL_FLOAT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_FLOAT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_INT64*. If the requested information does not fit into an *AIL_INT64*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodModifyResult

Synopsis

Modify or delete surface or planar surface results from a surface or planar surface 3D model finder result buffer.

Syntax

```
void M3dmodModifyResult(
    AIL_ID SrcResult3dmodId, //in
    AIL_INT64 SrcIndex,      //in
    AIL_ID DstResult3dmodId, //out
    AIL_INT64 DstIndex,      //in
    AIL_INT64 ModifyType,    //in
    AIL_DOUBLE ModifyParam,  //in
    AIL_INT64 ControlFlag    //in
)
```

Description

This function modifies or deletes surface or planar surface results and stores them in the destination result buffer. You can only modify results after calling M3dmodFind().

M3dmodModifyResult() is useful when reusing surface or planar surface results to speed up the next search (M3dmodControl() with M_REUSE_RESULT set to M_ENABLE); in this case, M3dmodFind() will search for occurrences at the stored positions before searching the rest of the target. **M3dmodModifyResult**() allows you to transform results if there was a known movement between scans, or to delete results for occurrences you don't expect to find in the next scan. Note that if you want to reuse the same results for multiple searches and you want the modification to be present for each search, you should perform the modifications on the result buffer with the original results, in-place (the same result buffer for the source and the destination). Then, before each search, you should copy the results to a separate result buffer, using M3dmodCopyResult() with M_RESULT, and use the separate result buffer for the search. If you want to reuse the original (unmodified) results in a future search, you must specify different result buffers for the source and destination to avoid overwriting the original results. Note that if you only want to reuse the results once, you do not need to copy the results to a separate result buffer; you can call M3dmodFind() with the original result buffer.

This function supports in-place processing; the source and destination result buffer identifiers can refer to the same result buffer.

Parameters

SrcResult3dmodId

Specifies the identifier of the source surface or planar surface 3D model finder result buffer from which to modify results. The surface or planar surface 3D model finder result buffer must have been allocated using M3dmodAllocResult() with M_FIND_SURFACE_RESULT or M_FIND_PLANAR_SURFACE_RESULT and must contain the results of a call to M3dmodFind().

SrcIndex

Specifies the index of the surface occurrence in the source surface or planar surface 3D model finder result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the index of the surface occurrence to modify																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_ALL	Specifies to modify the result for all surface occurrences.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* < M_NUMBER	Specifies the index of the surface occurrence for which to modify results.	c		e	f	g	a		i		j	k	l	m	n	b

DstResult3dmodId

Specifies the identifier of the destination surface or planar surface 3D model finder result buffer in which to store the modified results. The surface or planar surface 3D model finder result buffer must have been allocated using M3dmodAllocResult() with M_FIND_SURFACE_RESULT or M_FIND_PLANAR_SURFACE_RESULT and must be the same type of result buffer as the source.

DstIndex

Reserved for future expansion and must be set to **M_DEFAULT**.

ModifyType

Specifies the type of modification to perform.

See the Parameter associations section for possible values that can be specified.

ModifyParam

Specifies an attribute of the modification.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Parameter associations

The table below lists possible values for the **ModifyType** and **ModifyParam** parameters.

- For specifying the type of modification

Note that any unused parameters should be set to **M_DEFAULT**.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of modification																	
ModifyType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ModifyParam																
M_DELETE	Specifies to delete results for the specified surface occurrence from the surface or planar surface 3D model finder result buffer. This is useful if you don't need to reuse the results of all of the found occurrences. You should delete results for an occurrence if you don't expect to find an occurrence at the same position in the next scene.															

M_TRANSFORM	Specifies to translate or rotate the specified surface occurrence in the surface or planar surface 3D model finder result buffer. This is useful if there was a known movement between the scans. You should transform results for an occurrence if you expect the occurrence to be displaced in the next scene. Note that this modification type will invalidate all results related to the target point cloud, such as reserved points.	c		e	f	g	a		i		j	k	l	m	n	b
Transformation matrix object ID	Specifies the identifier of the transformation matrix object used to transform the surface occurrence. The transformation matrix object must have been previously allocated using M3dgeoAlloc() with M_TRANSFORMATION_MATRIX. You can use M3dgeoMatrixSetTransform() to generate the required transformation matrix.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodPreprocess

Synopsis

Preprocess a 3D model finder context.

Syntax

```
void M3dmodPreprocess(
    AIL_ID Context3dmodId, //in
    AIL_INT64 ControlFlag  //in
)
```

Description

This function preprocesses the specified 3D model finder context. It sets internal search settings so that future searches will be optimized for speed and robustness.

This function must be called before performing the first M3dmodFind() search. Call this function after all search settings have been set. When you save the 3D model finder context, the model's preprocessing changes are not stored with the model. Upon restoration, the model must be preprocessed again.

Note that if some of the model's search settings are changed after a call to **M3dmodPreprocess**(), the model must be preprocessed again.

Parameters

Context3dmodId

Specifies the identifier of the 3D model finder context to preprocess. The context must have been previously allocated using M3dmodAlloc() with M_FIND_BOX_CONTEXT, M_FIND_CYLINDER_CONTEXT, M_FIND_RECTANGULAR_PLANE_CONTEXT, M_FIND_SPHERE_CONTEXT, M_FIND_PLANAR_SURFACE_CONTEXT, or M_FIND_SURFACE_CONTEXT.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodRestore

Synopsis

Restore a 3D model finder context from disk.

Syntax

```
AIL_ID M3dmodRestore(
    AIL_CONST_TEXT_PTR FileName, //in
    AIL_ID SysId,                //in
    AIL_INT64 ControlFlag,       //in
    AIL_ID *Context3dmodIdPtr    //out
)
```

Description

This function restores a 3D model finder context that was previously saved to a file, using M3dmodSave() or M3dmodStream(). This function restores all of the 3D model finder context's settings that were in effect when it was saved.

When the restored 3D model finder context is no longer required, release it using M3dmodFree() unless M_UNIQUE_ID was specified during restoration; if M_UNIQUE_ID was specified, the smart identifier manages the 3D model finder context's lifetime and you must not manually free it.

Parameters

FileName

Specifies the name and path of the file from which to restore the 3D model finder context. The function handles (internally) the opening and closing of the file.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the file name and path																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INTERACTIVE	[This is only applicable to Windows] Opens the **File Open** dialog box from which you can interactively specify the drive, directory, and name of the file.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). Typically, 3D model finder files have an M3DMOD extension. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

SysId

Specifies the system on which to restore the 3D model finder context.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Context3dmodIdPtr

Specifies the address of the variable in which to write the 3D model finder context identifier or specifies the data type that the function should use to return the 3D model finder context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D model finder context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D model finder context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_3DMOD_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the 3D model finder context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b

Address in which to write the draw 3D model finder context identifier	Specifies the address of a draw *AIL_ID* in which to write the identifier of the allocated draw 3D model finder context. If restoration fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the find 3D model finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated find 3D model finder context. If restoration fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the 3D model finder context identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_3DMOD_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodSave

Synopsis

Save a 3D model finder context to a file.

Syntax

```
void M3dmodSave(
    AIL_CONST_TEXT_PTR FileName, //in
    AIL_ID Context3dmodId,       //in
    AIL_INT64 ControlFlag        //in
)
```

Description

This function saves all the information about a previously allocated 3D model finder context to disk.

To load a saved context, use either M3dmodRestore() or M3dmodStream().

Parameters

FileName

Specifies the name and path of the file in which to save the 3D model finder context; it is recommended that you use the M3DMOD file extension for easier use with other Aurora Imaging software products. The function internally handles the opening and closing of this file. If this file already exists, it will be overwritten.

This parameter can be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the file name and path																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INTERACTIVE	[This is only applicable to Windows] Opens the **File Save As** dialog box from which you can interactively specify the drive, directory, and name of the file.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). The recommended extension is M3DMOD. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Context3dmodId

Specifies the identifier of the 3D model finder context to save. The context must have been successfully allocated (using M3dmodAlloc()) prior to calling this function.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.

M3dmodStream

Synopsis

Load, restore, or save a 3D model finder context from/to a file or memory stream.

Syntax

```
void M3dmodStream(
    AIL_TEXT_PTR MemPtrOrFileName, //in-out
    AIL_ID SysId,                  //in
    AIL_INT64 Operation,           //in
    AIL_INT64 StreamType,          //in
    AIL_DOUBLE Version,            //in
    AIL_INT64 ControlFlag,         //in
    AIL_ID *Context3dmodIdPtr,     //in-out
    AIL_INT *SizeByteVarPtr        //out
)
```

Description

This function can load, restore, or save a 3D model finder context from/to a file or memory stream.

To inquire the number of bytes necessary to save a 3D model finder context to a memory stream, you should first call **M3dmodStream**() with M_INQUIRE_SIZE_BYTE.

The content saved to memory stream is equivalent to the content saved to file. This function is equivalent to a file saved using M3dmodSave().

You can use this and other Aurora Imaging Library stream functions, for example, to save all required Aurora Imaging Library objects, as well as any other custom data, for your application to a memory stream. Once in a memory stream, you can write the stream to a single file or transfer it over a network. You are responsible for concatenating the streams and for saving the stream to file.

All of the 3D model finder context's settings that were in effect when the 3D model finder context was saved will be restored. A loaded or restored 3D model finder context is not preprocessed, therefore you must call M3dmodPreprocess() before performing a search with M3dmodFind().

Using **M3dmodStream**(), you can choose to save a backwards-compatible version of the 3D model finder context, which will work using a version of Aurora Imaging Library that is up to one major release older than the current version (depending on which version is specified). However, all settings and features unique to the higher version will be ignored when restored using the lower version. Besides saving backwards-compatible versions, you can also load or restore 3D model finder contexts saved using any Aurora Imaging Library version with which this function was available. Settings that do not exist in the lower version will be filled with default values when the 3D model finder context is loaded or restored.

Parameters

MemPtrOrFileName
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_TEXT_PTR [optionally, in C++: a reference to an AIL_STRING]
- Optionally, in C++: a reference to a std::vector of type AIL_UINT8

Specifies the file or memory stream.

See the Parameter associations section for possible values that can be returned/specified.

SysId

Specifies the system on which to restore the 3D model finder context.

See the Parameter associations section for possible values that can be specified.

Operation

Specifies the operation to perform.

See the Parameter associations section for possible values that can be specified.

StreamType

Specifies the type of stream in which to store/from which to restore the 3D model finder context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of stream																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FILE	Specifies a file stream.	c		e	f	g	a		i		j	k	l	m	n	b
M_MEMORY	Specifies a memory stream. You are responsible for allocating a block of memory for the stream.	c		e	f	g	a		i		j	k	l	m	n	b

Version

Specifies the Aurora Imaging Library version of the 3D model finder context.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion. This parameter must be set to **M_DEFAULT**.

Context3dmodIdPtr

Specifies the address of the variable in which to write or from which to read the identifier of the 3D model finder context. If the M_RESTORE operation is not successful, **M_NULL** is returned.

See the Parameter associations section for possible values that can be returned/specified.

SizeByteVarPtr

Specifies the address of the variable in which to write the size of the 3D model finder context, in bytes.

If the size is not required, you can set this parameter to **M_NULL**.

> Note that the size of the 3D model finder context will vary depending on the Aurora Imaging Library version specified.

Parameter associations

The table below lists possible values for the **MemPtrOrFileName**, **SysId**, **Operation**, **Version**, and **Context3dmodIdPtr** parameters.

- For performing the stream operation.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For performing the stream operation.

Operation	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
MemPtrOrFileName - Possible values passed or returned																
SysId																
Version																
Context3dmodIdPtr - Possible values passed or returned																
M_INQUIRE_SIZE_BYTE	Inquires the number of bytes required to save a 3D model finder context to memory stream. This operation is not supported when the StreamType parameter is set to M_FILE.															
MemPtrOrFileName	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version	Specifies the Aurora Imaging Library version with which the 3D model finder context must be compatible when saved. The size of a 3D model finder context might differ from one Aurora Imaging Library version to another.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the current version of Aurora Imaging Library.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP5	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 5.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP6	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 6.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP7	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 7.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_110	Specifies the version as being Aurora Imaging Library 11.0.	c		e	f	g	a		i		j	k	l	m	n	b
Context3dmodIdPtr	Specifies the address of the variable containing the identifier of the 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b
M_LOAD	Loads the content of a specified file or memory stream into a previously allocated 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b

MemPtrOrFileName	Specifies the file or memory stream from which to load the 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. 3D model finder contexts typically have an M3DMOD file extension. The function handles (internally) the opening and closing of the file. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which the object is located, and the block of memory must contain the entire object. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version	This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
Context3dmodIdPtr	Specifies the address of the variable containing the identifier of the 3D model finder context. All previous settings are overwritten with those of the loaded 3D model finder context, or are set to the default value if the settings didn't exist in the version of Aurora Imaging Library from which the 3D model finder context is being loaded.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESTORE	Restores a 3D model finder context from a file or memory stream and assigns it an Aurora Imaging Library identifier.	c		e	f	g	a		i		j	k	l	m	n	b
MemPtrOrFileName	Specifies the file or memory stream from which to restore the 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b

	M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
	AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. 3D model finder contexts typically have an M3DMOD file extension. The function handles (internally) the opening and closing of the file. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
	MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which the object is located, and the block of memory must contain the entire object. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
SysId		Specifies the system on which to allocate the 3D model finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
	System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
Version		This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
Context3dmodIdPtr		Specifies the address of the variable in which to write the identifier of the 3D model finder context. If the operation is not successful, **M_NULL** is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE		Saves a 3D model finder context to a specified file or memory stream.	c		e	f	g	a		i		j	k	l	m	n	b
MemPtrOrFileName		Specifies the file or memory stream.	c		e	f	g	a		i		j	k	l	m	n	b

		M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
		AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. For easier use with other Aurora Imaging software products, when saving a 3D model context to a file, use the M3DMOD file extension. The function handles (internally) the opening and closing of the file. If the file already exists, it will be overwritten. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
		MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which to write, and the block of memory must be large enough to stream the entire object. To determine the required size, call this function with M_INQUIRE_SIZE_BYTE. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
	SysId		This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
	Version		Specifies the Aurora Imaging Library version with which the 3D model finder context must be compatible when saved.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the current version of Aurora Imaging Library.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP5	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 5.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP6	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 6.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP7	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 7.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_110	Specifies the version as being Aurora Imaging Library 11.0.	c		e	f	g	a		i		j	k	l	m	n	b

	Context3dmodIdPtr	Specifies the address of the variable containing the identifier of the 3D model finder context to save.	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmod11.lib.
DLL	Requires ail11.dll; ail3dmod11.dll.